

The normalized Erdős–Straus quartic, prime-local Fable fibres, and exact channel transfers in a fixed weighted conductor

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20 September 2026

Abstract

An ordered positive solution of the Erdős–Straus equation determines a normalized binary quartic whose four literal roots are the prime and its three denominators. On the distinct-root locus this coefficient target lies in the genuine degree-eight finite étale domain of the four-dimensional Fable map. Every positive integral witness at a prime $p \equiv 1 \pmod{12}$ lies in that locus, and explicit fixed- p bounds separate its discriminant and derivatives from zero and bound all inverse coordinates. The prime is intrinsic in the unmarked coefficients, an explicit hypersurface equation characterizes the algebraic image on the relevant open chart, and the two inverse states marking the prime are given in every source coordinate. At fixed complex $p \neq 0$, over the exact open base $\mathcal{B}_p = \{(A, C) : ACd_p \text{Disc}(g_p) \neq 0\}$, the signed cover splits into ranks $2 + 6$, has monodromy group of order 48, and has deck group $C_2 \times C_2$. At the defining prime, the original rank-eight integral order has normalization index p^2 in the exterior channel and p^9 in the middle channel, with every Smith factor, conductor, reduction kernel and local signed-label character made explicit. Three quadratic-character conditions select exactly 3456 reduced classes modulo 521220 on which the original odd receiving frame is nonsingular for every positive integral witness. The actual witness $(13; 4, 18, 468)$ also gives an exact finite-field descent example: no signed lift over $\mathbb{F}_{61}$, eight lifts over $\mathbb{F}_{61^2}$, and eight $\mathbb{F}_{61}$ -points on the specified nonsquare twist. We also retain, as a different map with a different fibre, the earlier reciprocal-cubic suspension: it lies on the nonproper hyperplane and has seven states. At the original prime its exterior and middle fibres have respectively ramified and unramified quadratic factors despite having the same three rational points; their root orders differ by one exact index- p quotient. A negative-four-square source transformation changes the arithmetic target and produces original same-prime middle states on its exact divisor domain. For every integer coefficient $t > 0$, the complete fixed-target divisor box splits into canonical, companion and finite branches; the finite branch has the sharp bound $h \leq t^2 - 3t + 1$. That bound is attained by actual exterior states at infinitely many hard primes, while every fixed $t \geq 4$ also occurs on infinitely many exterior states whose complete same-grade middle box is empty. Replacing the inverse square-root coordinate by its reciprocal gives a finite flat rank-eight completion whose double-root fibre is $\mathbb{C}[\eta]/(\eta^4)$; the original meromorphic pole remains explicit. An invertible marked odd-moment frame identifies the separate receiving divisor exactly. An explicit positive-real solution ray crosses that divisor at one exactly isolated simple parameter, where the odd receiving matrix has rank three while the quartic remains squarefree; for integral roots, a retained integer gives a sharp zero-or-inverse-bound dichotomy. On the completed signed root algebra, a perfect residue pairing factors the trace pairing through multiplication by $2\eta^3$, locating the exact noninvertible boundary map. A native moment Gram is obtained from residue duality by its second-kind polynomial multiplier, while an arbitrary ES root polynomial retains a displayed relation defect. Finally, invertible linear maps carry the coefficient targets, fibres and exceptional loci into one fixed Meixner–Pollaczek weighted-conductor four-plane. These exact bridges neither prove universal Erdős–Straus existence nor identify arithmetic roots with zeros of the Riemann zeta function.

1 The arithmetic source and the four-dimensional map

Retain an ordered positive integral solution

$$\frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}, \quad p, x, y, z \in \mathbb{Z}_{>0}. \quad (\text{EZ1})$$

Put

$$s_1 = x + y + z, \quad s_2 = xy + xz + yz, \quad s_3 = xyz, \quad S = p + s_1. \quad (\text{EZ2})$$

Multiplication of EZ1 by $pxyz$, with no division by a possibly vanishing coordinate, gives the exact relation

$$ps_2 = 4s_3. \quad (\text{EZ3})$$

In source coordinates $(a, y_0, z_0, w_0) \in \mathbb{C}^4$, let

$$\begin{aligned} P_0 &= a^3 z_0 + 2a^2 y_0 - ia, \\ P_2 &= -a^3 y_0^2 z_0 - 2ia^2 y_0 z_0 + a^2 w_0 - 2a^2 y_0^3 - 10ia y_0^2 + 3a z_0 + y_0, \\ P_3 &= 2a^3 y_0^3 z_0 + 6ia^2 y_0^2 z_0 + 2a^2 w_0 y_0 + 4a^2 y_0^4 \\ &\quad + 2ia w_0 - 4ia y_0^3 - 2a y_0 z_0 + 2iz_0 + 7y_0^2, \\ P_4 &= 2a^3 y_0^4 z_0 + 8ia^2 y_0^3 z_0 + a^2 w_0 y_0^2 + 4a^2 y_0^5 \\ &\quad + 2ia w_0 y_0 + 7ia y_0^4 - 10a y_0^2 z_0 - 4iy_0 z_0 - w_0 - 3y_0^3, \\ P(a, y_0, z_0, w_0) &= (P_0, P_2, P_3, P_4). \end{aligned} \quad (\text{EZ4})$$

For a target $u = (u_0, u_2, u_3, u_4)$, write

$$H_u(U, V) = u_0 U^4 + U^3 V + u_2 U^2 V^2 + u_3 U V^3 + u_4 V^4, \quad h_u(T) = H_u(T, 1). \quad (\text{EZ5})$$

The full inverse, image complement and nonproper-value locus of EZ4 are proved in the cumulative source [10, GF1–GF18]. We state every inverse formula used below, so the crosswalk does not depend on an unnamed analogy.

2 The normalized prime–denominator quartic

Definition 2.1 (Arithmetic quartic). Define

$$\mathcal{J}(p; x, y, z) = -\frac{1}{S}(U - pV)(U - xV)(U - yV)(U - zV). \quad (\text{EZ6})$$

The minus sign and the factor S^{-1} are retained: they make the coefficient of $U^3 V$ exactly one, as required by EZ5.

Theorem 2.2 (Exact coefficients and intrinsic prime). *The coefficient target $J_4(p; x, y, z) = (u_0, u_2, u_3, u_4)$ of EZ6 is*

$$\boxed{u_0 = -\frac{1}{S}, \quad u_2 = -\frac{ps_1 + s_2}{S}, \quad u_3 = \frac{ps_2 + s_3}{S} = \frac{5s_3}{S}, \quad u_4 = -\frac{ps_3}{S}.} \quad (\text{EZ7})$$

In particular $u_0 u_3 u_4 \neq 0$, and the prime survives the passage to the unmarked coefficient target:

$$\boxed{p = -\frac{5u_4}{u_3}.} \quad (\text{EZ8})$$

Proof. Expanding the product in EZ6 gives

$$-S^{-1}(U^4 - SU^3V + (ps_1 + s_2)U^2V^2 - (ps_2 + s_3)UV^3 + ps_3V^4).$$

This proves the first expressions in EZ7. Relation EZ3 gives $ps_2 + s_3 = 5s_3$. All source integers are positive, so the displayed denominators and numerators do not vanish. Dividing the last two coefficient identities gives EZ8. $\square$

Theorem 2.3 (Coefficient hypersurface and exact converse). *On the open target chart $D(u_0u_3u_4)$, every arithmetic quartic satisfies*

$$\boxed{G(u) = 625u_0u_4^3 - 125u_3u_4^2 + 25u_2u_3^2u_4 - 4u_3^4 = 0.} \quad (\text{EZ9})$$

Conversely, if $u_0u_3u_4 \neq 0$ and EZ9 holds, then $p = -5u_4/u_3$ is a root of h_u . If the other roots in an algebraic closure are denoted x, y, z , counted with multiplicity, then

$$p(xy + xz + yz) = 4xyz. \quad (\text{EZ10})$$

When all four roots are nonzero, EZ10 is equivalent to EZ1.

Proof. Direct substitution of $p = -5u_4/u_3$ into EZ5 gives

$$h_u(p) = \frac{u_4}{u_3^4}G(u).$$

The prefactor is nonzero on the stated chart, proving both directions of the root assertion. Factor $h_u(T) = u_0(T - p)(T - x)(T - y)(T - z)$. Vieta's formulas give $u_3 = -u_0(ps_2 + s_3)$ and $u_4 = u_0ps_3$, where now $s_2 = xy + xz + yz$, $s_3 = xyz$. Equation EZ8, which follows here from the definition of p , becomes $p = 5ps_3/(ps_2 + s_3)$. Since $p \neq 0$, this is exactly EZ10. Dividing EZ10 by $pxyz$ proves the last assertion. $\square$

Remark 2.4 (What the hypersurface does not encode). The equation $G = 0$ is the Zariski equation of the normalized reciprocal relation on this open chart. It does not force the four roots to be positive integers, make p prime, select an ordering of x, y, z , or impose any exterior, middle, congruence or divisibility gate. These arithmetic conditions remain part of the source. No admission criterion is being invented for the published Fable theorem; this is the exact image statement for the new ES map.

3 The exact quotient chart and its singular boundary

The converse in EZ9 can be sharpened from a root statement to an isomorphism of affine charts. Retain all four symmetric source coordinates and put

$$\begin{aligned} \mathcal{X}_{\text{sym}} &:= \text{Spec } \frac{\mathbb{C}[p, s_1, s_2, s_3, (Sp s_3)^{-1}]}{(ps_2 - 4s_3)}, \quad S = p + s_1, \\ \mathcal{Y}_{\text{ES}} &:= V(G) \cap D(u_0u_3u_4). \end{aligned} \quad (\text{EZ9a})$$

Theorem 3.1 (Exact symmetric quotient). *The coefficient map*

$$\Theta(p, s_1, s_2, s_3) = \left(-\frac{1}{S}, -\frac{ps_1 + s_2}{S}, \frac{5s_3}{S}, -\frac{ps_3}{S} \right) \quad (\text{EZ9b})$$

is an isomorphism $\mathcal{X}_{\text{sym}} \simeq \mathcal{Y}_{\text{ES}}$. Its inverse is

$$\boxed{p = -\frac{5u_4}{u_3}, \quad s_1 = -\frac{1}{u_0} + \frac{5u_4}{u_3}, \quad s_2 = \frac{4u_3^2}{25u_0u_4}, \quad s_3 = -\frac{u_3}{5u_0}.} \quad (\text{EZ9c})$$

Equivalently,

$$\mathcal{Y}_{\text{ES}} \simeq \text{Spec } \mathbb{C}[u_0^{\pm 1}, u_3^{\pm 1}, u_4^{\pm 1}] \simeq \mathbb{G}_m^3, \quad (\text{EZ9c}')$$

with

$$u_2 = \frac{5u_4}{u_3} - \frac{25u_0u_4^2}{u_3^2} + \frac{4u_3^2}{25u_4}.$$

If the ordered source with coordinates (p, x, y, z) is retained, its map to $\mathcal{Y}_{\text{ES}}$ is exactly the quotient by the action of S_3 permuting (x, y, z) . It is finite flat of degree six. Where the cubic with roots x, y, z is squarefree, it is finite étale and has six reduced ordered points in every geometric fibre.

Proof. On $\mathcal{X}_{\text{sym}}$, the relation $ps_2 = 4s_3$ gives EZ9 by the calculation in the proof above, and $Sp s_3 \neq 0$ makes all four fractions in EZ9b regular with $u_0u_3u_4 \neq 0$.

Conversely, define the four source coordinates by EZ9c. Directly,

$$p + s_1 = -u_0^{-1}, \quad ps_2 = 4s_3.$$

The first, third and fourth coordinates of EZ9b then return u_0, u_3, u_4 . For the second coordinate one obtains

$$-\frac{ps_1 + s_2}{p + s_1} - u_2 = -\frac{G(u)}{25u_3^2u_4},$$

which vanishes exactly on $\mathcal{Y}_{\text{ES}}$. Substituting EZ9b into EZ9c returns p, s_1, s_2, s_3 , so the two regular maps are inverse.

The invariant ring of $\mathbb{C}[x, y, z]$ under the displayed S_3 -action is $\mathbb{C}[s_1, s_2, s_3]$. The relation $ps_2 = 4s_3$ and the localized element $Sp s_3$ are invariant; since the characteristic is zero, taking invariants commutes with this invariant quotient and localization. Thus the ordered map factors through precisely $\mathcal{X}_{\text{sym}}$, not through a smaller unidentified object. The symmetric-polynomial ring is free of rank $|S_3| = 6$ over its invariant ring, and this remains true after the invariant quotient and localization; the ordered map is therefore finite flat of degree six. When the cubic discriminant is nonzero the action on ordered roots is free, giving the stated étale locus. Finally, use (S, p, s_3) as source coordinates: the relation solves $s_2 = 4s_3/p$ and $s_1 = S - p$. All three are units, so the source coordinate ring is a three-variable Laurent ring. Applying EZ9c proves EZ9c' and the displayed formula for u_2 . $\square$

The inverse also gives an exact arithmetic recognition statement. For $u \in \mathcal{Y}_{\text{ES}}(\mathbb{Q})$, form the four quantities EZ9c and the retained cubic

$$Q_u(T) = T^3 - s_1T^2 + s_2T - s_3. \quad (\text{EZ9d})$$

In literal target coordinates this is

$$Q_u(T) = T^3 + \left(\frac{1}{u_0} - \frac{5u_4}{u_3}\right)T^2 + \frac{4u_3^2}{25u_0u_4}T + \frac{u_3}{5u_0}.$$

Before imposing $G = 0$, polynomial division retains the exact residual

$$u_0 \left(T + \frac{5u_4}{u_3}\right) Q_u(T) - h_u(T) = -\frac{G(u)}{25u_3^2u_4}T^2. \quad (\text{EZ9d}')$$

Thus on the hypersurface $u_0^{-1}h_u(T) = (T + 5u_4/u_3)Q_u(T)$, with every scale retained. Then u comes from an Erdős–Straus solution with prime numerator p and positive integral denominators if and only if the recovered p is a positive prime integer and Q_u splits over $\mathbb{Z}$ as $(T-x)(T-y)(T-z)$ with $x, y, z > 0$. This criterion neither discards nor supplies any congruence or divisibility gate from the earlier programme.

Theorem 3.2 (Irreducibility, singular support and collision divisors). *The polynomial G is irreducible in $\mathbb{C}[u_0, u_2, u_3, u_4]$. The scheme-theoretic singular subscheme of $V(G)$ is cut out by G and*

$$\begin{aligned} \partial_{u_0} G &= 625u_4^3, & \partial_{u_2} G &= 25u_3^2u_4, \\ \partial_{u_3} G &= -125u_4^2 + 50u_2u_3u_4 - 16u_3^3, & \partial_{u_4} G &= 1875u_0u_4^2 - 250u_3u_4 + 25u_2u_3^2. \end{aligned} \quad (\text{EZ9e})$$

Its reduced support is exactly the plane $V(u_3, u_4)$. Consequently $\mathcal{Y}_{\text{ES}}$ is smooth, and every positive arithmetic image lies in its smooth locus with the retained signs

$$u_0 < 0, \quad u_2 < 0, \quad u_3 > 0, \quad u_4 < 0. \quad (\text{EZ9f})$$

On this chart the cubic collision divisor is $\text{Disc}(Q_u) = 0$, with the s_i given exactly by EZ9c. Collision of the distinguished root p with one of the three denominator roots is the different divisor

$$Q_u(p) = -\frac{1250u_0u_4^3 + 3u_3^4 - 125u_3u_4^2}{5u_0u_3^3} = 0. \quad (\text{EZ9g})$$

The full quartic discriminant factors without loss of scale as

$$\text{Disc}(h_u) = u_0^6 \text{Disc}(Q_u)Q_u(p)^2. \quad (\text{EZ9h})$$

Proof. View G as a degree-one polynomial in u_0 over the UFD $R = \mathbb{C}[u_2, u_3, u_4]$. Its leading coefficient is $625u_4^3$, while its constant coefficient is $u_3(-125u_4^2 + 25u_2u_3u_4 - 4u_3^3)$. These two coefficients are coprime: u_4 does not divide the latter coefficient, as its value at $u_4 = 0$ is $-4u_3^4$. The polynomial is therefore primitive and irreducible over $\text{Frac}(R)$, hence irreducible over R .

The four derivatives are EZ9e. At any geometric singular point the first one forces $u_4 = 0$, and then the third forces $u_3 = 0$. Conversely all four derivatives and G vanish on that plane. This proves the statement about the reduced support while retaining the nonreduced Jacobian ideal in EZ9e. The arithmetic signs follow term by term from EZ7 and positivity of p, x, y, z, S .

The cubic discriminant is the square of $(x - y)(x - z)(y - z)$. Direct substitution of EZ9c into EZ9d at $T = p = -5u_4/u_3$ gives EZ9g. Finally the root formula for the discriminant of $u_0(T - p)Q_u(T)$ gives EZ9h, retaining the leading factor u_0^6 . $\square$

For completeness, the nonreduced singular scheme can also be retained exactly. Write $A = u_0, B = u_2, C = u_3, D = u_4$, and divide the four generators in EZ9e only by nonzero constants. Its Jacobian ideal is

$$J = (D^3, C^2D, 16C^3 - 50BCD + 125D^2, BC^2 - 10CD + 75AD^2). \quad (\text{EZ9i})$$

The identity

$$-AG_A + BG_B + 2CG_C + 3DG_D = 8G$$

shows that adding G does not enlarge this ideal. Its irredundant primary decomposition is

$$\begin{aligned} J &= Q_P \cap Q_E, \\ Q_P &= (C, D)^3 + (2BCD - 5D^2, BC^2 - 10CD + 75AD^2), \\ Q_E &= (B, D^3, 16C^3 + 125D^2, 2CD - 15AD^2). \end{aligned} \quad (\text{EZ9j})$$

Thus the singular scheme is a thickened copy of the plane $V(C, D)$ with an embedded line supported on $V(B, C, D)$. One exact elimination check is that both J and $Q_P \cap Q_E$ have the Gröbner basis

$$(75AD^2 + BC^2 - 10CD, BC^3, 50BCD - 16C^3 - 125D^2, C^4, C^2D, CD^2, D^3).$$

Moreover $(C, D)^3 \subset Q_P \subset (C, D)$, so Q_P is (C, D) -primary. In Q_E , the relations give $D^3 = CD^2 = C^2D = C^4 = 0$; hence $(B, C, D)^4 \subset Q_E \subset (B, C, D)$, proving that Q_E is (B, C, D) -primary.

The reduced boundary of the torus chart EZ9c' inside $V(G)$ has the three components

$$V(A, C), \quad V(A, -125D^2 + 25BCD - 4C^3), \quad V(C, D), \quad (\text{EZ9k})$$

and every pairwise or triple intersection is the line $V(A, C, D)$. This follows by setting each of A, C, D equal to zero in the unreduced polynomial (G); no boundary component meets the sign chamber EZ9f.

The finite-flat quotient retains collision multiplicities. Over an algebraically closed field of characteristic zero, a denominator pattern $1 + 1 + 1$ gives six reduced orderings; a pattern $2 + 1$ gives three length-two fibres; and a triple root gives the length-six coinvariant fibre

$$\text{Spec } \frac{\mathbb{C}[\xi, \eta]}{(\xi^2 + \xi\eta + \eta^2, \xi\eta(\xi + \eta))}.$$

Hence sorting positive real roots is a semialgebraic section on the strict real chamber, not a rational ordering of the generic cubic.

The collision divisors are not empty on positive integral solutions. For example,

$$(p; x, y, z) = (3; 1, 6, 6), \quad (5; 2, 5, 10), \quad (2; 1, 2, 2)$$

satisfy EZ1; respectively they have a denominator collision, a collision of the distinguished root with a denominator, and both kinds. Thus an arbitrary positive solution need not have an eight-point fibre. These collision divisors lie inside the smooth chart $\mathcal{B}_{\text{ES}}$: they are ramification/fibre-degeneration loci for the root and signed-factor covers, not singularities of the coefficient hypersurface. The Turn 7 exterior theorem below proves that its own image avoids both divisors.

4 The eight states and the prime-marked pair

Let $t_1 = p, t_2 = x, t_3 = y, t_4 = z$, and for each root put

$$\Delta_j = \prod_{k \neq j} (t_j - t_k). \quad (1)$$

Theorem 4.1 (Complete fibre). *If p, x, y, z are pairwise distinct, then J_4 lies in the finite étale domain*

$$\mathcal{B} = \{u : u_0 \text{Disc}(H_u) \neq 0\} \quad (\text{EZ11})$$

of EZ4, and $P^{-1}(J_4)$ consists of exactly eight points. For every $j \in \{1, 2, 3, 4\}$ and each of the two signs, choose

$$a_{j,\sigma}^2 = -\frac{S}{\Delta_j}, \quad a_{j,-} = -a_{j,+}. \quad (\text{EZ12})$$

The corresponding source point is

$$\begin{aligned} q_{j,\sigma} &= (a_{j,\sigma}, y_{j,\sigma}, z_{j,\sigma}, w_{j,\sigma}), \\ y_{j,\sigma} &= -t_j - \frac{i}{a_{j,\sigma}}, \\ z_{j,\sigma} &= \frac{u_0}{a_{j,\sigma}^3} + \frac{2t_j}{a_{j,\sigma}} + \frac{3i}{a_{j,\sigma}^2}, \\ w_{j,\sigma} &= \frac{7it_j^2}{a_{j,\sigma}} + \frac{u_2 - 17t_j + u_0t_j^2}{a_{j,\sigma}^2} - \frac{13i}{a_{j,\sigma}^3} - \frac{2u_0}{a_{j,\sigma}^4}. \end{aligned} \quad (\text{EZ13})$$

These eight points are distinct and exhaust the fibre.

Proof. The leading coefficient is $u_0 = -S^{-1} \neq 0$, and

$$\text{Disc}(H_{J_4}) = S^{-6} \prod_{1 \leq j < k \leq 4} (t_j - t_k)^2. \quad (\text{EZ14})$$

Thus the distinctness hypothesis is exactly EZ11. At the root t_j ,

$$h'_{J_4}(t_j) = -S^{-1} \Delta_j,$$

so the inverse equation $a^2 h'(t_j) = 1$ is precisely EZ12. Comparing coefficients in a factorization $H = LC$, where $L = a(U - t_j V)$, gives EZ13. Substitution into EZ4 returns J_4 . Conversely, the complete inverse theorem for EZ4 says that on $u_0 \neq 0$ every source point selects a simple root $t = -b/a$ and a sign satisfying $a^2 h'(t) = 1$. Four simple roots and two signs therefore give exactly eight points, with no $a = 0$ chart. $\square$

Corollary 4.2 (The two prime-marked lifts). *Put*

$$\Delta_p = (p - x)(p - y)(p - z), \quad \lambda^2 = -\frac{S}{\Delta_p}, \quad \mu = -\frac{1}{\lambda S}. \quad (\text{EZ15})$$

Then

$$L = \lambda(U - pV), \quad C = \mu(U - xV)(U - yV)(U - zV) \quad (\text{EZ16})$$

satisfy

$$LC = \mathcal{J}(p; x, y, z), \quad \text{Res}(L, C) = 1, \quad ad + bc = 1, \quad 4af - be = 0. \quad (\text{EZ17})$$

The two choices of λ are exactly $q_{p,+}$, $q_{p,-}$, and the rational deck involution exchanges them. The other six fibre points mark x, y, z ; they are not additional prime-marked ES inverses.

Proof. The product identity follows from $\lambda\mu = -S^{-1}$. In the coefficient convention $L = aU + bV$, $C = cU^3 + dU^2V + eUV^2 + fV^3$, EZ16 gives

$$(a, b, c, d, e, f) = (\lambda, -\lambda p, \mu, -\mu s_1, \mu s_2, -\mu s_3).$$

The retained resultant orientation is

$$\text{Res}(L, C) = C(-b, a) = \lambda^3 \mu \Delta_p = -\lambda^2 \Delta_p / S = 1.$$

Furthermore $ad + bc = -\lambda\mu(s_1 + p) = 1$, and $4af - be = \lambda\mu(ps_2 - 4s_3) = 0$ by EZ3. The source formula in EZ13 is the inverse of these six factor coordinates. The deck statement is GF18 of [10]. $\square$

5 The complete Turn 7 exterior map

The exterior source retained in the Turn 7 work has

$$a = hrs, \quad u = hr^2, \quad R = 4a - p, \quad D = \frac{4u + 1}{R} = \frac{r + \kappa}{s}, \quad R\kappa = pr + s, \quad (\text{EZ18})$$

and ordered denominators

$$(x, y, z) = (hrs, hsk, phrk). \quad (\text{EZ19})$$

Theorem 5.1 (Exterior states land in the eight-sheet domain). *Every actual Turn 7 exterior state maps by EZ19 and EZ6 into $\mathcal{B}$. More precisely,*

$$x < p, qquad x < y < z, qquad z > p, qquad y \neq p, \quad (\text{EZ20})$$

so p, x, y, z are pairwise distinct and EZ14 is strictly positive.

Proof. The exterior inequalities give $p/4 < a = x < p/2$, hence $x < p$. The retained normalization gives $r < \kappa$, so $y/x = \kappa/r > 1$. Since $s = a/g \leq a < p/2$, while $pr \geq p$, we have $z/y = pr/s > 1$. Also $z/p = hr\kappa > 1$. Finally $p \nmid h$ and $p \nmid s$, because both divide the positive integer $a < p$. If $p \mid \kappa$, then $R\kappa = pr + s$ would imply $p \mid s$, a contradiction. Therefore $p \nmid h s \kappa = y$, so $y \neq p$. This proves EZ20 and the claim. $\square$

The marked shape curve retains

$$X = v, \quad Y = 2a, \quad R = X - \alpha, \quad D = X - \beta, \quad p = 2Y - R, \quad u = (RD - 1)/4. \quad (\text{EZ21})$$

On its stated integral image, the inverse

$$g = \gcd(a, u), \quad (h, r, s) = (g^2/u, u/g, a/g), \quad \kappa = (pr + s)/R \quad (\text{EZ22})$$

followed by EZ19 and EZ6 is the complete typed map from the Turn 7 marked cubic to J_4 . Algebraically, the coefficient target forgets the action of S_3 permuting x, y, z . On the actual strict real chamber EZ20, sorting the three non-prime roots restores $x < y < z$. One then reconstructs

$$R = 4x - p, \quad u = \frac{xz}{py} = \frac{x^2}{Ry - px}, \quad D = \frac{4u + 1}{R}, \quad v = \frac{x^2}{u}, \quad (X, Y) = (v, 2x), \quad (\text{EZ23})$$

and EZ22 restores every exterior marking. Outside the arithmetic chamber, there is no asserted global rational inverse from an unmarked root set to the ordered integral source.

6 The reciprocal suspension is a different exact bridge

For comparison, an ordered ES solution also gives reciprocal roots

$$t_1 = p/x, \quad t_2 = p/y, \quad t_3 = p/z, \quad t_1 + t_2 + t_3 = 4, \quad e_2 = \sum_{i < j} t_i t_j, \quad e_3 = t_1 t_2 t_3. \quad (\text{EZ24})$$

The normalized cubic

$$C_{\text{ES}}(U, V) = -\frac{1}{4} \prod_{j=1}^3 (U - t_j V)$$

has the canonical suspension

$$\mathfrak{S}(C_{\text{ES}}) = -4VC_{\text{ES}} = V \prod_{j=1}^3 (U - t_j V), \quad qquad J_3 = (0, -4, e_2, -e_3). \quad (\text{EZ25})$$

If the three roots are distinct, the fibre has exactly seven points: two over each finite root and the single infinity-chart point

$$q_\infty = \left(0, -4, \frac{i(112 - e_2)}{2}, -704 + 8e_2 + e_3 \right). \quad (\text{EZ26})$$

For $\Delta_j = \prod_{k \neq j} (t_j - t_k)$, the six finite points are EZ13 with $u_0 = 0, u_2 = -4, t_j$ as root and $a^2 = \Delta_j^{-1}$. Its discriminant is

$$\prod_{i < j} (t_i - t_j)^2 = \frac{p^6 (x - y)^2 (x - z)^2 (y - z)^2}{x^4 y^4 z^4}. \quad (\text{EZ27})$$

Unlike J_4 , every J_3 lies on the $u_0 = 0$ component of the exact nonproper locus

$$\mathcal{N}(P) = \{u : u_0 \text{ Disc}(H_u) = 0\}. \quad (\text{EZ28})$$

Thus the eight-state quartic and seven-state suspension retain different data, fibres and exceptional loci. The exact morphism between them is calculated below; neither is substituted for the other.

7 The exact elementary transform between the two quartics

The apparent change from eight to seven affine states does not make the two constructions unrelated. It records an exact marked-root elementary modification and a separately calculable square-class obstruction on the signed covers.

Theorem 7.1 (Marked-root transform to the reciprocal suspension). *Work over a characteristic-zero field containing i , and restrict to*

$$p s_3 S \prod_{i < j} (x_i - x_j) \prod_i (p - x_i) \neq 0. \quad (\text{EZ28A})$$

For $t_i = p/x_i$, let $f(T) = \prod_i (T - t_i) = T^3 - 4T^2 + e_2T - e_3$. Then

$$s_1 = \frac{pe_2}{e_3}, \quad s_2 = \frac{4p^2}{e_3}, \quad s_3 = \frac{p^3}{e_3}, \quad S = \frac{p(e_2 + e_3)}{e_3}, \quad (\text{EZ28B})$$

and the two binary quartics obey the literal identity

$$H_{\text{rec}}(U, V) = \frac{1}{u_4} \frac{V}{U - V} \mathcal{J}(p; pV, U) = V \prod_{i=1}^3 \left(U - \frac{p}{x_i} V \right). \quad (\text{EZ28C})$$

Here $\mathcal{J}(p; pV, U)$ means substitution $(U, V) \mapsto (pV, U)$ in EZ6; it is divisible by $U - V$ because the marked root p maps to 1.

On symmetric coefficient charts,

$$e_2 = \frac{125u_4^2(u_3 - 5u_0u_4)}{u_3^4}, \quad e_3 = \frac{625u_0u_4^3}{u_3^4}, \quad e_2 + e_3 = \frac{125u_4^2}{u_3^3}, \quad (\text{EZ28D})$$

$$u_0 = -\frac{e_3}{p(e_2 + e_3)}, \quad u_2 = -\frac{p(e_2 + 4)}{e_2 + e_3}, \quad u_3 = \frac{5p^2}{e_2 + e_3}, \quad u_4 = -\frac{p^3}{e_2 + e_3}. \quad (\text{EZ28E})$$

Put

$$\Delta_f = \text{Disc}(T^3 - 4T^2 + e_2T - e_3), \quad f(1) = e_2 - e_3 - 3, \\ \mathcal{Z}^\circ = \text{Spec } k[e_2, e_3, (e_3(e_2 + e_3)\Delta_f f(1))^{-1}].$$

The distinct-root arithmetic coefficient chart is

$$\boxed{\mathbb{G}_{m,p} \times \mathcal{Z}^\circ}. \quad (\text{EZ28F})$$

Projection to the reciprocal target forgets exactly the common scale p ; its scheme-theoretic fibre is $\mathbb{G}_m$. In quartic coordinates that scaling is

$$\lambda \cdot (u_0, u_2, u_3, u_4) = (\lambda^{-1}u_0, \lambda u_2, \lambda^2 u_3, \lambda^3 u_4), \quad (\text{EZ28G})$$

and e_2, e_3 are invariant.

Let $Q(X) = X^3 - s_1X^2 + s_2X - s_3$. There is a canonical isomorphism of degree-four projective root schemes

$$\{p\} \sqcup \text{Spec } k[X]/(Q) \xrightarrow{\sim} \{\infty\} \sqcup \text{Spec } k[T]/(f), \quad (\text{EZ28H})$$

given by $p \mapsto \infty$, $x_i \mapsto p/x_i$. On the finite coordinate rings it is

$$T \mapsto \frac{p}{s_3}(X^2 - s_1X + s_2), \quad X \mapsto \frac{p}{e_3}(T^2 - 4T + e_2). \quad (\text{EZ28I})$$

The collision divisors and discriminants transform as

$$Q(p) = -\frac{p^3}{e_3}f(1), \quad \text{Disc}(Q) = \frac{p^6}{e_3^4}\text{Disc}(f), \quad \text{Disc}(\mathcal{J}) = \frac{p^6 f(1)^2 \text{Disc}(f)}{(e_2 + e_3)^6}. \quad (\text{EZ28J})$$

No Möbius transformation realizes EZ28H: the unique one carrying the three distinct x_i to p/x_i is $T \mapsto p/T$, which sends p to 1, not to ∞ . Thus EZ28C is exactly inversion, deletion of the marked root 1, insertion of ∞ , and coefficient normalization.

Proof. The relation $ps_2 = 4s_3$ gives $\sum_i p/x_i = 4$, and the remaining formulas in EZ28B follow directly from $x_i = p/t_i$. Substitution in EZ6 gives

$$\mathcal{J}(p; pV, U) = \frac{p}{S}(U - V) \prod_i (pV - x_i U).$$

Since $\prod_i (pV - x_i U) = -s_3 \prod_i (U - (p/x_i)V)$ and $u_4 = -ps_3/S$, this proves EZ28C with its sign and scale.

Now $e_2 = p^2 s_1/s_3$, $e_3 = p^3/s_3$. Combining these identities with EZ8 and the inverse formulas EZ9b gives EZ28D. Conversely EZ28B substituted in EZ7 gives EZ28E. The two substitutions are inverse on EZ28A, proving EZ28F; simultaneous scaling of p, x_1, x_2, x_3 proves EZ28G.

Modulo Q ,

$$X^{-1} = \frac{X^2 - s_1 X + s_2}{s_3};$$

modulo f ,

$$T^{-1} = \frac{T^2 - 4T + e_2}{e_3}.$$

Multiplication by X and T , respectively, shows that the two maps in EZ28I are inverse, proving EZ28H. Finally

$$x_i - x_j = \frac{p(t_j - t_i)}{t_i t_j}, \quad p - x_i = \frac{p(t_i - 1)}{t_i}.$$

Multiplication of all retained differences gives EZ28J. Three-point uniqueness in PGL_2 proves the final obstruction. $\square$

Theorem 7.2 (Exact signed-cover correspondence and obstruction). *Let*

$$\mathcal{F}_8 = \{(r, A) : h_u(r) = 0, A^2 h'_u(r) = 1\},$$

with the remaining three Fable source coordinates recovered by EZ13, and let

$$\mathcal{F}_7 = \{q_\infty\} \sqcup \{(t, \xi) : f(t) = 0, \xi^2 f'(t) = 1\}.$$

Use EZ28H to form the fibre product

$$\mathcal{C} = \mathcal{F}_8 \times_{\mathcal{R}_4} \mathcal{F}_7. \tag{EZ28K}$$

It has fourteen geometric points: two over the prime/infinity branch and four over each denominator branch. The projection $\mathcal{C} \rightarrow \mathcal{F}_7$ is finite étale of degree two; the projection to $\mathcal{F}_8$ has degree one over the two prime-marked states and degree two over each denominator-marked state.

For $t_i = p/x_i$, the exact derivative twists are

$$\frac{h'_u(x_i)}{f'(t_i)} = \frac{x_i s_3 (p - x_i)}{S p^2} = \frac{p^2 (t_i - 1)}{t_i^2 (e_2 + e_3)}, \quad h'_u(p) = \frac{p^2 f(1)}{e_2 + e_3}. \tag{EZ28L}$$

There is generically no rational morphism $\mathcal{F}_8 \rightarrow \mathcal{F}_7$ covering EZ28H: the first ratio in EZ28L has odd valuation along the divisor $t_i = 1$, so it is not a square in the ordered-base function field.

After adjoining

$$c_i^2 = \frac{h'_u(x_i)}{f'(t_i)}, \tag{EZ28M}$$

the denominator branches are isomorphic by

$$(x_i, A_i) \longmapsto (t_i, \xi_i = c_i A_i),$$

followed by the full inverse coordinates

$$\begin{aligned} y'_i &= -t_i - \frac{i}{\xi_i}, & z'_i &= \frac{2t_i}{\xi_i} + \frac{3i}{\xi_i^2}, \\ w'_i &= \frac{7it_i^2}{\xi_i} + \frac{-4 - 17t_i}{\xi_i^2} - \frac{13i}{\xi_i^3}. \end{aligned}$$

The projective factor-incidence completion over ∞ has two signs $b = \pm i$. After adjoining $d_p^2 = h'_u(p)$, the prime states map to them by

$$b = \text{id}_p A_p. \quad (\text{EZ28N})$$

The affine chart $b = i + \xi y$ retains only $b = i$ when $\xi = 0$. Consequently the projective completion restores an eight-to-eight correspondence after the displayed quadratic trivializations. In the affine seven-state chart the induced surjection has a two-point fibre over q_∞ and singleton fibres elsewhere: it forgets exactly the prime sign.

Proof. Equation EZ28L follows by differentiating both products and substituting EZ28B. Over each denominator root, a signed map would have to identify the quadratic extensions generated by $h'_u(x_i)^{-1/2}$ and $f'(t_i)^{-1/2}$, which is equivalent to the ratio in EZ28L being a square. At the generic valuation associated with $t_i = 1$, its numerator has order one and $t_i^2(e_2 + e_3)$ has order zero. This proves the obstruction. After EZ28M, $(c_i A_i)^2 f'(t_i) = A_i^2 h'_u(x_i) = 1$, proving the finite branch map and its inverse.

For a projective infinity factor $L = bV$, the complementary cubic has leading coefficient b^{-1} , and the resultant condition is

$$C(-b, 0) = -b^2 = 1.$$

Thus $b = \pm i$. Since $(d_p A_p)^2 = 1$, EZ28N is a bijection between the two prime signs and the two infinity signs. The affine-chart equation fixes only the positive sign. Counting the independent signed choices over each root gives $2 + 3 \cdot 4 = 14$ points in EZ28K and the stated projection degrees. $\square$

Proposition 7.3 (Boundary resolution of the marked transform). *On the marked incidence space*

$$\mathfrak{J} = \{(H, p) : H(p, 1) = 0, p \neq 0, H \neq 0\},$$

the unnormalized map

$$(H, p) \mapsto V \frac{H(pV, U)}{U - V}$$

is regular. Its U^3V -coefficient is u_4 , so the projective binary form extends across $u_4 = 0$, whereas the affine normalization EZ28C exists exactly on $D(u_4)$. On the unmarked closure $V(G)$, the marked root is the rational map

$$u \mapsto [-5u_4 : u_3].$$

Its indeterminacy locus is $V(u_3, u_4)$, and its graph is the blow-up of the ideal (u_3, u_4) . The full reciprocal coefficient map has projective form

$$u \mapsto [u_3^4 : 125u_4^2(u_3 - 5u_0u_4) : 625u_0u_4^3], \quad (\text{EZ28O})$$

whose graph closure is the blow-up of the displayed three-generator base ideal.

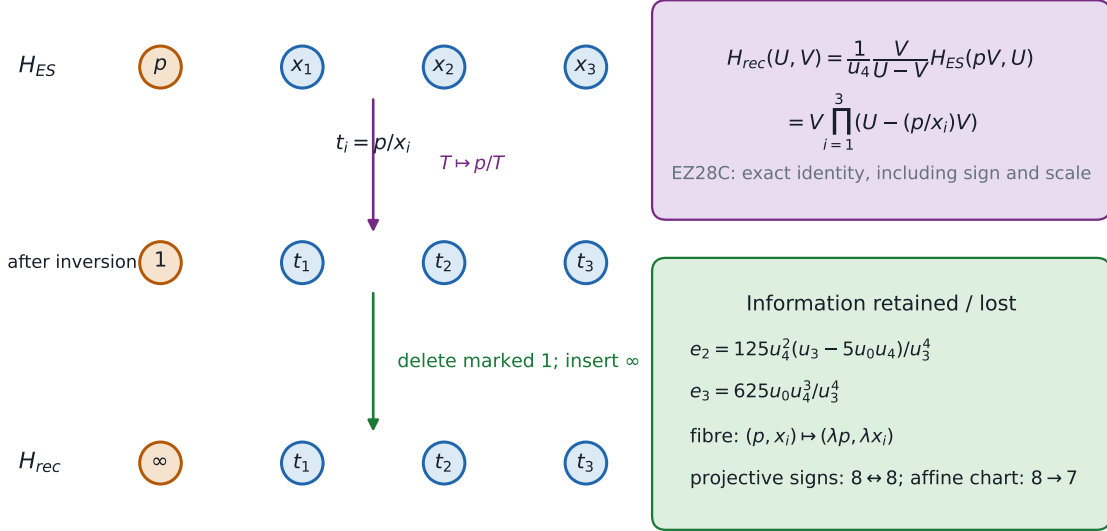
Proof. On $\mathfrak{J}$, polynomial division by the monic marked factor $U - V$ is regular in the coefficients; the leading coefficient is read directly from EZ28C. The unmarked-root formula is EZ8 in homogeneous coordinates, so the standard graph construction of a rational map to $\mathbb{P}^1$ is the blow-up of its base ideal. Clearing the denominators in EZ28D gives EZ28O, and the same graph construction proves the final statement. $\square$

This exact transform also reaches the integral calculation below. In E, $v_p(e_3) = 2$; in M, $v_p(e_3) = 1$. Therefore the inverse root-algebra formula

$$X = \frac{p}{e_3}(T^2 - 4T + e_2)$$

has coefficient valuation -1 in E and 0 in M. Exterior evaluations remain integral precisely because the root order supplies the congruence calculated in EZ28l. This is the exact morphism and its information loss; it is not the unrelated weighted analytic conductor of EZ29–35.

Exact elementary transform: literal roots to reciprocal suspension



Schematic of the proved projective root operation; horizontal spacing is not a metric.

Figure 1: The exact elementary transform EZ28C–EZ28O. The drawing records inversion $T \mapsto p/T$, deletion of the marked root 1, insertion of the root at infinity, the retained normalization, the common-scale fibre and the signed-cover information loss. Horizontal spacing is schematic and is not used as a metric. Reproduce with `tools/generate_20260920_illustrations.py`.

8 The prime-local seven-state fibre and its integral order

We now specialize EZ24–28 to the original shell coordinates rather than to an arbitrary reciprocal cubic. In this section A denotes the original shell denominator, so it is not the Fable source coordinate a of EZ4. Fix a prime $p \equiv 1 \pmod{4}$ and retain

$$\frac{p}{4} < A < \frac{p}{2}, \quad R = 4A - p, \quad u \mid A^2, \quad E : R \mid 4u + 1, \quad M : R \mid u + A. \quad (\text{EZ28a})$$

Put

$$g = \gcd(A, u), \quad (h, r, s) = \left(\frac{g^2}{u}, \frac{u}{g}, \frac{A}{g} \right), \quad A = hrs, \quad u = hr^2, \quad \gcd(r, s) = 1. \quad (\text{EZ28b})$$

The ordered denominator maps, including their channel labels, are

$$\begin{aligned} E : \quad \kappa &= \frac{pr + s}{R}, & (x_1, x_2, x_3) &= (A, h s \kappa, p h r \kappa), \\ M : \quad \lambda &= \frac{r + s}{R}, & (x_1, x_2, x_3) &= (A, p h s \lambda, p h r \lambda). \end{aligned} \quad (\text{EZ28c})$$

They are the retained Type-I/II coordinates of Elsholtz–Tao [1, Section 2, Propositions 2.2 and 2.6]; multiplication gives EZ1, and the original divisor is recovered respectively by

$$u = \frac{A^2}{Rx_2 - pA}, \quad u = \frac{pA^2}{Rx_2 - pA}. \quad (\text{EZ28d})$$

Lemma 8.1 (The original character survives the reciprocal map). *For every positive divisor $d \mid A^2$, with Jacobi symbol on the left,*

$$\left(\frac{d}{R}\right) = \left(\frac{d}{p}\right). \quad (\text{EZ28e})$$

Consequently an E state has $(h/p) = -1$, whereas an M state has $(rs/p) = -1$.

Proof. For every odd prime $\ell \mid A$, $R \equiv -p \pmod{\ell}$, while $R \equiv 3 \pmod{4}$ and $p \equiv 1 \pmod{4}$. Quadratic reciprocity therefore gives

$$\left(\frac{\ell}{R}\right) = \left(\frac{-1}{\ell}\right) \left(\frac{R}{\ell}\right) = \left(\frac{p}{\ell}\right) = \left(\frac{\ell}{p}\right).$$

If $2 \mid A$, then $R \equiv -p \pmod{8}$, and the supplementary law gives $(2/R) = (2/p)$. Multiplication with the original exponents proves EZ28e. In E , $u \equiv -1/4 \pmod{R}$, hence $(u/R) = -1$; since $u = hr^2$, $(h/p) = -1$. In M , $r \equiv -s \pmod{R}$, so $(r/p) = (r/R) = -(s/R) = -(s/p)$, proving the last claim. The reciprocity and supplementary laws used here are stated in Hatcher [2, Section 6.4]. $\square$

For the ordered reciprocal roots $t_i = p/x_i$, write $f(T) = \prod_i (T - t_i)$ and $d_i = f'(t_i)$. The roots are simple. In E , $\kappa - r = ((p - R)r + s)/R > 0$, $p\kappa > s$, and $pr > s$. In M , $p\lambda > r$, s ; equality $r = s$ would force $r = s = 1$, whence $R \mid 2$, impossible for the positive integer $R \equiv 3 \pmod{4}$.

Theorem 8.2 (Complete prime-local fibre algebra). *Choose and retain one embedding $i \mapsto i_p \in \mathbb{Q}_p$. The complete affine fibre of EZ4 over the reciprocal suspension EZ25 has coordinate algebra*

$$\mathcal{A}_{p,A,u} = \mathbb{Q}_p \times \prod_{i=1}^3 \mathbb{Q}_p[\xi_i]/(\xi_i^2 - d_i^{-1}). \quad (\text{EZ28f})$$

It is reduced of dimension seven and has seven geometric points. For an original state its field-factor type is

$$E : \mathcal{A}_{p,A,u} \simeq \mathbb{Q}_p^3 \times K_{\text{ram}}^2, \quad M : \mathcal{A}_{p,A,u} \simeq \mathbb{Q}_p^3 \times K_{\text{unr}}^2. \quad (\text{EZ28g})$$

Here the two ramified factors for a fixed E state are isomorphic, but the field is state-dependent and no canonical identification is asserted; K_{unr} is the unique unramified quadratic extension of $\mathbb{Q}_p$. In either channel there are exactly three $\mathbb{Q}_p$ -rational affine points.

Proof. Write the marked factor pair as $L = \xi U + bV$, $C = cU^3 + dU^2V + eUV^2 + f_0V^3$. If $\xi \neq 0$, the identity $LC = V \prod_j (U - t_j V)$ forces, for one unique label i ,

$$L = \xi(U - t_i V), \quad C = \xi^{-1} V \prod_{j \neq i} (U - t_j V).$$

Since $b = -\xi t_i$, the retained resultant condition is exactly

$$1 = C(-b, \xi) = \xi^2 \prod_{j \neq i} (t_i - t_j) = \xi^2 d_i.$$

Conversely either root of $\xi^2 = d_i^{-1}$ reconstructs the unique source point by EZ13. At $\xi = 0$, the four target equations successively force the single point EZ26. The source Jacobian is the constant -2 , a p -adic unit, so all factors are reduced. This proves EZ28f.

For E put $H = hrs\kappa$. In the original order

$$(t_1, t_2, t_3) = H^{-1}(p\kappa, pr, s), \quad (\text{EZ28h})$$

and direct subtraction gives

$$(d_1, d_2, d_3) = H^{-2}(p(\kappa - r)(p\kappa - s), -p(\kappa - r)(pr - s), (p\kappa - s)(pr - s)). \quad (\text{EZ28i})$$

The identity $pD + 1 = 4hr\kappa$ makes κ a unit. If $p \mid \kappa - r$, it would give $4hr^2 \equiv 1 \pmod{p}$, contradicting $(h/p) = -1$. Hence

$$(v_p(d_1), v_p(d_2), v_p(d_3)) = (1, 1, 0), \quad \frac{d_1}{d_2} \equiv -1, \quad d_3 \equiv 16 \pmod{p}. \quad (\text{EZ28j})$$

Because -1 is a square in $\mathbb{Q}_p$, the first two quadratic factors are isomorphic ramified fields and the third splits.

For M put $H = hrs\lambda$. Then

$$(t_1, t_2, t_3) = H^{-1}(p\lambda, r, s), \quad (d_1, d_2, d_3) \sim (rs, r(r - s), s(s - r)) \pmod{p}, \quad (\text{EZ28k})$$

where the omitted common factor H^{-2} is a square unit. By EZ28e, d_1 is a nonsquare unit, while $d_1 d_2 d_3 \sim -r^2 s^2 (r - s)^2$ is a square. Exactly one of labels 2, 3 is square and the other is nonsquare; the complement $r \leftrightarrow s$ exchanges those labels. Thus one finite quadratic splits and two give the same unramified quadratic field. Adding the infinity factor proves EZ28g and the count of three rational points. The square-unit lift and the ramified/unramified classification used here are the elementary Hensel and local-extension results in Milne [5, Chapter 7]; explicitly, a unit square root lifts by

$$b_{n+1} = b_n + p^n t, \quad t \equiv -(b_n^2 - z)p^{-n}(2b_n)^{-1} \pmod{p}.$$

□

Theorem 8.3 (Exact integral root order and one-prime loss). *Let $\mathcal{O}_E$ and $\mathcal{O}_M$ be the images of $\mathbb{Z}_p[T]/(f)$ under evaluation at the ordered roots. Then*

$$\boxed{\mathcal{O}_E = \{(z_1, z_2, z_3) \in \mathbb{Z}_p^3 : z_1 \equiv z_2 \pmod{p}\}, \quad \mathcal{O}_M = \mathbb{Z}_p^3.} \quad (\text{EZ28l})$$

The normalization quotient and conductor are

$$0 \longrightarrow \mathcal{O}_E \longrightarrow \mathbb{Z}_p^3 \xrightarrow{z_1 - z_2 \bmod p} \mathbb{F}_p \longrightarrow 0, \quad \mathfrak{f}_E = p\mathbb{Z}_p \times p\mathbb{Z}_p \times \mathbb{Z}_p. \quad (\text{EZ28m})$$

The sequence has no $\mathbb{Z}_p$ -linear right inverse. For any same-prime E and M states, the unique labelled quadratic interpolations $F(t_i^E) = t_i^M$ and $G(t_i^M) = t_i^E$ satisfy

$$(v_p(F_0), v_p(F_1), v_p(F_2)) = (0, -1, -1), \quad G \in \mathbb{Z}_p[T], \quad pF(T) \equiv cT(T - 4) \pmod{p} \quad (\text{EZ28n})$$

for a nonzero $c \in \mathbb{F}_p$. Their composition congruences are divisibilities in $\mathbb{Q}_p[T]$, not generally in $\mathbb{Z}_p[T]$.

Proof. In E, EZ28h–j give $v_p(t_1 - t_2) = 1$, the other two difference valuations zero, and $(t_1, t_2, t_3) \equiv (0, 0, 4) \pmod{p}$. Evaluation therefore forces the congruence in EZ28l. Conversely, for a tuple satisfying it, the polynomial

$$L(T) = z_1 + \frac{z_2 - z_1}{t_2 - t_1}(T - t_1)$$

is integral, and

$$L(T) + \frac{z_3 - L(t_3)}{(t_3 - t_1)(t_3 - t_2)}(T - t_1)(T - t_2)$$

is an integral interpolant. In M all three differences are units, so the Vandermonde interpolation is integral and surjective. The quotient map in EZ28m is therefore exact. An element of $\mathbb{Z}_p^3$ multiplies all of $\mathbb{Z}_p^3$ into $\mathcal{O}_E$ exactly when its first two coordinates are divisible by p , proving the conductor formula. A right inverse $\mathbb{F}_p \rightarrow \mathbb{Z}_p^3$ would have p -torsion image, but $\mathbb{Z}_p^3$ is torsion-free.

For interpolation write

$$F(T) = \sum_i t_i^M \frac{(T - t_j^E)(T - t_k^E)}{(t_i^E - t_j^E)(t_i^E - t_k^E)}.$$

The E denominators have valuations $(1, 1, 0)$, whereas the M roots have valuations $(1, 0, 0)$. Multiplication by p leaves only the $i = 2$ term modulo p , giving the last congruence in EZ28n and the linear and quadratic valuations -1 . The three contributions to the constant term have valuations $1, 0, 2$; the unique minimum cannot cancel, so $v_p(F_0) = 0$. All M difference denominators are units, proving $G \in \mathbb{Z}_p[T]$. Each composition fixes three distinct roots, hence $f_E \mid G(F(T)) - T$ and $f_M \mid F(G(T)) - T$ in $\mathbb{Q}_p[T]$. The exact denominator just proved prevents upgrading this assertion in general to $\mathbb{Z}_p[T]$. $\square$

Corollary 8.4 (The same rational point count hides different specialization). *For E,*

$$v_p(\sigma_2, \sigma_3, \text{Disc } f) = (1, 2, 2), \quad \bar{f} = T^2(T - 4). \quad (\text{EZ28o})$$

The four signed branches over t_1, t_2 have

$$(v_p(\xi), v_p(y_0), v_p(z_0), v_p(w_0)) = (-1/2, 1/2, 1, 1), \quad (\text{EZ28p})$$

so they escape the affine integral chart. The reduced special fibre consists exactly of

$$\begin{aligned} &(0, -4, 56i_p, -704), \\ &(1/4, -4 - 4i_p, 32 + 48i_p, -1152 - 384i_p), \\ &(-1/4, -4 + 4i_p, -32 + 48i_p, -1152 + 384i_p). \end{aligned} \quad (\text{EZ28q})$$

For M, $v_p(\sigma_2, \sigma_3, \text{Disc } f) = (0, 1, 0)$; all seven geometric branches are integral and remain distinct. In both channels exactly three special-fibre points are $\mathbb{F}_p$ -rational. The M roots have distinct reductions and all derivatives are units; their derivative residues need not be pairwise distinct.

Proof. For E, $\kappa + r = sD$, D is a unit, and the elementary symmetric expressions from EZ28h give the valuations in EZ28o. EZ28j and the inverse EZ13 have unique lowest-valuation terms $-i_p/\xi$, $3i_p/\xi^2$, and $-4/\xi^2$, proving EZ28p. Reduction has $t_3 = 4$, $d_3 = 16$, so $\xi = \pm 1/4$; substitution in EZ13 and EZ26 gives EZ28q. The double root zero has no finite inverse because $\xi^2 \bar{f}'(0) = 1$ is impossible. For M, EZ28k has three distinct root reductions and unit derivatives, proving the remaining assertions. For example, the valid state $p = 5, (A, u, R, h, r, s, \lambda, H) = (2, 1, 3, 1, 1, 2, 1, 2)$ has root residues $(0, 3, 1)$ but derivative residues $(3, 1, 3)$, so no stronger distinctness claim is made. $\square$

The conductor in EZ28m is the conductor of an integral root order in its normalization. It is not the fixed complex weighted operator of EZ29–35. The rational interpolation proves an exact morphism between already occupied channel targets; it does not create a missing source state.

9 A same-prime negative-square target change

Return to an original E state and put

$$D = \frac{4u+1}{R}, \quad v = hs^2, \quad \alpha = v - R, \quad \beta = v - D. \quad (\text{EZ28r})$$

The retained identities are

$$h \mid \alpha\beta - 1, \quad p \equiv \alpha \pmod{h}, \quad p\beta \equiv 1 \pmod{h}. \quad (\text{EZ28s})$$

Theorem 9.1 (Negative-four-square channel transfer). *Suppose $\alpha = -4c^2$ or $\beta = -4c^2$ for an integer $c > 0$. Then $h \equiv 3 \pmod{4}$ and $\gcd(h, c) = 1$. Put $j = (h+1)/4$. On the exact selected-word domain $c \mid j$, define*

$$Q = h, \quad U = \begin{cases} c^2, & \alpha = -4c^2, \\ (j/c)^2, & \beta = -4c^2, \end{cases} \quad R_M = \frac{p+4U}{h}, \quad A_M = \frac{p+R_M}{4}. \quad (\text{EZ28t})$$

Then (p, A_M, R_M, U) is an original M state. With $g_M = \gcd(j, U)$, its complete normalization and ordered denominators are

$$h_M = \frac{g_M^2}{U}, \quad r_M = \frac{U}{g_M}, \quad \lambda_M = \frac{j}{g_M}, \quad s_M = R_M \lambda_M - r_M, \quad (\text{EZ28u})$$

$$\boxed{(A_M, ph_M s_M \lambda_M, ph_M r_M \lambda_M)}. \quad (\text{EZ28v})$$

Writing $j = cm$ and $d = \gcd(c, m)$, the alpha branch has $(h_M, r_M, \lambda_M) = (d^2, c/d, m/d)$, while the beta branch has $(d^2, m/d, c/d)$. Thus $h_M = d^2$ is a square. In particular, every already-existing E state on $\alpha = -4$ or $\beta = -4$ returns to M, with no bound on the other shape coordinate. This last statement neither produces source states on those lines nor asserts that there are infinitely many.

Proof. If $\alpha = -4c^2$, then $R = hs^2 + 4c^2 \equiv 3 \pmod{4}$. If $\beta = -4c^2$, then $D = hs^2 + 4c^2$, and $RD = 4hr^2 + 1 \equiv 1 \pmod{4}$ with $R \equiv 3 \pmod{4}$, so $D \equiv 3 \pmod{4}$. In either case s is odd and $h \equiv 3 \pmod{4}$. Equation EZ28s makes h coprime to c .

In the alpha branch $h \mid p + 4c^2$. In the beta branch, $-4pc^2 \equiv 1 \pmod{h}$; since $4j \equiv 1 \pmod{h}$ and $c \mid j$, $h \mid p + 4(j/c)^2$. Hence R_M is integral. The original alpha inequality gives $4U < p - h < (h-1)p$; in beta, $4U \leq (h+1)^2/4 < 2h(h-1) < p(h-1)$. Thus $0 < R_M < p$. Moreover $hR_M \equiv 1 \pmod{4}$, so $R_M \equiv 3 \pmod{4}$ and $p/4 < A_M < p/2$.

Since $p = hR_M - 4U$,

$$A_M = jR_M - U, \quad A_M + U = jR_M,$$

and $U \mid j^2$ gives $U \mid A_M^2$. This is exactly the original M gate. Any common divisor of R_M, U divides p , while $0 < R_M < p$, so $\gcd(R_M, U) = 1$. It follows that $\gcd(A_M, U) = \gcd(j, U) = g_M$, and direct substitution proves EZ28u, $A_M = h_M r_M s_M$, $U = h_M r_M^2$, and $r_M + s_M = R_M \lambda_M$. These identities prove the original primitive normalization and EZ28v. The formulas with $j = cm$ follow by taking $d = \gcd(c, m)$. $\square$

The inverse fibres are also exact. In the alpha branch put $N = (p+4c^2)/h$. For every positive divisor $s \mid N$, set

$$k = N/s, \quad r = (s+k)/4, \quad R_s = hs^2 + 4c^2. \quad (\text{EZ28w})$$

The complete source fibre consists precisely of those divisors for which

$$s+k \equiv 0 \pmod{4}, \quad \gcd(r, s) = 1, \quad R_s < p, \quad \boxed{R_s \mid 4hr^2 + 1}. \quad (\text{EZ28x})$$

The boxed condition is the original E gate and cannot be omitted. With $D_s = (4hr^2 + 1)/R_s$, it gives $\kappa = sD_s - r = (pr + s)/R_s$; conversely these formulas reconstruct the entire original E state and $\alpha = -4c^2$. At $p = 5209, h = 95, c = 2$, the candidate $s = 5, k = 11, r = 4$ passes the other tests but has $R_s = 2391 \nmid 6081 = 4hr^2 + 1$, proving that the gate is material.

In the beta branch put

$$F_0 = \frac{4c^2p + 1}{h}, \quad D_s = hs^2 + 4c^2, \quad \Delta_s = s^2(D_s^2 - p) - F_0. \quad (\text{EZ28y})$$

For $1 \leq s \leq \lfloor (p-1)/(2h) \rfloor$, retain exactly the positive squares $k^2 = \Delta_s$ for which $sD_s - k$ is positive and even, put $r = (sD_s - k)/2$, $\kappa = (sD_s + k)/2$, and retain the original coprimality and first-half inequalities. The identity $4hr\kappa = pD_s + 1$ then proves both directions, including the original E divisibility; there is no omitted gate in this branch.

Finally, equality of full selected targets first recovers the common $Q = 4h_M r_M \lambda_M - 1 = h$. For this fixed h , an alpha label c_1 and beta label c_2 select the same U exactly when

$$c_1 c_2 = j. \quad (\text{EZ28z})$$

Different sources in one fibre remain distinct until a stated pushforward; the kernel of that pushforward is generated by their same-target differences. The complete independent review and corrected build are preserved in [9].

10 The complete fixed-cofactor middle fibre

The two selected square words above belong to a larger finite fibre. Keeping the original E-state variables fixed, suppose $h \equiv 3 \pmod{4}$, and put

$$j = \frac{h+1}{4}, \quad \mathcal{U}_{p,h} = \{U \in \mathbb{Z}_{>0} : U \mid j^2, h \mid p + 4U\}. \quad (\text{EZ28P})$$

Theorem 10.1 (Complete fixed- Q middle fibre). *For each $U \in \mathcal{U}_{p,h}$, define*

$$\begin{aligned} R_U &= \frac{p+4U}{h}, & A_U &= \frac{p+R_U}{4}, & g_U &= \gcd(j, U), \\ h_U &= \frac{g_U^2}{U}, & r_U &= \frac{U}{g_U}, & \lambda_U &= \frac{j}{g_U}, & s_U &= R_U \lambda_U - r_U. \end{aligned} \quad (\text{EZ28Q})$$

Then $(A_U, R_U, U, h_U, r_U, s_U, \lambda_U)$ is an original M state at p , and its normalized cofactor is

$$Q_U = 4h_U r_U \lambda_U - 1 = h, \quad (\text{EZ28R})$$

and its ordered ES denominators are

$$\boxed{(A_U, ph_U s_U \lambda_U, ph_U r_U \lambda_U)}. \quad (\text{EZ28S})$$

The assignment is a bijection from $\mathcal{U}_{p,h}$ onto all original M states at p whose normalized cofactor is $Q = h$.

Proof. An original E state has $h \leq A < p/2$, so $p > 2h$; here $h \geq 3$. Since $U \mid j^2$,

$$4U \leq 4j^2 = \frac{(h+1)^2}{4} < 2h(h-1) < p(h-1).$$

Hence $0 < R_U < p$. Moreover $hR_U = p + 4U \equiv 1 \pmod{4}$; because $p \equiv 1 \pmod{4}$ and $h \equiv 3 \pmod{4}$, $R_U \equiv 3 \pmod{4}$. Thus A_U is integral and $p/4 < A_U < p/2$.

Prime by prime, $U \mid j^2$ implies $g_U^2/U \in \mathbb{Z}_{>0}$. The definitions give

$$h_U r_U = g_U, \quad h_U r_U^2 = U, \quad h_U r_U \lambda_U = j,$$

and

$$R_U j - U = \frac{R_U(h+1)}{4} - U = \frac{p+R_U}{4} = A_U.$$

Consequently $h_U r_U s_U = A_U$, $r_U + s_U = R_U \lambda_U$, $A_U + U = R_U j$, and $U \mid A_U^2$. Also $\gcd(r_U, \lambda_U) = 1$. If a prime divided both r_U and s_U , it would divide R_U , then $hR_U - 4U = p$; this is impossible because $0 < R_U < p$. Hence $\gcd(r_U, s_U) = 1$. These are precisely the original M gates, and EZ28R and EZ28S follow by substitution.

Conversely, let an original M state have cofactor $Q = h$, written

$$A_M = h_M r_M s_M, \quad U = h_M r_M^2, \quad \lambda_M = \frac{r_M + s_M}{R_M}.$$

Then

$$j = \frac{h+1}{4} = h_M r_M \lambda_M, \quad qquad \frac{j^2}{U} = h_M \lambda_M^2 \in \mathbb{Z}_{>0},$$

so $U \mid j^2$. Furthermore

$$A_M + U = h_M r_M (r_M + s_M) = R_M j, \quad qquad p + 4U = 4(A_M + U) - R_M = R_M h.$$

Thus $U \in \mathcal{U}_{p,h}$. The primitive normalization gives $\gcd(j, U) = h_M r_M$, and EZ28Q recovers $(h_M, r_M, \lambda_M, s_M)$ exactly. The original middle divisor U separates distinct states, proving bijectivity. $\square$

Corollary 10.2 (Exact negative-square slices and emptiness). *If the existing E state has $\alpha = -4c^2$ or $\beta = -4c^2$, then*

$$\begin{aligned} \alpha = -4c^2 : \quad \mathcal{U}_{p,h} &= \{U \mid j^2 : U \equiv c^2 \pmod{h}\}, \\ \beta = -4c^2 : \quad \mathcal{U}_{p,h} &= \{U \mid j^2 : c^2 U \equiv j^2 \pmod{h}\}. \end{aligned} \quad (\text{EZ28T})$$

The selected words $U = c^2$ and $U = (j/c)^2$ exist exactly when $c \mid j$, but that divisibility is not necessary for the complete fibre to be nonempty. With

$$\Delta_h = \{U \bmod h : U \mid j^2\},$$

the exact criterion is

$$\boxed{\mathcal{U}_{p,h} = \emptyset \iff -p4^{-1} \pmod{h} \notin \Delta_h.} \quad (\text{EZ28U})$$

Proof. In the alpha branch, $p \equiv -4c^2 \pmod{h}$, so the defining congruence $p + 4U \equiv 0 \pmod{h}$ is equivalent to $U \equiv c^2 \pmod{h}$. In the beta branch, $-4c^2 p \equiv 1 \pmod{h}$. Multiplying the defining congruence by $4c^2$ and using $4j \equiv 1 \pmod{h}$ yields $c^2 U \equiv j^2 \pmod{h}$; every step is reversible because $\gcd(4c^2, h) = 1$. Formula EZ28U is the defining congruence solved for $U \bmod h$. $\square$

Let $\mathcal{E}_p$ be the original E-state set at a hard prime. After the proved diagonal return and every fixed- Q return, the exact residual source is

$$\boxed{\mathcal{E}_p^{\text{res}} = \{e \in \mathcal{E}_p : R \neq D, \quad [h \not\equiv 3 \pmod{4} \text{ or } \mathcal{U}_{p,h} = \emptyset]\}.} \quad (\text{EZ28V})$$

The shape-rigidity theorem of [8] states that radius at most seven is the diagonal profile $(\alpha, \beta) = (-1, -1)$, while the only nonsingular radius-eight profile is $(8, -4, 11)$. For the latter, $j = 3$ and $U = 9 \in \mathcal{U}_{p,11}$, so EZ28Q gives

$$(h_U, r_U, s_U, \lambda_U) = \left(1, 3, \frac{p+3}{11}, 1\right), \quad A_U = \frac{3(p+3)}{11}, \quad R_U = \frac{p+36}{11}. \quad (\text{EZ28W})$$

Therefore every state remaining in EZ28V has $\max(|\alpha|, |\beta|) \geq 9$. If such a residual state still has a negative-square coordinate, EZ28T fails for every divisor $U \mid j^2$, a strict finite exclusion stronger than failure of the selected word.

This improvement is strict. At the certified prime $p = 41161$, the E state

$$(A, R, u, h, r, s, \kappa, D, \alpha, \beta) = (10318, 111, 9678284, 11, 938, 1, 347829, 348767, -100, -348756)$$

has $j = 3$, $c = 5 \nmid j$, but $U = 3 \in \mathcal{U}_{41161,11}$. The resulting M state and exact decomposition are

$$(A_U, R_U, U, h_U, r_U, s_U, \lambda_U) = (11226, 3743, 3, 3, 1, 3742, 1), \quad (\text{EZ28X})$$

$$\boxed{\frac{4}{41161}} = \boxed{\frac{1}{11226} + \frac{1}{462073386} + \frac{1}{123483}}.$$

Primality follows from the Lucas witness 22: $22^{41160} \equiv 1$ modulo 41161, while for $q = 2, 3, 5, 7$ the residues $22^{41160/q}$ are 41160, 40429, 26424, 13707, each with predecessor coprime to 41161. Conversely, $p = 3049, h = 31, j = 8, c = 6$ has $\{U \bmod 31 : U \mid 64\} = \{1, 2, 4, 8, 16\}$ but $c^2 \equiv 5 \pmod{31}$; its alpha slice is empty. Thus a negative-square coordinate alone does not force a fixed-(Q) return. The executable exact comparison with every recorded original M state is retained in [9].

11 The precise continuation

This tranche completes a fixed-target divisor problem in the original Type-II cofactor box. It does not assert occupancy at every prime. Starting from an actual exterior state with a negative-four-integer shape, it gives the entire same-grade middle return, including every noncanonical divisor and its inverse. The noncanonical repair has a sharp finite bound depending only on the shape coefficient. That bound is realized by original exterior states at infinitely many hard primes. Conversely, for every fixed shape coefficient $t \geq 4$, an explicit reduced prime progression gives original exterior states whose entire same-grade middle box is empty. Their known exterior solutions remain present.

The Type-I/II coordinate identities and the square-divisor cofactor criterion are antecedents [1, §2, Propositions 2.2 and 2.6]. The preceding source [9, negative-four-square transfer] supplied $t = c^2$ when $c \mid (h+1)/4$. Here the arithmetic restriction is neither omitted nor made into an unspecified admissibility hypothesis: it is solved exactly as a divisor problem. No historical-priority determination is claimed.

12 Original coordinates and a general negative shape

Let $p \equiv 1 \pmod{4}$ be prime. An original exterior state consists of

$$p/4 < a < p/2, \quad R = 4a - p, \quad u \mid a^2, \quad R \mid 4u + 1.$$

Retain $a = \prod_l l^{e_l}$, the full box $-e_l \leq \beta_l \leq e_l$, and $u = \prod_l l^{e_l + \beta_l}$. Set

$$g = \gcd(a, u), \quad (h, r, s) = (g^2/u, u/g, a/g), \quad \kappa = (pr + s)/R.$$

Then $a = hrs$, $u = hr^2$, $\gcd(r, s) = 1$, and the ordered denominators are

$$(x, y, z) = (a, h s \kappa, p h r \kappa), \quad \frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}. \quad (1)$$

Multiplication proves the identity, and its ordered inverse is $u = a^2/(Ry - pa)$. All normalization factors are positive integers. Let

$$v = h s^2 = a^2/u, \quad D = (4u + 1)/R = (r + \kappa)/s, \quad \alpha = v - R, \quad \beta = v - D.$$

The symbol β here is not an exponent β_l . The original identities are

$$\begin{aligned} RD &= 4hr^2 + 1, & pD + 1 &= 4hr\kappa, & r + \kappa &= sD, \\ h &\mid \alpha\beta - 1, & p &\equiv \alpha \pmod{h}, & p\beta &\equiv 1 \pmod{h}. \end{aligned} \quad (2)$$

For every original divisor $d \mid a^2$, reciprocity gives $(d/R) = (d/p)$. For an odd prime $l \mid a$, use $R \equiv -p \pmod{l}$ and $R \equiv 3 \pmod{4}$; for 2 use the supplementary law. The exterior target $u \equiv -1/4 \pmod{R}$ therefore gives $(h/p) = (u/p) = -1$. These are the classical laws, used in their original factor coordinates [2, §6.4].

Suppose one ordered shape coordinate is $-4t$, $t \geq 1$. Since $R, D \equiv 3 \pmod{4}$,

$$h \equiv 3 \pmod{4}, \quad s \text{ odd}, \quad \gcd(h, t) = 1, \quad (t/h) = 1. \quad (3)$$

Indeed hs^2 is 3 modulo four. Equation (2) makes h coprime to $4t$. For α , reciprocity gives $(h/p) = (p/h) = (-4t/h) = -(t/h)$; the inverse congruence gives the same result for β . Also $p > 2h$ because $a \geq h$.

Put $j = (h + 1)/4$ and define the full original target set

$$\mathcal{A}_t(j) = \{W > 0 : W \mid j^2, W \equiv t \pmod{4j - 1}\}. \quad (4)$$

There is no primality condition on $h = 4j - 1$ in (4).

Theorem 12.1 (Complete same-grade channel return). *For every original exterior state as above and every $W \in \mathcal{A}_t(j)$, set*

$$U = \begin{cases} W, & \alpha = -4t, \\ j^2/W, & \beta = -4t, \end{cases} \quad Q = h, \quad R_M = (p + 4U)/h, \quad a_M = (p + R_M)/4. \quad (5)$$

These are exactly the original cofactor- h middle choices, with the indicated ordered shape branch retained. In particular $U \mid j^2$, $0 < R_M < p$, and

$$g_M = \gcd(j, U), \quad h_M = g_M^2/U, \quad r_M = U/g_M, \quad \lambda_M = j/g_M, \quad s_M = R_M \lambda_M - r_M \quad (6)$$

returns the original middle state and its ordered denominators

$$(a_M, ph_M s_M \lambda_M, ph_M r_M \lambda_M).$$

Proof. In the α case $p \equiv -4t \pmod{h}$, so the cofactor congruence is exactly $W \equiv t$. In the β case $p \equiv -(4t)^{-1} \pmod{h}$ and $j^2 \equiv 16^{-1} \pmod{h}$, so $U = j^2/W$ satisfies $p + 4U \equiv 0$ precisely when $W \equiv t$. Every divisor involved is a unit modulo h . Since $U \leq j^2$, $h \geq 3$ and $p > 2h$,

$$4U \leq (h + 1)^2/4 < (h - 1)p.$$

Thus $R_M < p$, and its positive integer value is 3 modulo four. Prime valuations in $U \mid j^2$ give $h_M r_M \lambda_M = j$, $h_M r_M^2 = U$, and $\gcd(r_M, \lambda_M) = 1$. Directly

$$a_M = R_M j - U = h_M r_M s_M > 0.$$

A common prime of r_M and R_M would divide p by $hR_M = p + 4U$, whereas $r_M \leq j < p$. Hence $\gcd(r_M, s_M) = 1$. These identities prove the middle normalization, the first-half range, and the reciprocal identity. Conversely a cofactor- h middle word has $U \mid j^2$ and the same gate; taking $W = U$ or j^2/U recovers (4). Neither $U < a_M$ nor $r_M < s_M$ is imposed: the ordered orientation is kept, or complemented explicitly afterward. $\square$

13 An exact disjoint classification of every available divisor

The next theorem is pure integer arithmetic. In particular, t need not be a square and h need not be prime.

Theorem 13.1 (Fixed-target capacity completion). *Let $t, j \geq 1$ and $h = 4j - 1 > t$. The set $\mathcal{A}_t(j)$ is the disjoint union of:*

1. the canonical word $W = t$, present exactly when $t \mid j^2$;
2. the companion word $W = 4tj$, present exactly when $4t \mid j$;
3. the finite words described by

$$1 \leq \ell \leq \left\lfloor \frac{t-2}{4} \right\rfloor, \quad w \mid \ell^2, \quad n = \frac{t-w}{4\ell+1} \in \mathbb{Z}_{>0}, \quad (7)$$

$$V = \ell^2/w, \quad j = \ell + 4nV, \quad h = 4j - 1 > t, \quad W = j^2/V = t + nh. \quad (8)$$

In the last item the displayed j must equal the input j . Each word occurs once. The inverse, for $W \neq t$, is

$$n = (W - t)/h, \quad V = j^2/W, \quad \ell = j - 4nV. \quad (9)$$

For $\ell = 0$ it is the companion. Otherwise $w = \ell^2/V$ is the unique label in (7).

Proof. If $W \neq t$, then $W = t + nh$ for some $n \geq 1$, because $0 < t < h$. Let $V = j^2/W$. Reduction of $16WV = 16j^2 = (h+1)^2$ modulo h gives

$$16tV = 1 + kh$$

for an integer $k \geq 1$. Modulo four, $k \equiv 1$, so $k = 4\ell + 1$ with $\ell \geq 0$. Comparing the two exact products gives

$$h + 2 = k + 16nV, \quad j = \ell + 4nV, \quad t = (4\ell + 1)n + \frac{\ell^2}{V}. \quad (10)$$

If $\ell = 0$, then $n = t$, $j = 4tV$, and $W = 4tj$. This is equivalent to $4t \mid j$ and coexists with the canonical divisor. If $\ell > 0$, (10) proves $V \mid \ell^2$, and $w = \ell^2/V \geq 1$. Therefore $t \geq 4\ell + 2$, giving (7). All labels are recovered by (9). Conversely (7)–(8) give

$$j^2/V = w + 8\ell n + 16n^2V = t + n(4j - 1),$$

an integer divisor of j^2 in the required class. The three cases are disjoint by $W = t$, $\ell = 0$, or $\ell > 0$. $\square$

The small-grade domain $h \leq t$ is separately finite: $j \leq \lfloor (t+1)/4 \rfloor$. Literal enumeration there is retained; the theorem is not asserted on that domain. A reduced residue $t \bmod h$ may be used to test a word, but it must not replace the original shape coefficient in a bound.

Theorem 13.2 (Sharp finite-repair bound). *Every finite word in Theorem 13.1 satisfies*

$$\boxed{h \leq t^2 - 3t + 1.} \quad (11)$$

Equality holds exactly when $w = n = 1$ and $t = 4\ell + 2$. In particular, if the canonical divisor is absent but $\mathcal{A}_t(j)$ is nonempty, (11) holds. For $h > \max(t, t^2 - 3t + 1)$ the exact count is

$$\boxed{|\mathcal{A}_t(j)| = \mathbf{1}_{t \mid j^2} + \mathbf{1}_{4t \mid j}.} \quad (12)$$

Proof. Use $n = (t - w)/(4\ell + 1)$ to write

$$h + 1 = 4\ell + \frac{16\ell^2(t - w)}{(4\ell + 1)w} \leq 4\ell + \frac{16\ell^2(t - 1)}{4\ell + 1}.$$

The last expression is strictly increasing in real $\ell \geq 0$. Since $\ell \leq (t - 2)/4$, it is at most $(t - 2) + (t - 2)^2 = t^2 - 3t + 2$. Equality requires both $w = 1$ and $\ell = (t - 2)/4$, and then $n = 1$. The companion cannot repair a missing canonical word because $4t \mid j$ implies $t \mid j^2$. The classification proves (12). $\square$

For every $\ell \geq 1$ the arithmetic equality family is

$$t = 4\ell + 2, \quad j = \ell(4\ell + 1), \quad h = 4j - 1, \quad W = (4\ell + 1)^2, \quad V = \ell^2. \quad (13)$$

Here $t \nmid j^2$: the odd number $2\ell + 1 > 1$ is coprime to both factors of j . Thus (13) is a genuine repair, not a second name for the canonical word. Not every finite word is a repair: at

$$t = 289, \quad j = 1224, \quad h = 4895, \quad \mathcal{A}_t(j) = \{289, 5184\},$$

the canonical word and a finite word coexist. The latter has $(n, V, \ell, w) = (1, 289, 68, 16)$. Its coefficient is two, even though its occupied residue is a single class.

The complete finite search uses at most

$$\sum_{\ell \leq (t-2)/4} \tau_{\text{div}}(\ell^2) \leq L(1 + \log L)^2, \quad L = \lfloor (t - 2)/4 \rfloor \geq 1. \quad (14)$$

Indeed $\tau_{\text{div}}(\ell^2) \leq d_3(\ell)$ primewise, and $\sum_{abc \leq L} 1 \leq L(\sum_{a \leq L} 1/a)^2$. This counts candidate labels under unit-cost integer arithmetic; it is not a bit-complexity estimate that includes integer factorization. It is a finite classification in the original integers, not an assertion that a successful word always exists.

There is also a strict smaller-parameter return on each finite word. Put $g_0 = \gcd(\ell, V)$ and

$$H_0 = g_0^2/V, \quad r_0 = V/g_0, \quad \lambda_0 = \ell/g_0.$$

Then $\ell = H_0 r_0 \lambda_0$, $V = H_0 r_0^2$, and $\gcd(r_0, \lambda_0) = 1$. The middle normalization in (6) is

$$\begin{array}{c|cc} & h_M & r_M & \lambda_M \\ \hline \alpha & H_0 & \lambda_0 + 4nr_0 & r_0 \\ \beta & H_0 & r_0 & \lambda_0 + 4nr_0. \end{array} \quad (15)$$

In particular $h_M \leq \ell \leq (t - 2)/4$. This decreases a parameter of the finite capacity reduction; it is not a well-founded proof of initial ES occupancy.

14 Entire shape lines with forced original availability

Theorem 14.1 (Automatic negative-shape returns). *Let $p \equiv 1 \pmod{24}$ be prime and start from an original exterior state. The canonical word is always available on the three lines*

$$\alpha = -4t, \quad t \in \{1, 2, 3\},$$

and on the nine lines

$$\boxed{\beta = -4t, \quad t \mid 36.} \quad (16)$$

Every such line has the original middle return (5)–(6), with no bound on the other shape coordinate. The coefficient t need not be a square.

Proof. For α , (3) gives $(2/h) = 1$ at $t = 2$; together with $h \equiv 3 \pmod{4}$, this forces $h \equiv 7 \pmod{8}$, hence $2 \mid j$. At $t = 3$, the supplementary quadratic law gives $h \equiv 11 \pmod{12}$ and $3 \mid j$. The case $t = 1$ is literal.

For β , use the stronger actual equations, not just their Jacobi projection. Here $D = hs^2 + 4t$ and $r + \kappa = sD$ is odd. Thus $r\kappa$ is even. Since $pD + 1 = 4hr\kappa$ and $p \equiv 1 \pmod{8}$, necessarily $D \equiv 7 \pmod{8}$, whence

$$h \equiv 7 - 4t \pmod{8}.$$

If $2 \mid t$, this gives $2 \mid j$. If $3 \mid t$, all equalities in the rest of this paragraph are modulo three. The case $h = 0$ makes $D = 0$ and contradicts $pD + 1 = 4hr\kappa$. If $h = 1$ and $s = 0$, the same equation would give $1 = -r^2$, impossible. If $h = 1$ and $s = \pm 1$, then $D = 1$, $\kappa = s - r$, and $2 = r(s - r)$; the right side has only values zero and one. Thus $h = 2 \pmod{3}$, so $3 \mid j$. Every prime exponent of $t \mid 36$ is at most two. Therefore these two forced divisibilities imply $t \mid j^2$, and Theorem 12.1 applies. $\square$

For the canonical $W = t$, the squarefree part of h_M is exactly that of t . This follows from $h_M = g_M^2/U$ and, in the inverse branch, $U = j^2/t$. The preceding square-target theorem is the special case in which this part is one. We do not claim that (16) lists every possible automatic β coefficient. Actual complete failures at $t = 5, 8, 27$ are given below.

15 The quadratic bound is sharp on actual hard-prime sources

Theorem 15.1 (Prime realization of the finite-repair boundary). *For every $t = 6 + 420z$, $z \geq 0$, infinitely many primes $p \equiv 1 \pmod{840}$ have an original exterior state with $\alpha = -4t$,*

$$h = t^2 - 3t + 1, \quad R = t^2 + t + 1, \quad s = 1,$$

whose canonical cofactor word is unavailable but whose finite word in (13) returns an original middle state. Thus the bound (11) is attained within the original prime arithmetic, for unbounded t .

Proof. Put $\ell = (t - 2)/4$, $j = \ell(4\ell + 1)$, so $h = 4j - 1$ and the word is $W = (t - 1)^2$. The two integers h, R are coprime, since their difference is $4t$ and $h \equiv R \equiv 1 \pmod{t}$. For the displayed progression of t , $\gcd(hR, 210) = 1$. Also $t^3 \equiv 1 \pmod{R}$. Consequently

$$r_0 = (t + 2R)/4 \in \mathbb{Z}, \quad 16tr_0^2 \equiv 1 \pmod{R}.$$

Since $h \equiv -4t \pmod{R}$, the latter is exactly $R \mid 4hr_0^2 + 1$. Choose r by the explicit compatible CRT conditions

$$r \equiv r_0 \pmod{R}, \quad r \equiv \frac{R+1}{4}h^{-1} \pmod{210}.$$

Then $p = 4hr - R$ varies in a progression of modulus $840hR$, is 1 modulo 840, and is coprime to the entire modulus: modulo h it is $-R$ and modulo R it is $4hr_0$. Dirichlet's theorem supplies infinitely many prime members [3, Theorem 18.1]. For all sufficiently large positive members, $a = hr$, $u = hr^2$ is an original first-half exterior state; $R < 2hr$ verifies the only upper range issue. Its shape is $h - R = -4t$. Formula (13) proves canonical unavailability and supplies the sharp finite repair. The target is reconstructed by Theorem 12.1. Prime values are supplied by a reduced *linear* progression; no assertion about primes represented by a higher-degree polynomial has been inserted. $\square$

A small explicit original member, in another hard class, is

$$\boxed{p = 825241, \quad (a, R, u, h, r, s, \kappa) = (206321, 43, 2240439739, 19, 10859, 1, 208402140).} \quad (17)$$

Its coefficient is $t = 6$, $j = 5$, and the entire capacity box is $\{1, 5, 25\}$. Only $25 \equiv 6 \pmod{19}$ succeeds. The original middle return is

$$(a_M, R_M, U, h_M, r_M, s_M, \lambda_M) = (217170, 43439, 25, 1, 5, 43434, 1), \quad (18)$$

with ordered denominators $(217170, 825241 \cdot 43434, 825241 \cdot 5)$. This is a nonsquare coefficient and a genuinely noncanonical return.

16 Every other alpha coefficient has infinite original same-grade obstructions

Lemma 16.1 (Exact limitation of the character-only budget). *For $t \geq 1$, let the product run over every prime, including 2:*

$$K(t) = \prod_{l^e \parallel t} l^{\lceil e/2 \rceil}, \quad M = 4K(t).$$

For odd A coprime to t , write $\chi_t(A) = (t/A)$ for the quadratic Kronecker–Jacobi character determined by the squarefree part of t ; its conductor divides M . Among units $A \pmod{M}$ with $A \equiv 3 \pmod{4}$ and $\chi_t(A) = 1$, the implication $A \equiv -1 \pmod{M}$ holds for all such A exactly when $t = 1, 2, 3$.

Proof. The Jacobi character is well-defined modulo M by reciprocity, including the factor at two. It takes value one at -1 . If t is a square, it is trivial, and the indicated set has $\varphi(M)/2$ elements. If t is nonsquare, its restriction to the units 1 modulo four is nontrivial: at an odd prime appearing to odd exponent choose a nonsquare by CRT, with the other coordinates one; if only the factor two has odd exponent, choose the coordinate 5 modulo eight. Thus it is independent of the mod-four sign character, and the indicated set has $\varphi(M)/4$ elements. A multiple of four with totient two is $M = 4$; those with totient four are 8, 12. The square/non-square alternatives give exactly $t = 1, 2, 3$. In each singleton case the member is -1 . $\square$

Theorem 16.2 (An actual obstruction for every fixed $t \geq 4$). *For every integer $t \geq 4$, there are infinitely many hard primes possessing an original exterior state with $\alpha = -4t$ but no middle state at cofactor equal to that state’s original grade h , even after the entire divisor box $\text{Div}(j^2)$ is tested. These primes already have the constructed exterior solution. This is not an Erdős–Straus counterexample.*

Proof. Choose a unit class $A \pmod{4K(t)}$ from the preceding lemma other than -1 . Then $A \equiv 3 \pmod{4}$ and $(t/A) = 1$. Lift it by CRT to a reduced class modulo $\text{lcm}(4K(t), 840)$ such that, for $l = 3, 5, 7$, both R and $R - 4t$ are units modulo l . If $l \mid K(t)$ this is already true, because $l \mid t$ and A is a unit. Otherwise one need only avoid $0, 4t \pmod{l}$, which leaves a choice even for $l = 3$. Choose an odd lift modulo eight as well. Dirichlet’s theorem gives a prime R in this class as large as required. Set $h = R - 4t$ and choose it so that

$$h > \max(4t, t^2 - 3t + 1).$$

Then $h \equiv 3 \pmod{4}$, $\gcd(h, 210) = \gcd(h, R) = 1$, and $h \not\equiv -1 \pmod{4K(t)}$. Thus $t \nmid j^2$ for $j = (h + 1)/4$. Since $\chi_t(R) = (t/R) = 1$, choose q with $q^2 \equiv t \pmod{R}$ and set $r_0 \equiv (4q)^{-1} \pmod{R}$. Then

$$16tr_0^2 \equiv 1 \pmod{R}.$$

Solve

$$r \equiv r_0 \pmod{R}, \quad r \equiv \frac{R+1}{4} h^{-1} \pmod{210}.$$

Again $p = 4hr - R$ is a reduced progression modulo $840hR$, with $p \equiv 1 \pmod{840}$. Dirichlet gives infinitely many prime members. Take $a = hr$, $u = hr^2$, $s = 1$. Then $R \mid 4u + 1$, $\gcd(p, a) = 1$, and $p > 2a$ follows already from $h > 4t$ and $r \geq 1$. The original shape is $\alpha = h - R = -4t$. Theorem 13.2 now gives $\mathcal{A}_t(j) = \emptyset$: neither the canonical word nor its companion is available and the finite-exception range has been passed. Theorem 12.1 identifies this with failure of the complete cofactor- h middle source. $\square$

For $t = 4$ one explicit construction is

$$R = 59, \quad h = 43, \quad j = 11, \quad r = 8415 + 12390n, \quad \boxed{p = 1447321 + 2131080n.} \quad (19)$$

The progression is reduced and 1 modulo 840. At $n = 3$, $p = 7840561$ is prime, with

$$(a, R, u, h, r, s, \kappa) = (1960155, 59, 89353665675, 43, 45585, 1, 6057830054).$$

The three capacity divisors are 1, 11, 121, of residues 1, 11, 35 modulo 43, not four. Every prime in (19) nevertheless has its displayed exterior solution.

The beta automatic theorem has genuine limits too. The following are original hard-prime sources, and complete enumeration of their small j^2 boxes proves failure at their old grade:

p	a	u	h	r	s	κ	$t \ (\beta = -4t)$
346201	158815	43864703	23	1381	5	1654	8
593401	188100	39710000	11	1900	9	7091	27
2727841	929071	464321099	11	6497	13	17930	5.

(20)

The case $h \leq t$ in the middle row is checked literally, not attributed to Theorem 13.1. No infinite beta obstruction theorem is asserted.

17 Original inverse fibres and a necessary prose correction

For fixed p, h, t on an α branch, put $N = (p + 4t)/h$. Its complete original fibre is

$$s \mid N, \quad s + N/s \equiv 0 \pmod{4}, \quad r = (s + N/s)/4, \quad a = hrs, \quad u = hr^2,$$

$$\boxed{\gcd(r, s) = 1, \quad p/4 < a < p/2, \quad R = 4a - p \mid 4hr^2 + 1.} \quad (21)$$

The shape equation is $4rs - s^2 = N$; the extra gate in (21) is necessary and not implied by this factorization. For a beta branch, put $F = (4tp + 1)/h$ and enumerate

$$1 \leq s \leq \lfloor (p - 1)/(2h) \rfloor, \quad D = hs^2 + 4t, \quad \Delta_s = s^2(D^2 - p) - F.$$

Keep the positive squares $k^2 = \Delta_s$ with $sD - k$ positive even, and set $r = (sD - k)/2$, $\kappa = sD - r$, retaining the original coprimality, range and gate. These are equivalences from $4hr(sD - r) = pD + 1$ and $k = \kappa - r > 0$.

The predecessor's prose alpha-fibre formula omitted the final gate in (21). Its executable inverse-fibre enumeration did test that gate. At $p = 5209, h = 95, t = 4$, the displayed divisor equation has $N = 55$. The choice $s = 5, r = 4$ satisfies its factorization, coprimality, shape and first-half conditions, but gives

$$a = 1900, \quad R = 2391, \quad u = 1520, \quad (4u + 1) \bmod R = 1299.$$

It is not an exterior state. This correction changes the textual claimed inverse domain, not the predecessor's forward theorem or its executable fibre filter. The old source is preserved, and `CORRECTION.md` identifies the affected passage and file hash.

For the new maps, retain $(p, \text{original E state, side, } t, j, W)$ and its canonical/companion/finite label. The M output recovers $Q = h$ and j ; (21) and its beta counterpart recover precisely the possible E sources once t and the side remain. The finite labels are inverted by (9). Across two ordered branches, alpha word W_1 and beta word W_2 give the same M word exactly when $W_1 W_2 = j^2$. Multiple original sources above that word are not identified.

On the free abelian group of these labelled sources the coefficient pushforward sums each actual target fibre. Its kernel is generated by same-target differences. No division by a fibre size is used to prove positivity. Reducing a positive even coefficient modulo two can yield a supported zero; it does not remove its original words. The two-word example following (13) is an explicit control.

18 Verification scope and the remaining milestone

The verifier exhausts (4) for $t \leq 120$, $j \leq 5000$, $4j - 1 > t$, and compares every integer word with Theorem 13.1. It separately verifies all finite profiles and one hundred members of the sharp arithmetic family, the original prime-realizable progressions, and the character-only classification through 120. It also enumerates every original first-half E/M divisor at primes $p \equiv 1 \pmod{4}$ through 3000, retaining actual factorization and channel records, and checks the returned original states. The fixed larger examples use complete trial-division primality checks. Their empty cofactor boxes are exhausted.

The independent program imports neither the main verifier nor predecessor code. It uses literal divisors of j^2 , coprime primitive r, s sources, primewise Euler-symbol evaluation, and a complementary-divisor parameter for the finite profiles. Normal and optimized command outcomes, hashes and scopes belong to the execution receipt. Multiple runs are not independent mathematical review.

The completed unbounded statements here concern the *fixed-target capacity problem* and the original shape lines in Theorem 14.1. The obstruction in Theorem 16.2 proves why increasing the old-grade divisor search cannot repair every negative-square alpha source. It does not exclude a different original middle cofactor or an exterior solution, which the construction already supplies. No initial state has been forced at an arbitrary hard prime. Turn 7's universal existence milestone remains outstanding; no Turn 8 witness guarantee, Lean build, new global verification range or priority claim follows.

19 The exact bridge to a fixed weighted conductor

Fix one reference arithmetic quartet with distinct roots $\rho_{*,1}, \dots, \rho_{*,4}$, its original exponential polynomial, first nonzero moment μ_v , period frame and Taylor unit. Let

$$V_* = (\rho_{*,j}^n)_{0 \leq n \leq 3, 1 \leq j \leq 4}, \quad (B_{v,*})_{kj} = \begin{cases} \binom{v+j}{k} \mu_{v+j-k}, & 0 \leq k \leq j \leq 3, \\ 0, & k > j, \end{cases} \quad (\text{EZ29})$$

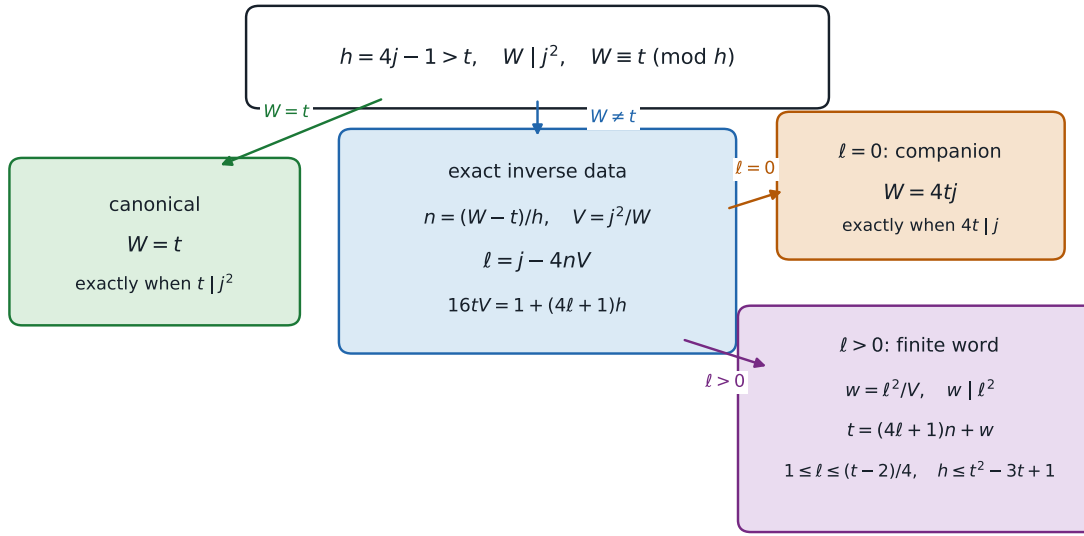
and retain the source frame

$$K = \begin{pmatrix} 0 & i & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & -1 & -1 - \sqrt{2}i & 1 - \sqrt{2}i \\ i/2 & -3i & 2\sqrt{2} + 6i & -2\sqrt{2} + 6i \\ 0 & 13 & -34 - 19\sqrt{2}i & 34 - 19\sqrt{2}i \end{pmatrix}. \quad (\text{EZ30})$$

Define

$$\Phi_* = (V_*^\top)^{-1} K^{-1}, \quad \Psi_* = B_{v,*} (V_*^\top)^{-1} K^{-1}. \quad (\text{EZ31})$$

Exact inverse classification of a fixed-target cofactor word



The three outputs are disjoint and the displayed formulas recover every label. The quadratic bound applies only to the finite branch.

Figure 2: The exact inverse in Theorem 13.1. Starting from a word W , equations (9)–(10) recover n, V, ℓ , while the signs and domains separate the canonical, companion and finite branches. The sharp grade bound belongs only to the finite branch.

Sharp capacity boundary and its nonsquare channel return

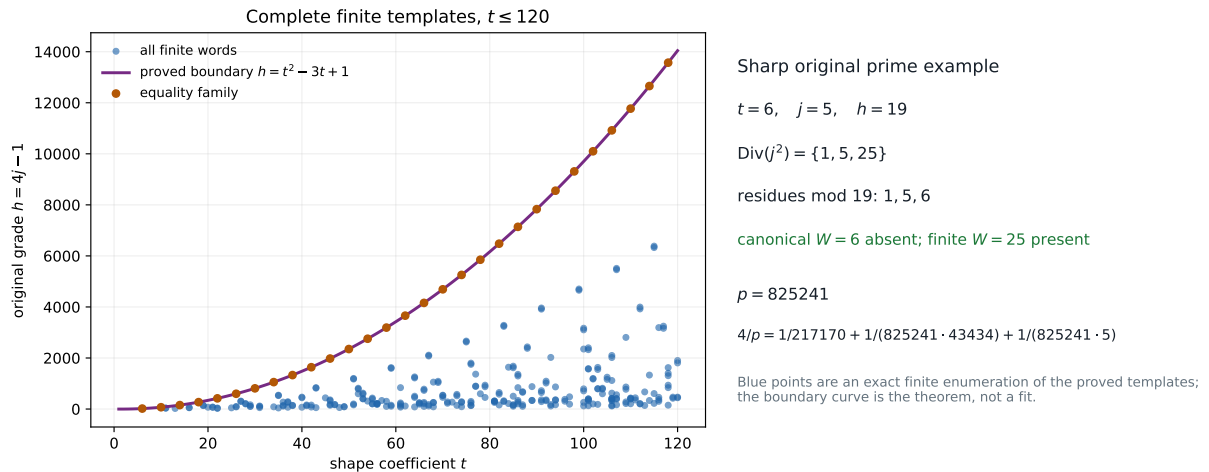


Figure 3: Every finite template with $t \leq 120$, plotted against the proved boundary in Theorem 13.2. The right panel retains the exact $p = 825241$ nonsquare return from equations (17)–(18). The points are a reproducible enumeration by `tools/generate_20260920_illustrations.py`; the curve is the proved bound, not an empirical fit.

The columns of K are the auxiliary labelled source frame; they are not identified with either p, x, y, z or the reference roots. The original weighted conductor is $T_{A,*}|_{\mathcal{U}_*}$, and the exact square is

$$T_{A,*}\Phi_* = \Psi_*. \quad (\text{EZ32})$$

The receiving map Ψ_* is not itself the conductor.

Theorem 19.1 (Conductor transport of the ES quartic). *All three maps in EZ32 are invertible and have zero kernel. Put*

$$P_{\mathcal{U},*} = \Phi_* P \Phi_*^{-1}, \quad P_{\mathcal{W},*} = \Psi_* P \Psi_*^{-1}. \quad (\text{EZ33})$$

For every target u and source point q with $P(q) = u$,

$$P_{\mathcal{U},*}(\Phi_* q) = \Phi_* u, \quad q \text{quad} P_{\mathcal{W},*}(\Psi_* q) = \Psi_* u, \quad q \text{quad} T_{A,*} P_{\mathcal{U},*} = P_{\mathcal{W},*} T_{A,*}. \quad (\text{EZ34})$$

In particular the eight points EZ13 above J_4 , their deck involution, and the exceptional set EZ28 pass bijectively to the fixed conductor planes.

Proof. Distinct reference roots make V_* invertible. Moreover

$$\det B_{v,*} = \mu_v^4 \prod_{j=0}^3 \binom{v+j}{v} \neq 0, \quad \det K = 77\sqrt{2}i/2.$$

Thus EZ31 is invertible and EZ32 is the exact conductor identity proved in [10, FC26–FC35]. The first two identities in EZ34 are the definitions of conjugacy. Inserting EZ32 between them proves the last one. Invertible linear maps preserve fibres, escaping sequences, convergence and deck equations, proving the final assertion. $\square$

The coefficient hypersurface itself also transports, with no loss of its arithmetic coordinates. Define the two inverse matrices

$$R_{\mathcal{U}} := \Phi_*^{-1} = K V_*^T, \quad R_{\mathcal{W}} := \Psi_*^{-1} = K V_*^T B_{v,*}^{-1}. \quad (\text{EZ34a})$$

For $\mathcal{C} \in \{\mathcal{U}, \mathcal{W}\}$, let $\ell_0^{\mathcal{C}}, \ell_2^{\mathcal{C}}, \ell_3^{\mathcal{C}}, \ell_4^{\mathcal{C}}$ be the four row covectors of $R_{\mathcal{C}}$, in the retained coefficient order (u_0, u_2, u_3, u_4) .

Theorem 19.2 (Exact equation in the fixed zeta receiving coordinates). *In either fixed conductor plane, the transported arithmetic image is the irreducible quartic hypersurface*

$$\begin{aligned} G_{\mathcal{C}}(q) := & 625\ell_0^{\mathcal{C}}(q)\ell_4^{\mathcal{C}}(q)^3 - 125\ell_3^{\mathcal{C}}(q)\ell_4^{\mathcal{C}}(q)^2 \\ & + 25\ell_2^{\mathcal{C}}(q)\ell_3^{\mathcal{C}}(q)^2\ell_4^{\mathcal{C}}(q) - 4\ell_3^{\mathcal{C}}(q)^4 = 0. \end{aligned} \quad (\text{EZ34b})$$

On the open chart where $\ell_0^{\mathcal{C}}\ell_3^{\mathcal{C}}\ell_4^{\mathcal{C}} \neq 0$, its exact inverse arithmetic coordinates are

$$\begin{aligned} p &= -5\frac{\ell_4^{\mathcal{C}}}{\ell_3^{\mathcal{C}}}, & s_1 &= -\frac{1}{\ell_0^{\mathcal{C}}} + 5\frac{\ell_4^{\mathcal{C}}}{\ell_3^{\mathcal{C}}}, \\ s_2 &= \frac{4(\ell_3^{\mathcal{C}})^2}{25\ell_0^{\mathcal{C}}\ell_4^{\mathcal{C}}}, & s_3 &= -\frac{\ell_3^{\mathcal{C}}}{5\ell_0^{\mathcal{C}}}. \end{aligned} \quad (\text{EZ34c})$$

The reduced singular support of EZ34b is the codimension-two linear plane

$$\ell_3^{\mathcal{C}} = \ell_4^{\mathcal{C}} = 0. \quad (\text{EZ34d})$$

Finally, the original conductor carries these equations and recovered coordinates identically:

$$G_{\mathcal{W}}(T_{A,*}q) = G_{\mathcal{U}}(q), \quad \ell_j^{\mathcal{W}}(T_{A,*}q) = \ell_j^{\mathcal{U}}(q) \quad (j = 0, 2, 3, 4). \quad (\text{EZ34e})$$

Proof. For a coefficient column u , EZ34a gives $R_{\mathcal{U}}\Phi_*u = u$ and $R_{\mathcal{W}}\Psi_*u = u$. Thus each retained covector recovers the corresponding literal coefficient, and substituting these four identities into EZ9 and EZ9c proves EZ34b–EZ34c. An invertible linear change of coordinates preserves irreducibility and carries the reduced singular plane $u_3 = u_4 = 0$ exactly to EZ34d. From $T_{A,*} = \Psi_*\Phi_*^{-1}$ one has $R_{\mathcal{W}}T_{A,*} = R_{\mathcal{U}}$; applying each row and then the quartic polynomial proves EZ34e. $\square$

We now write every receiver covector in EZ34b without hiding the inverse matrix. Use the fixed Time order (M, N, T, E), write $m_j = \mu_{v+j}$, and put

$$\alpha_2 = \binom{v+2}{2}, \quad \alpha_3 = \binom{v+3}{3}, \quad \beta_{23} = \binom{v+3}{2}.$$

For a receiver coefficient column $\mathbf{r} = (r_0, r_1, r_2, r_3)^\top$, define successively

$$\begin{aligned} c_3 &= \frac{r_3}{\alpha_3 m_0}, \\ c_2 &= \frac{r_2 - \beta_{23} m_1 c_3}{\alpha_2 m_0}, \\ c_1 &= \frac{r_1 - (v+2)m_1 c_2 - (v+3)m_2 c_3}{(v+1)m_0}, \\ c_0 &= \frac{r_0 - m_1 c_1 - m_2 c_2 - m_3 c_3}{m_0}, \end{aligned} \tag{EZ34f}$$

$$\xi_a(\mathbf{r}) = c_0 + c_1 \rho_{*,a} + c_2 \rho_{*,a}^2 + c_3 \rho_{*,a}^3 \quad (a = \text{M, N, T, E}). \tag{EZ34g}$$

The four literal coefficient-recovery forms are

$$\begin{aligned} U_0 &= i\xi_N + \frac{\xi_T + \xi_E}{\sqrt{2}}, \\ U_2 &= -\xi_N + (-1 - \sqrt{2}i)\xi_T + (1 - \sqrt{2}i)\xi_E, \\ U_3 &= \frac{i}{2}\xi_M - 3i\xi_N + (2\sqrt{2} + 6i)\xi_T + (-2\sqrt{2} + 6i)\xi_E, \\ U_4 &= 13\xi_N + (-34 - 19\sqrt{2}i)\xi_T + (34 - 19\sqrt{2}i)\xi_E. \end{aligned} \tag{EZ34h}$$

Indeed EZ34f is forward substitution in $B_{v,*}c = \mathbf{r}$, EZ34g is $V_*^\top c$, and EZ34h is multiplication by the four rows of K . Hence $U_j(\mathbf{r}) = \ell_j^{\mathcal{W}}(\mathbf{r})$ exactly. In particular

$$F_*(\mathbf{r}) := 625U_0U_4^3 - 125U_3U_4^2 + 25U_2U_3^2U_4 - 4U_3^4 \tag{EZ34i}$$

is the fully expanded-by-rows form of $G_{\mathcal{W}}$; no coefficient has been fitted to an ES example. It is nonzero and irreducible, has total degree four and retains the nonzero cubic term $-125U_3U_4^2$.

Theorem 19.3 (Prime-marked receiver pair and polynomial selector). *Assume*

$$F_*(\mathbf{r}) = 0, qquad U_0U_3U_4 \text{ Disc}(H_{\mathbf{r}}) \neq 0, qquad p = -\frac{5U_4}{U_3}, \tag{EZ34j}$$

where

$$H_{\mathbf{r}}(U, V) = U_0U^4 + U^3V + U_2U^2V^2 + U_3UV^3 + U_4V^4.$$

Then

$$\begin{aligned} H_{\mathbf{r}}(U, V) &= (U - pV)Q_{\mathbf{r},p}(U, V), \\ Q_{\mathbf{r},p}(U, V) &= U_0U^3 + (1 + pU_0)U^2V \\ &\quad + (U_2 + p + p^2U_0)UV^2 + (U_3 + pU_2 + p^2 + p^3U_0)V^3. \end{aligned}$$

Put

$$\kappa = 4U_0p^3 + 3p^2 + 2U_2p + U_3 = h'_r(p).$$

It is nonzero by the discriminant hypothesis. For each of the two choices $A^2\kappa = 1$, define

$$L_A = A(U - pV), \quad C_A = A^{-1}Q_{r,p}.$$

Their six factor coordinates are

$$(a, b, c, d, e, f) = \left(A, -Ap, \frac{U_0}{A}, \frac{1+pU_0}{A}, \frac{U_2+p+p^2U_0}{A}, \frac{U_3+pU_2+p^2+p^3U_0}{A} \right). \quad (\text{EZ34k})$$

They satisfy, without a changed resultant convention,

$$L_AC_A = H_r, \quad \text{Res}(L_A, C_A) = 1, \quad ad + bc = 1, \quad 4af - be = -\frac{F_*(\mathbf{r})}{U_3^3} = 0. \quad (\text{EZ34l})$$

The corresponding original source points are

$$q_A(\mathbf{r}) = \left(A, -p - \frac{i}{A}, \frac{U_0}{A^3} + \frac{2p}{A} + \frac{3i}{A^2}, \right. \\ \left. \frac{7ip^2}{A} + \frac{U_2 - 17p + U_0p^2}{A^2} - \frac{13i}{A^3} - \frac{2U_0}{A^4} \right), \quad (\text{EZ34m})$$

and the two receiver-source points are $z_A = \Psi_* q_A$.

More intrinsically, let z be any of the eight points over $\mathbf{r}$ under $P_{\mathcal{W},*}$, and decode $(A(z), Y(z), Z(z), \Omega(z))^T = \Psi_*^{-1}z$. Then (z) is one of the two points marking (p) if and only if

$$5A(z)U_4(\mathbf{r}) - A(z)Y(z)U_3(\mathbf{r}) - iU_3(\mathbf{r}) = 0. \quad (\text{EZ34n})$$

Thus EZ34n selects exactly two states from the eight-point receiver fibre.

Proof. Multiplication of $(U - pV)Q_{r,p}$ reproduces the first four coefficients of H_r . Its last residual is $h_r(p) = U_4F_*(\mathbf{r})/U_3^4$, so it vanishes under EZ34j. The resultant orientation retained in the source is

$$\text{Res}(A(U - pV), A^{-1}Q) = A^2Q(p, 1) = A^2h'_r(p) = 1.$$

Direct substitution of EZ34k gives $ad + bc = 1$ and

$$4af - be = 4U_3 + 5pU_2 + 5p^2 + 5p^3U_0 = -F_*(\mathbf{r})/U_3^3,$$

proving EZ34l. Formula EZ34m is the full inverse of the Fable factor chart, already checked in EZ13, with no coordinate suppressed.

For the selector, the marked root decoded from a source point is exactly $t(z) = -Y(z) - i/A(z)$. Equating it with $p = -5U_4/U_3$ and multiplying by $A(z)U_3$ gives EZ34n. Distinctness of the four roots gives two signs over each root, so precisely two of the eight states satisfy it. $\square$

Equations EZ34b and EZ34i are literal polynomial constraints in the fixed receiving coordinates. They do not make the reference roots $\rho_{*,j}$, the moments μ_{v+j} , or any zeros of a zeta or L -function equal to p, x, y, z . Their role is confined to the proved invertible coordinate matrices EZ29–EZ34h.

For fixed $s > 0$ and centre c_* , the receiving plane retains the norm

$$\|q\|_{\Gamma,*}^2 = \int_{\mathbb{R}} |q(c_* + iy)|^2 \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy. \quad (\text{EZ35})$$

Its Meixner–Pollaczek weight, mass, recurrence and generating function are the exact specialization of DLMF 18.19.8–9, 18.22.8 and 18.23.7 [14]. This metric makes the bridge quantitative; it does not make the four ES roots into zeta zeros.

20 Exact three-dimensional Fable–Erdős–Straus transport

The canonical project reader contains a four-dimensional raw-cell carrier and an exact linear transport to the Erdős–Straus-aligned Lorentz coordinates (T, X, Y, W) [7, Theorem on the exact spatial quarter altitude of the transported raw-cell tetrahedron]. The following coordinate statement is reproduced here because it is the three-dimensional geometry actually used by the present bridge.

Theorem 20.1 (Tetrahedron transport). *In the ordered raw-cell basis $(E_{11}, E_{12}, E_{21}, E_{22})$, put*

$$\begin{aligned} v_* &= (0, 0, 1, 0)^\top, & v_0 &= (2, 2, 0, 2)^\top, \\ v_+ &= (2 + \sqrt{3}, -1, 0, 2 - \sqrt{3})^\top, & v_- &= (2 - \sqrt{3}, -1, 0, 2 + \sqrt{3})^\top. \end{aligned}$$

The retained raw-cell to Lorentz map is

$$\Phi_\Delta = \begin{pmatrix} 5/8 & 0 & 3/4 & 5/8 \\ 0 & 1 & 0 & 0 \\ 3/8 & 0 & 5/4 & 3/8 \\ 1/2 & 0 & 0 & -1/2 \end{pmatrix}, \quad \det \Phi_\Delta = -\frac{1}{2}. \quad (\text{EZ35A})$$

It sends these four vertices to

$$\begin{aligned} z_* &= (3/4, 0, 5/4, 0)^\top, & z_0 &= (5/2, 2, 3/2, 0)^\top, \\ z_+ &= (5/2, -1, 3/2, \sqrt{3})^\top, & z_- &= (5/2, -1, 3/2, -\sqrt{3})^\top. \end{aligned} \quad (\text{EZ35B})$$

For $q_L(T, X, Y, W) = X^2 + Y^2 + W^2 - T^2$,

$$q_L(z_*) = 1, \quad q_L(z_0) = q_L(z_+) = q_L(z_-) = 0. \quad (\text{EZ35C})$$

After the spatial projection $\pi_{\text{sp}}(T, X, Y, W) = (X, Y, W)$, write $p_a = \pi_{\text{sp}}(z_a)$. Then

$$\frac{p_0 + p_+ + p_-}{3} - p_* = (0, 1/4, 0), \quad (\text{EZ35D})$$

every face edge has length $2\sqrt{3}$, every apex edge has length $\sqrt{65}/4$, and the Euclidean volume is $\sqrt{3}/4$.

Proof. Multiplication of the matrix in EZ35A by the four displayed raw vertices gives EZ35B, retaining every coordinate and sign. Substitution in q_L gives EZ35C. The spatial vertices are

$$p_* = (0, 5/4, 0), \quad p_0 = (2, 3/2, 0), \quad p_+ = (-1, 3/2, \sqrt{3}), \quad p_- = (-1, 3/2, -\sqrt{3}).$$

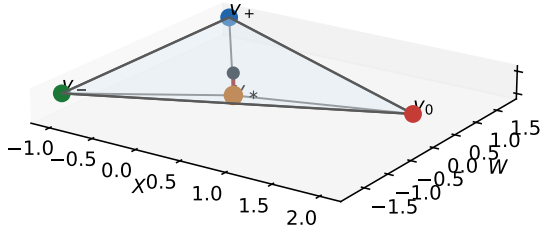
Their face centroid is $(0, 3/2, 0)$, which proves EZ35D. Direct squared distances are 12 between distinct face vertices and $65/16$ from the apex to each face vertex. Finally,

$$\frac{1}{6} \left| \det \begin{pmatrix} p_0 - p_* & p_+ - p_* & p_- - p_* \end{pmatrix} \right| = \frac{\sqrt{3}}{4}.$$

These calculations prove the asserted metric data. They also show why this tetrahedron is not to be silently replaced by a regular Euclidean one. $\square$

Fable raw cells transported into the Erdős–Straus spatial slice

Exact spatial tetrahedron



$$p_* = (0, 5/4, 0)$$

$$p_0 = (2, 3/2, 0)$$

$$p_+ = (-1, 3/2, \sqrt{3})$$

$$p_- = (-1, 3/2, -\sqrt{3})$$

$$\tilde{p}_{\text{face}} - p_* = (0, 1/4, 0)$$

$$\text{face edges: } 2\sqrt{3}$$

$$\text{apex edges: } \sqrt{65}/4$$

$$\text{volume: } \sqrt{3}/4$$

These are the retained spatial coordinates after the exact raw-cell to ES-aligned Lorentz transport; no Euclidean regularity is asserted.

Figure 4: The actual spatial projection in EZ35B–EZ35D. The three null vertices lie in the plane $Y = 3/2$; the spacelike apex lies at $Y = 5/4$. The face centroid and apex have identical X, W coordinates and differ by the exact quarter. Axis coordinates and all metric quantities are retained; the drawing does not assert Euclidean regularity. The figure source is `tools/generate_20260920_illustrations.py`; the exact coordinate certificate is the local `verify_fable_es_raw_tetrahedron.py`, reproduced from the cited canonical source calculation.

21 The odd receiving divisor pulled back to the Erdős–Straus locus

This section applies the odd receiving-frame calculation of the fixed weighted-conductor source to the literal Erdős–Straus coefficient map. The nonlinear Fable map, the invertible linear conductor, and the odd receiving matrix are three different maps. Only the third map is considered here. The inverse branches and their original source coordinates are those of [10, GF5–GF12]; the determinant divisor and its ES pullback are calculated below.

For a target $u = (A, B, C, D)$, put

$$h_u(r) = Ar^4 + r^3 + Br^2 + Cr + D, \quad d(r) = h'_u(r).$$

On the open locus where $A \neq 0$ and h_u has four distinct roots $r_1, \dots, r_4$, choose ξ_j with $\xi_j^2 d(r_j) = 1$. The odd columns of the eight signed inverse states are

$$o_j = \xi_j \begin{pmatrix} 1 \\ -id(r_j) \\ Ad(r_j)^2 + 2r_j d(r_j) \\ i(7r_j^2 d(r_j) - 13d(r_j)^2) \end{pmatrix}, \quad O = [o_1 \ \cdots \ o_4]. \quad (\text{EZ37})$$

Changing an individual square-root sign changes the corresponding column sign; therefore the determinant zero locus is independent of these choices.

Proposition 21.1 (Exact odd receiving divisor). *Define*

$$\begin{aligned} \mathfrak{D}_{\text{odd}}(A, B, C, D) := & 20(3C - B^2) + A(87B^3 - 282BC - 51D) \\ & + A^2(-28B^4 + 22B^2C + 431C^2 + 204BD) \\ & + A^3(224B^2D - 168BC^2 - 856CD) - 448A^4D^2. \end{aligned} \quad (\text{EZ38})$$

Then

$$\det O = \varepsilon_{\text{br}} \frac{4\mathfrak{D}_{\text{odd}}(A, B, C, D)}{A^3}, \quad \varepsilon_{\text{br}} \in \{1, -1\}. \quad (\text{EZ39})$$

Proof. Reduce the four component polynomials in EZ37 modulo h_u in the ordered basis $1, r, r^2, r^3$. The first reduced polynomial is 1. Removing its row and column leaves the coefficient matrix

$$R = \begin{pmatrix} 2B & 3 & 4A \\ 4ABC - 8AD - 7C & 4AB^2 - 2AC - 16A^2D - 5B & 4AB - 8A^2C - 3 \\ 20C/A + 76D - 52BC & 20B/A + 5C - 52B^2 + 208AD & 20/A - 66B + 104AC \end{pmatrix}.$$

Direct expansion, without changing coordinates, gives $\det R = 4\mathfrak{D}_{\text{odd}}/A$. Let $V = \prod_{j < k} (r_k - r_j)$. Since

$$\prod_j d(r_j) = A^4 V^2, \quad \left(V \prod_j \xi_j \right)^2 = A^{-4},$$

one has $V \prod_j \xi_j = \varepsilon_{\text{br}} A^{-2}$. Multiplying the coefficient determinant, the evaluation Vandermonde and the four retained square-root factors proves EZ39. $\square$

Return to the ordered Erdős–Straus source EZ1–EZ3. For compactness write

$$S = p + s_1, \quad b = ps_1 + s_2, \quad c = ps_2 + s_3, \quad d = ps_3. \quad (\text{EZ40})$$

The normalized coefficient map EZ7 is

$$\Theta(p, s_1, s_2, s_3) = \left(-\frac{1}{S}, -\frac{b}{S}, \frac{c}{S}, -\frac{d}{S} \right). \quad (\text{EZ41})$$

Its domain in this section is the ordered algebraic ES source

$$X_{\text{ES}}^{\text{ord}} = \{ps_2 = 4s_3, Sp s_3 \Delta \neq 0\},$$

where Δ is the discriminant of the four literal roots. The codomain of Θ is the coefficient affine space $\mathbb{A}_{A,B,C,D}^4$, and $\mathfrak{D}_{\text{odd}} : \mathbb{A}^4 \rightarrow \mathbb{A}^1$.

Theorem 21.2 (Exact pullback and scale-free irreducible divisor). *The composite $\mathfrak{D}_{\text{odd}} \circ \Theta$ is*

$$\mathfrak{D}_{\text{odd}} \circ \Theta = \frac{\tilde{P}}{S^6}, \quad (\text{EZ42})$$

where

$$\begin{aligned} \tilde{P} = & 60cS^5 - (20b^2 + 51d)S^4 - 282bcS^3 \\ & + (87b^3 + 431c^2 + 204bd)S^2 + (22b^2c - 856cd)S \\ & - 28b^4 + 224b^2d - 168bc^2 - 448d^2. \end{aligned}$$

On the ES relation, put

$$T = \frac{s_1}{p}, \quad Z = \frac{s_3}{p^3}. \quad (\text{EZ43})$$

Then

$$\mathfrak{D}_{\text{odd}} \circ \Theta = \frac{p^2 \Phi(T, Z)}{(1+T)^6}, \quad (\text{EZ44})$$

with

$$\begin{aligned} \Phi(T, Z) = & -7168Z^4 + 32(174T^2 + 179T - 184)Z^3 \\ & + (-320T^4 - 2744T^3 - 705T^2 + 3350T + 903)Z^2 \\ & + (140T^5 + 443T^4 - 440T^3 - 390T^2 - 70T + 249)Z \\ & - T^2(T-1)^2(20T^2 + 33T + 20). \end{aligned} \quad (\text{EZ45})$$

The polynomial Φ is primitive and irreducible over $\mathbb{Q}$.

Proof. Substitution of EZ41 into every retained term of EZ38 and collection over the common denominator S^6 gives EZ42. On $ps_2 = 4s_3$, the quantities in EZ40 become

$$S = p(1+T), \quad b = p^2(T+4Z), \quad c = 5p^3Z, \quad d = p^4Z.$$

Substitution into EZ42 and collection of the factor $p^2/(1+T)^6$ gives EZ44–EZ45 term for term. The exact checker factors Φ independently with SymPy and python-flint. Both return a single primitive factor. Its reduction modulo 11 is again one degree-six factor, so Gauss's lemma also certifies irreducibility over $\mathbb{Q}$. $\square$

The supplied coefficient point from the receiving-frame calculation is not on the ES coefficient hypersurface. Indeed, for $u_* = (1, 0, -60/431, 0)$, the polynomial G of EZ9 gives

$$G(u_*) = -\frac{51840000}{34507149121} \neq 0. \quad (\text{EZ46})$$

It also has $A > 0, D = 0$, whereas every positive ES image in EZ7 has $A < 0, D < 0$. This proves a specific nonidentity; it does not make the two divisors unrelated, because EZ42–EZ45 are their exact pullback relation.

Theorem 21.3 (Positive-real intersection and exact arithmetic residue). *The divisor $V(\Phi)$ meets the positive-real, pairwise-distinct ES locus. More precisely, set $p = 1$ and*

$$\frac{p}{x} = \frac{1}{4} - \delta, \quad \frac{p}{y} = \frac{1}{4} + \delta, \quad \frac{p}{z} = \frac{7}{2}, \quad (\text{EZ47})$$

where $w = \delta^2$. Then $4/p = 1/x + 1/y + 1/z$, and

$$\mathfrak{D}_{\text{odd}} \circ \Theta = -\frac{8Q(w)}{(144w - 65)^6},$$

where

$$\begin{aligned} Q(w) = & 319186534400w^6 + 12383961481216w^5 - 26262983737344w^4 \\ & - 1257389244416w^3 + 1383887871744w^2 + 2670303991200w - 90309375. \end{aligned}$$

One has $Q(10^{-6}) < 0 < Q(10^{-4})$. Consequently some $w_0 \in (10^{-6}, 10^{-4})$ gives a positive, pairwise-distinct real ES point on the divisor.

For positive rational ES shapes, intersection with this divisor is exactly the following splitting problem:

$$\begin{aligned} \Phi(T, Z) &= 0, \\ X^3 - TX^2 + 4ZX - Z &\text{ splits into three distinct positive rational roots,} \\ &\text{none equal to 1.} \end{aligned} \quad (\text{EZ48})$$

If the three roots in EZ48, written in lowest terms, have common denominator L , the shape lifts to a positive integral ES solution at every positive integer scale p divisible by L . It lifts with p prime exactly when $L = 1$ or L itself is prime. No existence or nonexistence result for such a prime-integral point is asserted here.

Proof. The three reciprocals in EZ47 sum to 4. For $10^{-6} < w < 10^{-4}$, one has $0 < \delta < 1/100$, so all three are positive and distinct. Direct substitution into EZ38 gives the displayed rational function. Its denominator is positive on the closed interval and the exact endpoint numerators are

$$\begin{aligned} Q(10^{-6}) &= -\frac{208947824537547554186600178130584239}{238418579101562500000000000000} < 0, \\ Q(10^{-4}) &= \frac{421368746134524395585536}{2384185791015625} > 0. \end{aligned}$$

The intermediate value theorem proves the real intersection.

For the arithmetic statement, the roots of the cubic in EZ48 are precisely $x/p, y/p, z/p$: their first and third elementary symmetric functions are $T = s_1/p$ and $Z = s_3/p^3$, while EZ3 gives the middle elementary symmetric function $s_2/p^2 = 4Z$. Positive integral denominators give the stated rational splitting. Conversely, if L is the least common denominator of the three roots, every multiple p of L makes x, y, z integral, and $s_2/p^2 = 4s_3/p^3$ gives the ES equation. Such a multiple can be prime exactly when $L = 1$, in which case any prime works, or L is prime, in which case the only prime multiple is $p = L$. Distinctness and exclusion of the root 1 are exactly the retained root-collision and prime-equals-denominator exclusions. This proves every assertion. $\square$

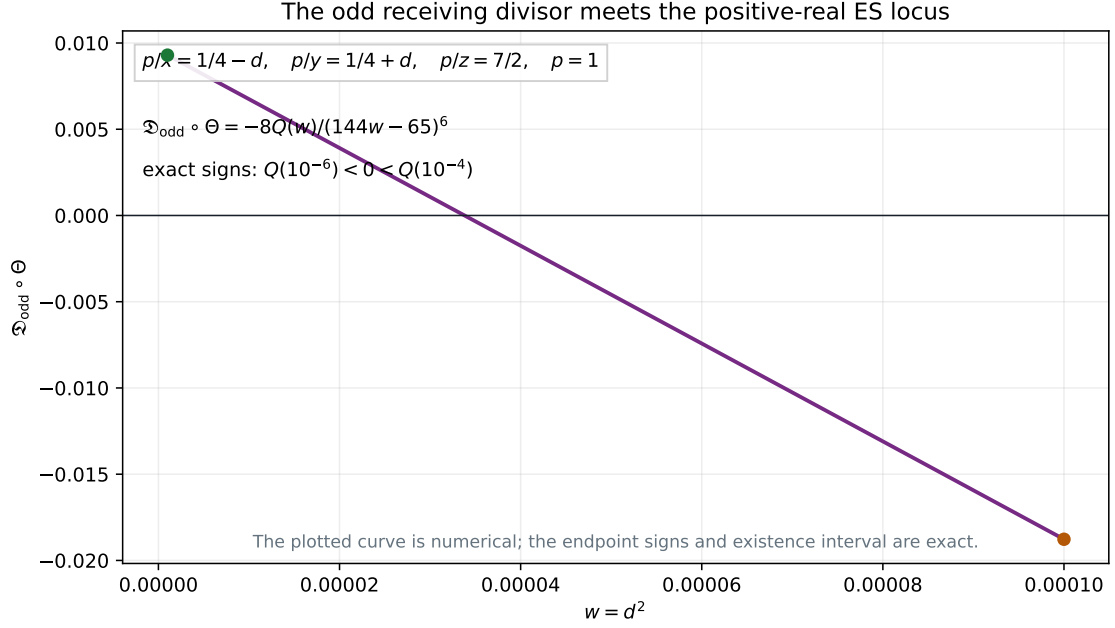


Figure 5: The exact one-parameter section EZ47 of the positive-real ES locus. The curve is rendered numerically, while the two endpoint signs printed in the proof are exact rational inequalities. Their sign change proves an intersection with the receiving divisor; it does not prove a rational or integral intersection. Reproduce with `tools/generate_20260920_illustrations.py` and certify with `verify_odd_receiver_es_pullback.py`.

22 Every hard-prime witness lies in the eight-sheet domain

The earlier exterior-state theorem can be strengthened without an exterior channel hypothesis. Throughout this section p is a positive prime with $p \equiv 1 \pmod{12}$, and

$$\frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}, \quad x, y, z \in \mathbb{Z}_{>0}. \quad (\text{EZ49})$$

Theorem 22.1 (Distinctness of every hard-prime witness). *The four integers p, x, y, z in EZ49 are pairwise distinct. Consequently the normalized literal-root quartic EZ6 has nonzero discriminant, and its Fable inverse has eight distinct signed states.*

Proof. If one denominator equals p , relabel it as $x = p$. Then

$$\frac{3}{p} = \frac{1}{y} + \frac{1}{z}, \quad (3y - p)(3z - p) = p^2.$$

Positivity of the reciprocal equation gives $y, z > p/3$, so both factors are positive integers. Since p is prime, each factor belongs to $\{1, p, p^2\}$. Every element of this set is $1 \pmod{3}$, because $p \equiv 1 \pmod{3}$, whereas $3y - p \equiv 2 \pmod{3}$. This is impossible.

If two denominators coincide, relabel them as $x = y = a$. Positivity gives $a > p/2$. Put $d = 2a - p > 0$. Exact rearrangement of EZ49 gives

$$4dz = p(d + p),$$

so $d \mid p^2$. The three possibilities $d = 1, p, p^2$ give, respectively,

$$z = \frac{p(p+1)}{4}, \quad z = \frac{p}{2}, \quad z = \frac{p+1}{4}.$$

None is integral when $p \equiv 1 \pmod{4}$. Symmetry excludes every collision of these two types. Thus all four literal roots are distinct. At every simple root t , the equation $\xi^2 h'_u(t) = 1$ has two nonzero solutions; the root and the sign distinguish the resulting eight source points. $\square$

Theorem 22.2 (Fixed-input separation and inverse bounds). *Relabel the denominators so that $x < y < z$, and define*

$$R = 4x - p, \quad S_R = px = \frac{p(p+R)}{4}, \quad s_p = \frac{p(p+3)}{4}, \quad B_p = p + s_p(s_p + 1). \quad (\text{EZ50})$$

Then

$$\frac{p}{4} < x < \frac{3p}{4}, \quad 3 \leq R \leq 2p - 3, \quad (\text{EZ51})$$

and

$$(Ry - S_R)(Rz - S_R) = S_R^2, \quad z \leq \frac{s_p(s_p + 1)}{3}, \quad S = p + x + y + z \leq B_p. \quad (\text{EZ52})$$

For $t_1, t_2, t_3, t_4 = p, x, y, z$,

$$\boxed{\text{Disc}(h_u) = S^{-6} \prod_{j < k} (t_j - t_k)^2 \geq \frac{144}{B_p^6}.} \quad (\text{EZ53})$$

At every selected root t_j ,

$$\frac{2}{S} \leq |h'_u(t_j)| \leq S^2, \quad \frac{1}{S} \leq |\xi_j| \leq \sqrt{\frac{S}{2}}. \quad (\text{EZ54})$$

The complete inverse point $q_j = (\xi_j, y_j, z_j, w_j)$, with the coordinates of EZ13 retained, satisfies

$$|y_j| \leq 2S, \quad |z_j| \leq 6S^2, \quad |w_j| \leq 41S^3. \quad (\text{EZ55})$$

If G is any specified positive-definite Hermitian Gram matrix in these four source coordinates and $\|G\|_2$ is its spectral norm, then

$$\boxed{\|q_j\|_G^2 \leq \|G\|_2 \left(\frac{S}{2} + 4S^2 + 36S^4 + 1681S^6 \right).} \quad (\text{EZ56})$$

Every right-hand side can be made a function of p alone by replacing S with B_p ; the lower bounds become $2/B_p$ and $1/B_p$.

Proof. Since x is the smallest denominator, $1/x < 4/p < 3/x$, which proves the first inequalities in EZ51. Because $p \equiv 1 \pmod{4}$, the positive integer $R = 4x - p$ is $3 \pmod{4}$. The strict upper bound on x then gives $3 \leq R \leq 2p - 3$.

The remaining two reciprocals satisfy

$$\frac{1}{y} + \frac{1}{z} = \frac{R}{S_R}.$$

Multiplication gives the first identity in EZ52. Both factors on its left are positive: each of $1/y$ and $1/z$ is strictly smaller than their sum R/S_R . Hence $Rz - S_R \leq S_R^2$, and

$$z \leq \frac{S_R(S_R + 1)}{R}.$$

The following subtraction is exact:

$$\frac{S_R(S_R + 1)}{R} - \frac{s_p(s_p + 1)}{3} = \frac{p^2(R - 3)}{16} \left(1 - \frac{p^2 + 4}{3R} \right).$$

It is nonpositive on EZ51 because

$$p^2 + 4 - 3R \geq p^2 + 4 - 3(2p - 3) = (p - 3)^2 + 4 > 0.$$

This proves the bound on z . Since $x + y + z \leq 3z$, it also proves $S \leq B_p$.

For four ordered distinct integers, their six positive pairwise gaps are at least 1, 1, 1, 2, 2, 3; their product is at least 12. The normalized leading coefficient is $-S^{-1}$, so the quartic discriminant is the first expression in EZ53 and the asserted lower bound follows.

At a root,

$$h'_u(t_j) = -S^{-1} \prod_{\ell \neq j} (t_j - t_\ell).$$

The three signed differences are distinct nonzero integers, and their absolute product is at least 2. Each absolute difference is smaller than S . This proves the derivative bounds. Taking absolute values in $\xi_j^2 h'_u(t_j) = 1$ proves the bounds on ξ_j .

Put $v = |\xi_j|^{-1} \leq S$ and $t = t_j \leq S$. The literal coefficient bounds are $|A| = S^{-1}$ and $|B| < S$. Applying them term by term to EZ13 gives

$$|y_j| \leq t + v \leq 2S,$$

$$|z_j| \leq |A|v^3 + 2tv + 3v^2 \leq 6S^2,$$

and

$$|w_j| \leq 7t^2v + (|B| + 17t + |A|t^2)v^2 + 13v^3 + 2|A|v^4 \leq 41S^3.$$

Finally $q_j^* G q_j \leq \|G\|_2 \|q_j\|_2^2$, and EZ54–55 give EZ56. $\square$

The preceding theorem is a compactness statement only after p has been fixed. Its bounds grow with p , and it neither proves that a witness exists nor bounds the separate odd receiving divisor away from zero.

23 The fixed-input signed cover

Fix $p \in \mathbb{C}^\times$. On the coefficient hypersurface, retain both the intrinsic-root relation $p = -5D/C$ and EZ9. On $ACD \neq 0$, these two relations are equivalent to

$$D = -\frac{pC}{5}, \quad B = -Ap^2 - p - \frac{4C}{5p}. \quad (\text{EZ57})$$

Substitution gives

$$h_u(r) = (r - p)g_p(r), \quad g_p(r) = Ar^3 + (1 + Ap)r^2 - \frac{4C}{5p}r + \frac{C}{5}, \quad (\text{EZ58})$$

and

$$d_p = g_p(p) = 2Ap^3 + p^2 - \frac{3C}{5}, \quad \text{Disc}(h_u) = d_p^2 \text{Disc}(g_p). \quad (\text{EZ59})$$

Let

$$\mathcal{B}_p = \{(A, C) : ACd_p \text{Disc}(g_p) \neq 0\},$$

and let $\mathcal{R}_p$ be its coordinate ring. The factor C is retained because this is the intrinsic-prime ES chart. The exact squarefreeness boundary at fixed $p \neq 0$ is $Ad_p \text{Disc}(g_p) = 0$.

Theorem 23.1 (Rank 2 + 6 decomposition). *In $\mathcal{R}_p[r]/(h_u)$,*

$$e_p(r) = \frac{g_p(r)}{d_p} \quad (\text{EZ60})$$

is the idempotent equal to one at the marked root p and zero at the three denominator roots. The signed inverse algebra splits as

$$\boxed{\mathcal{S}_p \simeq \mathcal{R}_p[\xi]/(\xi^2 d_p - 1) \times (\mathcal{R}_p[r]/(g_p)) [\xi]/(\xi^2(r-p)g'_p(r) - 1)} \quad (\text{EZ61})$$

The two factors have ranks two and six over $\mathcal{R}_p$.

Proof. Substituting the first identity of EZ57 into EZ9 gives

$$-C^3(5Ap^3 + 5p^2 + 5pB + 4C) = 0,$$

which gives the second identity of EZ57 because $pC \neq 0$. Direct multiplication proves EZ58. The resultant product formula gives EZ59. Chinese remaindering by the coprime factors $r - p$ and g_p gives the idempotent EZ60. Finally

$$h'_u(p) = d_p, \quad h'_u(r) = (r-p)g'_p(r) \quad \text{on } g_p(r) = 0,$$

and substituting these identities into $\xi^2 h'_u(r) = 1$ proves EZ61. $\square$

Theorem 23.2 (Order-48 monodromy and exact deck group). *Label the marked root by 0 and the denominator roots by 1, 2, 3. The monodromy image on the eight signed states over $\mathcal{B}_p$ is*

$$\boxed{G_p = \left\{ (\sigma, \pi) \in \{\pm 1\}^4 \rtimes S_4 : \pi(0) = 0, \prod_{j=0}^3 \sigma_j = \text{sgn}(\pi) \right\}}, \quad |G_p| = 48. \quad (\text{EZ62})$$

The deck group of the disconnected rank-2 + 6 cover is

$$\boxed{\text{Deck}(\mathcal{S}_p/\mathcal{B}_p) \simeq C_2 \times C_2.} \quad (\text{EZ63})$$

Proof. Let $\Delta = \prod_{j < k} (t_k - t_j)$. Since

$$\prod_j h'_u(t_j) = A^4 \Delta^2,$$

the retained signed invariant

$$\chi = A^2 \Delta \prod_j \xi_j \in \{1, -1\}$$

is locally constant. Continuation by (σ, π) therefore forces $\prod_j \sigma_j = \text{sgn}(\pi)$. The marked root cannot move, which proves the upper bound in EZ62.

Define the ordered denominator space X_p by choosing $(x, y) \in \mathbb{C}^2$, putting

$$z = \frac{pxy}{4xy - p(x+y)}, \quad (\text{EZ64})$$

and deleting the loci on which a denominator is zero, two of p, x, y, z collide, the denominator in EZ64 vanishes, or $S = p + x + y + z$ vanishes. This is the complement of finitely many complex algebraic curves in $\mathbb{C}^2$, and it is nonempty. It is also path connected. Given two retained points, the complex affine line through them cannot be contained in any deleted curve, because both endpoints lie outside that curve. It therefore meets each deleted algebraic curve in finitely many points. A complex line with finitely many points removed is path connected, so the two retained points are joined inside X_p .

There is an exact coefficient map

$$q_p : X_p \longrightarrow \mathcal{B}_p, \quad q_p(x, y) = \left(-\frac{1}{p + x + y + z}, \frac{5xyz}{p + x + y + z} \right) = (A, C).$$

It is invariant under all six reorderings of (x, y, z) . The reciprocal equation and the two displayed coefficients give the literal identity

$$A \prod_{t \in \{p, x, y, z\}} (r - t) = (r - p)g_p(r).$$

The deletions in the definition of X_p then give $ACd_p \text{ Disc}(g_p) \neq 0$, so q_p lands in $\mathcal{B}_p$. Conversely, for $(A, C) \in \mathcal{B}_p$, the unordered denominator triple is exactly the root set of

$$\frac{g_p(r)}{A} = r^3 + \left(p + \frac{1}{A}\right)r^2 - \frac{4C}{5pA}r + \frac{C}{5A}.$$

Thus its elementary symmetric functions are

$$x + y + z = -p - \frac{1}{A}, \quad xy + xz + yz = -\frac{4C}{5pA}, \quad xyz = -\frac{C}{5A}.$$

They obey $p(xy + xz + yz) = 4xyz$, which is precisely the reciprocal Erdős–Straus equation and gives EZ64. The factors defining $\mathcal{B}_p$ say exactly that these roots are nonzero, mutually distinct, different from p , and have $S \neq 0$. Hence every fibre of q_p is the free six-element ordering orbit, and the elementary-symmetric inverse proves the exact quotient

$$X_p/S_3 \xrightarrow{\sim} \mathcal{B}_p.$$

Since X_p is connected, this finite S_3 -cover has unsigned monodromy equal to all of S_3 .

The sign subgroup is realized by two exact local families. First put

$$x = p(2 + w), \quad y = p(2 - w), \quad z = p \frac{4 - w^2}{12 - 4w^2}.$$

For $0 < |w| \leq 1/10$, only $x = y$ occurs inside the puncture. The two coefficient and derivative factors are

$$A = -\frac{4(w^2 - 3)}{p(21w^2 - 64)}, \quad C = -\frac{5p^2(w - 2)^2(w + 2)^2}{21w^2 - 64},$$

$$h'_u(p) = \frac{p^2(w - 1)(w + 1)(3w^2 - 8)}{21w^2 - 64},$$

$$h'_u(x) = -\frac{2p^2w(w + 1)(w + 2)(4w^2 - w - 10)}{21w^2 - 64},$$

$$h'_u(y) = \frac{2p^2w(w - 2)(w - 1)(4w^2 + w - 10)}{21w^2 - 64}.$$

$$h'_u(z) = -\frac{p^2(w - 2)(w + 2)(3w^2 - 8)(4w^2 - w - 10)(4w^2 + w - 10)}{16(w^2 - 3)^2(21w^2 - 64)}.$$

On $|w| \leq 1/10$, the estimates $|21w^2| < 64$, $|3w^2| < 8$, and $|4w^2 \pm w| < 10$, together with $|w| < 1 < 2$, prove that every displayed base factor and every derivative factor other than the single w -factor in $h'_u(x), h'_u(y)$ is nonzero. Thus the path stays in $\mathcal{B}_p$ away from its central collision. A half-circle exchanges x, y ; its square fixes the labels and flips exactly their two square-root signs.

Next put

$$x = p(1 + w), \quad y = \frac{2p}{5}, \quad z = \frac{2p(1 + w)}{1 + 3w}.$$

For $0 < |w| \leq 1/100$, only $p = x$ occurs inside the puncture, and

$$A = -\frac{5(3w + 1)}{p(15w^2 + 51w + 22)}, \quad C = \frac{20p^2(w + 1)^2}{15w^2 + 51w + 22},$$

$$h'_u(p) = \frac{3p^2w(w - 1)}{15w^2 + 51w + 22},$$

$$h'_u(x) = -\frac{p^2w(w + 1)(3w - 1)(5w + 3)}{15w^2 + 51w + 22}.$$

$$h'_u(y) = \frac{12p^2(w + 2)(5w + 3)}{25(15w^2 + 51w + 22)},$$

$$h'_u(z) = -\frac{4p^2(w - 1)(w + 1)(w + 2)(3w - 1)}{(3w + 1)^2(15w^2 + 51w + 22)}.$$

For $|w| \leq 1/100$, $|15w^2 + 51w| < 22$; all remaining displayed linear factors are separated from zero except the single w -factor in $h'_u(p), h'_u(x)$. Hence this path also stays in $\mathcal{B}_p$ away from its central collision. A full circle fixes every root label and flips exactly the signs over p, x . Denominator pair flips span the even-sign plane on the three denominator labels; adjoining this prime-denominator flip spans all eight even sign vectors. With the six denominator permutations, the realized image has $8 \cdot 6 = 48$ elements and attains the upper bound.

The two connected sheet-orbits have sizes two and six. On the six denominator sheets, the induced action is the full signed permutation group $C_2^3 \rtimes S_3$, whose centralizer is generated by simultaneous denominator sign reversal. The prime-pair centralizer is C_2 . The two deck maps can be written

$$\xi \mapsto (1 - 2e_p)\xi, \quad \xi \mapsto -\xi.$$

They independently reverse the prime pair and the denominator sheets, and their products exhaust the centralizer. This proves EZ63. $\square$

Corollary 23.3 (Correct order-48 orbit metric). *In even/odd coordinates for each signed root pair, the eight-dimensional representation is*

$$\mathbf{1}^{\oplus 2} \oplus V_2 \oplus \chi_{\det} \oplus V_3, \tag{EZ65}$$

where V_2 is the standard two-dimensional S_3 -module, V_3 is the three-dimensional signed-permutation module, and

$$\chi_{\det}(\sigma, \pi) = \text{sgn}(\pi) \prod_{i=1}^3 \sigma_i = \sigma_0.$$

Let $E = [e_p, E_d]$ and $O = [o_p, O_d]$ be the original even and odd source columns, and let G be the specified receiving form. Averaging the pulled back Gram matrix over G_p , in the corresponding orthonormal decomposition, gives

$$\boxed{H_{\text{triv}} \oplus \lambda_2 I_2 \oplus \lambda_p \oplus \lambda_3 I_3}, \quad (\text{EZ66})$$

where

$$H_{\text{triv}} = \begin{bmatrix} e_p & E_d \mathbf{1}/\sqrt{3} \end{bmatrix}^* G \begin{bmatrix} e_p & E_d \mathbf{1}/\sqrt{3} \end{bmatrix},$$

$$\lambda_2 = \frac{1}{2} \text{tr}(P_2 E_d^* G E_d), \quad P_2 = I_3 - \frac{1}{3} \mathbf{1} \mathbf{1}^*, \quad \lambda_p = o_p^* G o_p, \quad \lambda_3 = \frac{1}{3} \text{tr}(O_d^* G O_d).$$

In particular, the mutual Gram term between the prime-even state and the denominator-even sum survives; the two-dimensional block cannot be replaced by two unrelated scalars.

Proof. The even denominator permutation module is $\mathbf{1} \oplus V_2$, while the prime-even line is another trivial module. The prime-odd line transforms by χ_{det} , and the three denominator-odd coordinates form V_3 . Averaging kills maps between inequivalent summands, retains the full Gram form on the two equivalent trivial copies, and gives the normalized traces on the irreducible V_2, V_3 blocks. The one-dimensional χ_{det} block is $o_p^* G o_p$. These are exactly the four blocks in EZ66. $\square$

Fixed- p signed cover: rank 2 + 6, monodromy order 48

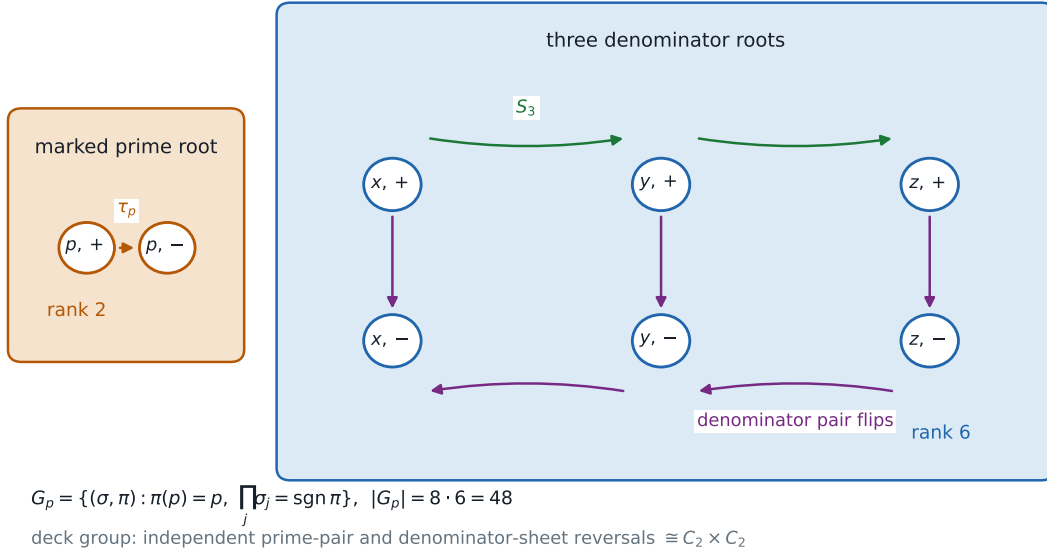


Figure 6: The proved fixed-input decomposition. The marked prime produces a rank-two component and the three denominator roots a rank-six component. Denominator permutations, denominator pair flips, and one prime–denominator flip generate the order-48 monodromy; the two displayed deck reversals commute with it. This is a sheet-action diagram, not a metric embedding. Reproduce it with `tools/generate_20260920_illustrations.py`; exact group enumeration is in `verify_fixed_input_boundary.py`.

24 Finite completion of the inverse-derivative boundary

Work over

$$\mathcal{R}_0 = \mathbb{C}[A, A^{-1}, B, C, D], \quad \mathcal{E} = \mathcal{R}_0[r]/(h_u).$$

Introduce the reciprocal signed coordinate $\eta = \xi^{-1}$ and define

$$\boxed{\widehat{\mathcal{S}} = \mathcal{E}[\eta]/(\eta^2 - h'_u(r))}. \quad (\text{EZ67})$$

Theorem 24.1 (Finite flat rank-eight completion). *The algebra EZ67 is free of rank eight over $\mathcal{R}_0$, with basis*

$$1, r, r^2, r^3, \eta, \eta r, \eta r^2, \eta r^3. \quad (\text{EZ68})$$

It recovers the original signed inverse exactly on $\eta \neq 0$:

$$\boxed{\mathcal{E}[\xi]/(\xi^2 h'_u - 1) \simeq \widehat{\mathcal{S}}[\eta^{-1}], \quad \eta = \xi h'_u = \xi^{-1}, \quad \xi = \eta^{-1}}. \quad (\text{EZ69})$$

Thus its spectrum is finite flat of degree eight and proper over $\text{Spec } \mathcal{R}_0$. Its ramification Jacobian in the (r, η) variables is

$$\det \begin{pmatrix} h'_u(r) & 0 \\ -h''_u(r) & 2\eta \end{pmatrix} = 2\eta h'_u(r) = 2\eta^3. \quad (\text{EZ70})$$

Proof. Because A is a unit, $A^{-1}h_u$ is monic of degree four; hence $\mathcal{E}$ is free of rank four. The relation in EZ67 is monic of degree two in η , which proves EZ68. Inverting η and setting $\xi = \eta^{-1}$ gives $\xi^2 h'_u = 1$. Conversely, in the original signed algebra, $\eta = \xi h'_u$ is the inverse of ξ and satisfies $\eta^2 = h'_u$. This proves EZ69. A finite morphism is proper [15]. Direct differentiation gives EZ70; away from $\eta = 0$, its determinant is a unit, agreeing with the standard derivative-unit criterion for an étale algebra [16]. $\square$

The completion is over $A \neq 0$. It does not repair the independent infinity-chart boundary $A = 0$.

Proposition 24.2 (The original meromorphic pole is retained). *In the completed coordinates, the original inverse point is*

$$q = \begin{pmatrix} \eta^{-1} \\ -r - i\eta \\ A\eta^3 + 2r\eta + 3i\eta^2 \\ 7ir^2\eta + (B - 17r + Ar^2)\eta^2 - 13i\eta^3 - 2A\eta^4 \end{pmatrix}. \quad (\text{EZ71})$$

*Let a path approach a fixed finite coefficient target of $\text{Spec } \mathcal{R}_0$, with $\eta \rightarrow 0$; equivalently the coefficients and r remain bounded and A stays nonzero. For any fixed injective receiving map F , with positive target form Q and $G = F^*QF$,*

$$\boxed{|\eta| \|Fq\|_Q \longrightarrow \sqrt{G_{11}} > 0}. \quad (\text{EZ72})$$

Proof. Formula EZ71 is EZ13 after substituting $\xi = \eta^{-1}$, with every term retained. Under the stated finite-target hypothesis, $\eta q \rightarrow (1, 0, 0, 0)^T$. Taking the norm induced by G gives EZ72. Thus the finite algebra adds the missing boundary state but does not turn the original meromorphic norm pole into a bounded point. $\square$

Theorem 24.3 (Exact nonreduced fibre at a multiple root). *Suppose*

$$h_u(r) = (r - r_0)^m g(r), \quad g(r_0) = g_0 \neq 0.$$

With $\varepsilon = r - r_0$, the completed local signed fibre is

$$\boxed{\widehat{\mathcal{S}}_{r_0} = \mathbb{C}[\varepsilon, \eta]/(\varepsilon^m, \eta^2 - mg_0\varepsilon^{m-1}), \quad \dim_{\mathbb{C}} \widehat{\mathcal{S}}_{r_0} = 2m}. \quad (\text{EZ73})$$

For a double root,

$$\varepsilon = \frac{\eta^2}{2g_0}, \quad \boxed{\widehat{\mathcal{S}}_{r_0} \simeq \mathbb{C}[\eta]/(\eta^4)}. \quad (\text{EZ74})$$

Multiplication by ε has two Jordan blocks of size m .

Proof. The factor g is a unit in the local root algebra, so $(h_u) = (\varepsilon^m)$. Modulo ε^m ,

$$h'_u(r) = mg_0\varepsilon^{m-1}.$$

This proves the presentation in EZ73. It is free on $1, \eta$ over $\mathbb{C}[\varepsilon]/(\varepsilon^m)$, proving the dimension and the two Jordan blocks. When $m = 2$, eliminating ε gives EZ74. $\square$

Theorem 24.4 (Exact local ES-paired-root isomorphism). *The integral ES boundary witness*

$$\frac{4}{5} = \frac{1}{5} + \frac{1}{2} + \frac{1}{10}$$

has

$$h_{\text{ES}}(r) = -\frac{1}{22}(r-5)^2(r-2)(r-10), \quad \eta_{\text{ES}}^2 = \frac{15}{11}\varepsilon_{\text{ES}}.$$

More generally, the rational family

$$(p; x, y, z) = \left(p; p, \frac{2p}{5}, 2p\right) \quad (\text{EZ75})$$

has a double root at p with $\eta_{\text{ES}}^2 = (3p/11)\varepsilon_{\text{ES}}$.

For the auxiliary paired-root quartic

$$h_{\text{pair}}(r) = -\frac{1}{2}\left((r - \frac{1}{2})^2 + \gamma^2\right)^2, \quad r_{\pm} = \frac{1}{2} \pm i\gamma,$$

one has $\eta_{\text{pair}}^2 = 4\gamma^2\varepsilon_{\text{pair}}$ at either double root. If $p\gamma \neq 0$ and

$$c^2 = \frac{3p}{44\gamma^2},$$

then

$$\boxed{\eta_{\text{ES}} \longmapsto c\eta_{\text{pair}}, \quad \varepsilon_{\text{ES}} \longmapsto \varepsilon_{\text{pair}}} \quad (\text{EZ76})$$

is an isomorphism of the two length-four local algebras. It intertwines centred multiplication. On uncentred coordinates,

$$r_{\text{ES}} \longmapsto r_{\text{pair}} + (p - r_{\pm}).$$

In the bases $1, \eta, \eta^2, \eta^3$, its matrix is $D_c = \text{diag}(1, c, c^2, c^3)$. Therefore the pullback of a target Hermitian form G_{pair} is $D_c^* G_{\text{pair}} D_c$, whose determinant is $|c|^{12} \det G_{\text{pair}}$.

Proof. All displayed local equations follow by evaluating the residual factor g at the double root and applying EZ73. In EZ75 the normalized residual value is $g_0 = 3p/22$; at $r_{\pm}$ it is $g_0 = 2\gamma^2$. The defining relation is preserved by EZ76 because

$$c^2(4\gamma^2) = \frac{3p}{11}.$$

The inverse uses c^{-1} . The statements about centred and uncentred multiplication follow from $r_{\text{ES}} = p + \varepsilon_{\text{ES}}$ and $r_{\text{pair}} = r_{\pm} + \varepsilon_{\text{pair}}$. The basis matrix and metric determinant are then direct. $\square$

This is a local algebra and action morphism. It does not identify the global fibres, their native metrics, or a varying family of zeta zeros. It also does not make the nilpotent action isometric. Indeed let

$$N = M_\varepsilon = (2g_0)^{-1} M_{\eta^2}$$

in EZ74. Then $N \neq 0$ and $N^2 = 0$. For every positive Hermitian form G , choose v with $Nv = w \neq 0$. In the ordered pair (w, v) , the compression of $N^*G + GN$ has matrix

$$\begin{pmatrix} 0 & \|w\|_G^2 \\ \|w\|_G^2 & 2\operatorname{Re}\langle w, v \rangle_G \end{pmatrix},$$

whose determinant is $-\|w\|_G^4 < 0$. Thus it has both signs.

Finite inverse-derivative completion at a double root

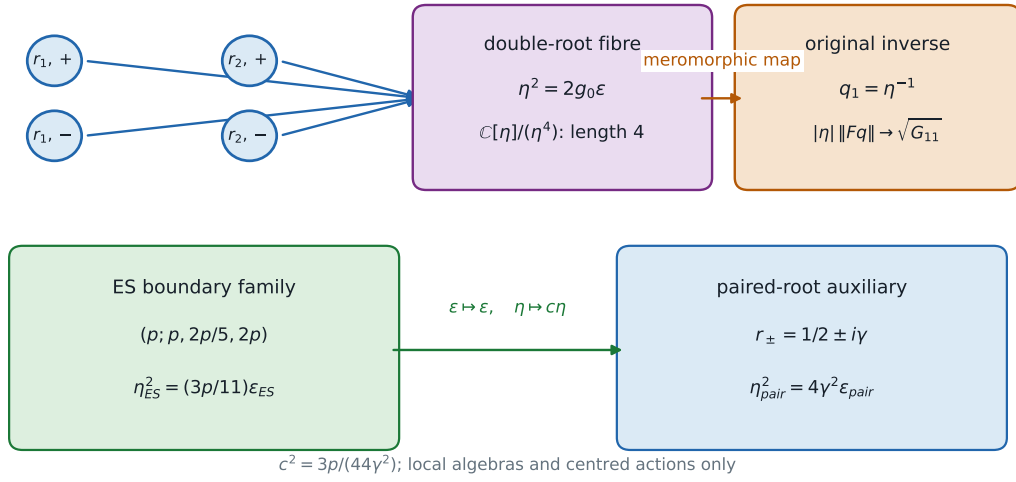


Figure 7: The finite completion and its exact double-root fibre. On $\eta \neq 0$, $\xi = \eta^{-1}$ recovers the original inverse. At a double root, four nearby signed states coalesce into the length-four algebra $\mathbb{C}[\eta]/(\eta^4)$, while the original first source coordinate keeps its η^{-1} pole. The lower arrow is the typed local algebra map EZ76; it does not identify the global ES and paired-root constructions. Reproduce with `tools/generate_20260920_illustrations.py`.

25 An invertible marked odd-moment frame

On the ordered signed-root cover of the squarefree $A \neq 0$ coefficient domain, retain the labels $r_1, \dots, r_4$ and signs $\xi_j^2 h'_u(r_j) = 1$, and define

$$U = \begin{pmatrix} 1 & 1 & 1 & 1 \\ r_1 & r_2 & r_3 & r_4 \\ r_1^2 & r_2^2 & r_3^2 & r_4^2 \\ r_1^3 & r_2^3 & r_3^3 & r_4^3 \end{pmatrix} \operatorname{diag}(\xi_1, \xi_2, \xi_3, \xi_4). \quad (\text{EZ77})$$

Theorem 25.1 (Odd-frame factorization and complete inverse). *One has*

$$\boxed{\det U = \frac{\chi}{A^2}, \quad \chi \in \{1, -1\}.} \quad (\text{EZ78})$$

For each of the four component polynomials

$$1, \quad -ih'_u, \quad A(h'_u)^2 + 2rh'_u, \quad i(7r^2h'_u - 13(h'_u)^2),$$

reduce modulo h_u , and let the corresponding ascending coefficient rows in the ordered basis $1, r, r^2, r^3$ be the rows of V_h . If O is the original odd evaluation matrix EZ37, then

$$\boxed{O = V_h U.} \quad (\text{EZ79})$$

For $m = U\beta$, the inverse is

$$\boxed{\begin{aligned} \beta_j = \xi_j [& (C + Br_j + r_j^2 + Ar_j^3)m_0 \\ & + (B + r_j + Ar_j^2)m_1 + (1 + Ar_j)m_2 + Am_3]. \end{aligned}} \quad (\text{EZ80})$$

Proof. The first matrix in EZ77 is the labelled Vandermonde matrix. If $\Delta = \prod_{j < k} (r_k - r_j)$, then

$$\prod_j h'_u(r_j) = A^4 \Delta^2, \quad \left(\Delta \prod_j \xi_j \right)^2 = A^{-4},$$

which proves EZ78. Evaluating the four reduced component polynomials at the four roots and multiplying each column by its retained ξ_j gives EZ79.

For the inverse, synthetic division gives

$$\frac{h_u(r)}{r - r_j} = Ar^3 + (1 + Ar_j)r^2 + (B + r_j + Ar_j^2)r + (C + Br_j + r_j^2 + Ar_j^3).$$

The corresponding Lagrange polynomial is this quotient divided by $h'_u(r_j)$. Since $1/(\xi_j h'_u(r_j)) = \xi_j$, applying the Lagrange coefficient functional to m_0, m_1, m_2, m_3 gives EZ80. $\square$

Although U itself lives on the marked signed cover, EZ78–79 show that the additional unmarked zero divisor of the original odd evaluation is exactly $\det V_h = 0$, away from $A \text{ Disc}(h_u) = 0$. At that divisor, moments cannot be reconstructed from the already-collapsed data $O\beta$; they must be retained before applying V_h .

The original integral signed completion

The next proof works in the literal root basis (p, x, y, z) over $\mathbb{Z}_p$. It retains the original arithmetic lattice rather than replacing a finite flat rank-eight algebra by an abstract vector space.

The original integral signed order and its exact normalization

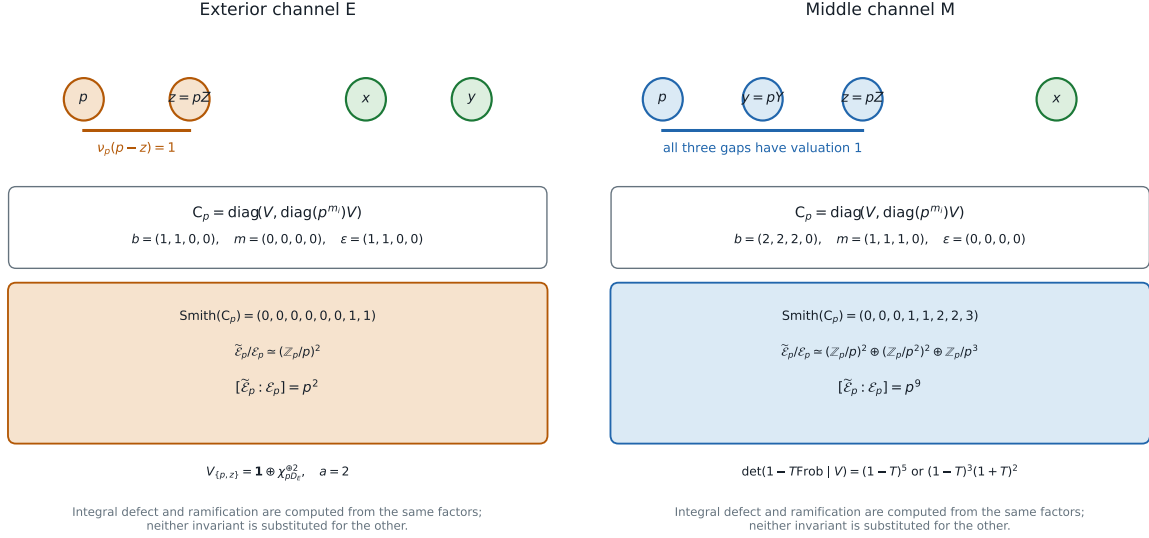


Figure 8: The exact normalization matrix and all Smith factors in the exterior and middle channels. The matrix, inverse lattice, quotient modules and specialization kernels are proved in the following sections.

26 Question, inherited objects, and the unresolved implication

The preceding capacity theorem [13] completes a fixed-target divisor problem. It proves that searching additional divisors at a fixed grade cannot repair every exterior source. Its universal existence obligation is still open: for a prescribed hard prime, produce an original exterior or middle state. We do not assume that obligation in the present calculation.

The newly supplied ES-Zeta work [11, §§1–3, equations (8)–(24)] uses the *literal* four roots (p, x, y, z) and completes its signed inverse by $\eta = \xi^{-1}$. Its monic equations give a finite flat rank-eight algebra. The task here is to determine what happens to that particular algebra over $\mathbb{Z}_p$, with its original coordinate lattice, rather than replacing it by an arbitrary rank-eight vector space. The literal quartic is different from the reciprocal cubic in [12]; the exact inversion and its derivative twist are given in Section 32.1.

The Type-I/II identities are classical [1, §2, Propositions 2.2 and 2.6]. Quadratic character restrictions are also antecedents; in particular compare [4, Corollary 1.3 and Appendix A]. All calculations needed for the local and integral results below are proved explicitly. No historical-priority claim is made for these specializations or for the standard local algebra tools.

27 Original coordinates and the literal polynomial

Fix a prime $p \equiv 1 \pmod{12}$. Thus $p \geq 13$, and both -1 and 3 are squares modulo p . Set $K = \mathbb{Q}_p$, $\mathfrak{o} = \mathbb{Z}_p$, and normalize $v_p(p) = 1$. An original state retains

$$p/4 < a < p/2, \quad R = 4a - p, \quad u \mid a^2, \quad E : R \mid 4u + 1, \quad M : R \mid u + a. \quad (\text{IC1})$$

For $a = \prod_l l^{e_l}$ its original exponent vector is $\beta_l = v_l(u) - e_l \in [-e_l, e_l]$. Let

$$g = \gcd(a, u), \quad (h, r, s) = (g^2/u, u/g, a/g).$$

Then $a = hrs$, $u = hr^2$, and $\gcd(r, s) = 1$. The exact ordered returns are

$$\begin{aligned} E : \quad & \kappa = (pr + s)/R, \quad (x, y, z) = (a, hsk, phr\kappa), \\ M : \quad & \lambda = (r + s)/R, \quad (x, y, z) = (a, phs\lambda, phr\lambda). \end{aligned} \tag{IC2}$$

Multiplication proves $4/p = 1/x + 1/y + 1/z$. The ordered divisor inverses are $a^2/(Ry - pa)$ and $pa^2/(Ry - pa)$, respectively. The middle complement $u \mapsto a^2/u$ exchanges the last two denominators and is kept as an orientation, not suppressed as a duplicate.

Lemma 27.1 (Original character and unit facts). *Every factor in (IC2) is a p -unit. For each $d \mid a^2$, $\left(\frac{d}{R}\right) = \left(\frac{d}{p}\right)$, with Jacobi symbol on the left. Consequently*

$$E : \left(\frac{h}{p}\right) = -1, \quad M : \left(\frac{rs}{p}\right) = -1. \tag{IC3}$$

In the exterior case, $\kappa - r$ and $\kappa + r$ are p -units.

Proof. For an odd $l \mid a$, reciprocity and $R \equiv -p \pmod{l}$ give $\left(\frac{l}{R}\right) = \left(\frac{-1}{l}\right) \left(\frac{R}{l}\right) = \left(\frac{p}{l}\right) = \left(\frac{l}{p}\right)$. If $2 \mid a$, use $R \equiv -p \pmod{8}$ and the supplementary law. Multiplication over the original exponents proves the divisor identity. The E target is $-1/4$ modulo $R \equiv 3 \pmod{4}$, so $\left(\frac{u}{p}\right) = -1$ and $u = hr^2$ gives the first assertion. The M relation $r \equiv -s \pmod{R}$ gives the second.

The factors h, r, s are below p . In E put $D = (4u + 1)/R = (r + \kappa)/s$; the original identities give $pD + 1 = 4hr\kappa$, so κ is a unit too. If $p \mid \kappa - r$, this identity gives $4hr^2 \equiv 1 \pmod{p}$, contradicting $\left(\frac{h}{p}\right) = -1$. If $p \mid \kappa + r$, then $p \mid D$ and $4u + 1 = RD$ gives the equally impossible nonresidue $u \equiv -1/4$. For M, $r + s \leq rs + 1 \leq a + 1 < p$, so $\lambda < p$. $\square$

The fixed, ordered root vector and scale are

$$\begin{aligned} (\ell_0, \ell_1, \ell_2, \ell_3) &= (p, x, y, z), \quad S = p + x + y + z, \\ \mathcal{H}(T) &= -S^{-1} \prod_{i=0}^3 (T - \ell_i) = AT^4 + T^3 + BT^2 + CT + D. \end{aligned} \tag{IC4}$$

Here D is a quartic coefficient, not the original exterior cofactor D . Vieta and the reciprocal identity give

$$A = -S^{-1}, \quad C = 5xyz/S, \quad D = -pxyz/S, \quad D = -pC/5, \quad B = -Ap^2 - p - 4C/(5p). \tag{IC5}$$

These are the literal coefficient equations in [11, §2.1]; the prime and every root label are retained. Define $d_i = \mathcal{H}'(\ell_i)$.

Theorem 27.2 (Original prime-local root clusters). *The roots are distinct, $S \in \mathfrak{o}^\times$, and the following assertions hold. For E write $(x, y, z) = (a, b, pc)$. Then $a, b, a - b, a + b$ are units, $c \equiv 1/4 \pmod{p}$, and*

$$(v_p(d_0), v_p(d_1), v_p(d_2), v_p(d_3)) = (1, 0, 0, 1), \quad v_p \text{Disc}(\mathcal{H}) = 2. \tag{IC6}$$

For M write $(x, y, z) = (a, pb, pc)$. Then $b, c, b - c, b - 1, c - 1$ are units,

$$\begin{aligned} 4bc &\equiv b + c \pmod{p}, \quad \left(\frac{bc}{p}\right) = -1, \\ (v_p(d_0), v_p(d_1), v_p(d_2), v_p(d_3)) &= (2, 0, 2, 2), \quad v_p \text{Disc}(\mathcal{H}) = 6. \end{aligned} \tag{IC7}$$

In coefficient order (A, B, C, D) the valuations are respectively $(0, 0, 1, 2)$ and $(0, 1, 2, 3)$.

Proof. For E, $c = hr\kappa$ and $4c = pD + 1$. The remaining unit statements follow from Lemma 27.1 and $a \pm b = hs(r \pm \kappa)$. Thus exactly one pair of roots, p and pc , has difference of valuation one; all other differences are units. In particular they are all distinct, and $S \equiv a + b$ is a unit. The derivative and discriminant formulas prove (IC6). The coefficient $B \equiv -ab/S$ is a unit, and (IC5) gives the other valuations.

For M, $b = hs\lambda$, $c = hr\lambda$. The unit and distinctness of b, c follow from $r \neq s$, $r, s < p$; equality would force $r = s = 1$ and $R \mid 2$. The ES equation gives $4bc = pbc/a + b + c$, and (IC3) gives $\left(\frac{bc}{p}\right) = -1$. If $b = 1 \pmod{p}$, then $c = 1/3$, making bc a square since $\left(\frac{3}{p}\right) = 1$. The same excludes $c = 1$. The three roots p, pb, pc therefore have pairwise valuation-one differences; the singleton a is separated by units. Now $S \equiv a$, giving (IC7). In B the leading term is $-pa(1 + b + c)/S$, and $1 + b + c \equiv 1 + 4bc$ cannot vanish, since $bc = -1/4$ would be a square. Hence $v_p(B) = 1$. The remaining values follow from (IC5). $\square$

In particular any two original E and M coefficient targets at the same prime have distance exactly one in the maximum p -adic norm on (A, B, C, D) : the B -coordinate is a unit on E and divisible by p on M, while all coordinates are integral. An infinitesimal target deformation preserving reduction is not such a channel change. No comparison with a complex Gamma metric is inferred.

28 The rank-eight generic algebra and its original integral order

The supplied completion specializes, with its exact original basis, to

$$\mathcal{O} = \mathfrak{o}[T]/(\mathcal{H}), \quad \widehat{\mathcal{S}} = \mathcal{O}[\eta]/(\eta^2 - \mathcal{H}'(T)). \quad (\text{IC8})$$

The leading coefficient A is a unit, so both equations can be made monic. Thus $\widehat{\mathcal{S}}$ is free over $\mathfrak{o}$ on

$$1, T, T^2, T^3, \eta, \eta T, \eta T^2, \eta T^3. \quad (\text{IC9})$$

After inverting η it is the original signed inverse, with $\xi = \eta^{-1}$ and $\xi^2 \mathcal{H}'(T) = 1$. The full physical Fabel coordinates, inherited from [11, equation (21)], are

$$\left(\eta^{-1}, -T - i\eta, A\eta^3 + 2T\eta + 3i\eta^2, 7iT^2\eta + (B - 17T + AT^2)\eta^2 - 13i\eta^3 - 2A\eta^4\right). \quad (\text{IC10})$$

Either embedding of i in K may be chosen and is kept fixed. No complex period coefficient is presumed to have a p -adic descent.

Over K , root evaluation gives exactly

$$\widehat{\mathcal{S}}_K \simeq \prod_{i=0}^3 K[\eta_i]/(\eta_i^2 - d_i). \quad (\text{IC11})$$

There is no infinity factor: $A \neq 0$ and all four roots are finite. This is not the seven-dimensional reciprocal-cubic algebra.

Theorem 28.1 (Generic local signed types). *An E state has either*

$$K^4 \times K_{\text{ram}}^2 \quad \text{or} \quad K_{\text{unr}}^2 \times K_{\text{ram}}^2, \quad (\text{IC12})$$

and consequently four or zero K-rational signed inverse points. An M state has either

$$K^8 \quad \text{or} \quad K^4 \times K_{\text{unr}}^2, \quad (\text{IC13})$$

and consequently eight or four such points. Repeated field factors retain their root labels; they are not identified.

Proof. For E the two valuation-one derivatives have square ratio: after dividing by p their ratio is $-1 \pmod{p}$. They define the same ramified quadratic extension. At the other two roots the derivative residues are

$$-\frac{a^2(a-b)}{a+b}, \quad -\frac{b^2(b-a)}{a+b}.$$

Their ratio is a square, so either both split or both define the unramified quadratic extension. This proves (IC12).

For M, divide the three cluster derivatives by p^2 . Their unit residues are

$$(1-b)(1-c), \quad (b-1)(b-c), \quad (c-1)(c-b). \quad (\text{IC14})$$

Their product is the negative of a nonzero square. Since -1 is a square, either all three or exactly one are squares. At the singleton, $d_1 \equiv -a^2$ is a square. This proves (IC13).

For the local facts, a square unit modulo the odd prime has unique compatible lifts after a sign is chosen: if $z_n^2 \equiv d \pmod{p^n}$, set $z_{n+1} = z_n + p^n k$, with $k \equiv -(z_n^2 - d)/p^n (2z_n)^{-1} \pmod{p}$. A nonsquare unit has irreducible separable quadratic reduction and gives the unramified quadratic extension. An odd-valuation class gives a ramified quadratic. These are also the standard constructions in [5, Chapter 7, Newton's lemma and local extensions]. $\square$

At the single hard prime $p = 1009$, all four alternatives occur. The respective original (a, u) and ordered denominators are

channel, number of K -points	(a, u)	(x, y, z)	
$E, 0$	$(253, 11)$	$(253, 87032, 3818056)$	
$E, 4$	$(253, 23)$	$(253, 86020, 7890380)$	(IC15)
$M, 8$	$(253, 11)$	$(253, 2042216, 88792)$	
$M, 4$	$(253, 23)$	$(253, 1021108, 92828)$	

These are four independently checked original states. In particular a K -rational point on the literal signed fibre is not a necessary condition for an ES witness. Every row still has eight geometric characteristic-zero points.

29 Exact normalization, conductors and invariant factors

Put $k_i = \lfloor v_p(d_i)/2 \rfloor$ and

$$\theta_i = \eta_i/p^{k_i}, \quad \delta_i = d_i/p^{2k_i}, \quad N_i = \mathfrak{o}[\theta_i]/(\theta_i^2 - \delta_i), \quad \mathcal{N} = \prod_i N_i. \quad (\text{IC16})$$

Each δ_i is a unit or has valuation one. Thus N_i is the maximal integral order in its generic factor. For a field factor, integrality of $a + b\theta_i$ first gives $a \in \mathfrak{o}$ by its trace. Its norm then gives $2v_p(b) + v_p(\delta_i) \geq 0$, forcing $b \in \mathfrak{o}$ because $v_p(b)$ is integral. For a split unit factor use the two values $a \pm b\sqrt{\delta_i}$ and the unit $2\sqrt{\delta_i}$. This also proves maximality without an unspecified integral basis.

The precise normalization map is

$$f(T) + \eta g(T) \mapsto (f(\ell_i) + p^{k_i} g(\ell_i) \theta_i)_{i=0}^3, \quad \deg f, \deg g < 4. \quad (\text{IC17})$$

It is injective: over K , evaluate at all distinct labelled roots. All its inverse coefficients are provided by the corresponding Lagrange polynomials, so its integral image can be tested without changing the lattice.

Theorem 29.1 (Complete original integral defect). *For E,*

$$\mathcal{N}/\widehat{\mathcal{S}} \simeq (\mathfrak{o}/p)^2, \quad [\mathcal{N} : \widehat{\mathcal{S}}] = p^2, \quad \mathfrak{f}_E = pN_0 \times N_1 \times N_2 \times pN_3. \quad (\text{IC18})$$

For M ,

$$\begin{aligned}\mathcal{N}/\widehat{\mathcal{S}} &\simeq (\mathfrak{o}/p)^2 \oplus (\mathfrak{o}/p^2)^2 \oplus \mathfrak{o}/p^3, & [\mathcal{N} : \widehat{\mathcal{S}}] &= p^9, \\ \mathfrak{f}_M &= p^3 N_0 \times N_1 \times p^3 N_2 \times p^3 N_3.\end{aligned}\tag{IC19}$$

Here $\mathfrak{f} = \{n \in \mathcal{N} : n\mathcal{N} \subseteq \widehat{\mathcal{S}}\}$ is the conductor ideal in this finite algebra. It is not the analytic weighted conductor of the Zeta programme. In the eight-dimensional original basis, the Smith exponents are

$$E : (0, 0, 0, 0, 0, 0, 1, 1), \quad M : (0, 0, 0, 1, 1, 2, 2, 3).\tag{IC20}$$

Proof. Unit differences split off every noncolliding root integrally, by the polynomial CRT. For E , the remaining root lattice on the pair p, pc is exactly

$$L_2 = \{(z_0, z_3) \in \mathfrak{o}^2 : z_0 - z_3 \in p\mathfrak{o}\}.$$

A linear interpolation proves sufficiency, since $p - pc$ is p times a unit. Both coefficients f and g in (IC17) give a copy of L_2 : here $k_i = 0$. This proves the quotient and exponents. A normal element supported at just one of these two roots is in the original order precisely when its coefficients of 1 and θ_i both lie in $p\mathfrak{o}$. The resulting pN_i is an ideal and is maximal with this property, giving (IC18).

For M , order the cluster as p, pb, pc and use the Newton polynomial basis $1, T-p, (T-p)(T-pb)$. Evaluation gives diagonal factors $1, p, p^2$ times units, and integral row operations remove the lower entries. Thus its root lattice L_3 has quotient $\mathfrak{o}/p \oplus \mathfrak{o}/p^2$. The two coefficients in (IC17) give $L_3 \oplus pL_3$. Their exponents are $(0, 1, 2)$ and $(1, 2, 3)$; the singleton contributes two zeros. This proves (IC19)–(IC20).

For the conductor, a vector supported at one of the three roots has a unique Lagrange interpolation of degree at most two. Its leading coefficient has denominator of exact order p^2 . Hence a normal value $a + b\theta_i$ supported there belongs to $\widehat{\mathcal{S}}$ precisely when $a \in p^2\mathfrak{o}$ and $b \in p^3\mathfrak{o}$. For it to lie in the conductor, multiplication by the unit θ_i must also belong to the order. This forces $a, b \in p^3\mathfrak{o}$, and that condition is sufficient because $1, \theta_i$ is the full normal basis. The singleton is already maximal. This proves the displayed conductor. $\square$

The root orders alone have indices p (E) and p^3 (M), with Smith exponents $(0, 0, 0, 1)$ and $(0, 0, 1, 2)$. A further exact check is the trace discriminant:

$$v_p \text{Disc}(\widehat{\mathcal{S}}) = 2v_p \text{Disc}(\mathcal{O}) + \sum_i v_p(4d_i) = 6 \text{ } (E), \quad 18 \text{ } (M).$$

The normal algebra has discriminant valuation two (E) or zero (M). The square of the change-of-basis determinant again gives index exponents two and nine. Neither determinant is a density or positivity assertion.

Corollary 29.2 (The original actions on the finite defect module). *Let $Q = N/\widehat{\mathcal{S}}$, a module rather than a quotient ring. Multiplication by the original T and η descends to it. Their exact nilpotence indices are*

$$E : (1, 2), \quad M : (3, 3),\tag{IC20a}$$

respectively. Thus the finite quotient retains nonzero action, even though rational extension removes it.

Proof. The quotient is supported only at the colliding roots. For E , T is divisible by p there and $\eta^2 = d_i$ is also divisible by p , so the conductor kills both actions. Multiplying a supported normal constant by η still has a coefficient not divisible by p and is nonzero in Q . For M , both T^3 and η^3 have a factor p^3 at each colliding normal factor. But their squares, applied to a supported normal θ_i , have a θ_i -coefficient of exact valuation two. The membership criterion in the proof of Theorem 29.1 requires three, so both squared actions remain nonzero. All the normal root quotients are units, precluding cancellation. $\square$

Complete inverse coordinates. For a normal vector $a_i + b_i\theta_i$, interpolate the four values a_i to obtain $f(T)$, and the values b_i/p^{k_i} to obtain $g(T)$. Membership in the original order means all eight coefficients of f, g are in $\mathfrak{o}$. Thus (IC17) has an explicit partial integral inverse and a total K -linear inverse. The quotient in (IC18) or (IC19) is retained when the integral inverse fails. There is no $\mathfrak{o}$ -linear splitting of its surjection from $\mathcal{N}$, since that would map a nonzero torsion module into a torsion-free one.

30 Boundary fibres and a sharp finite lifting obstruction

At E, the colliding factor of the special fibre is

$$\mathbb{F}_p[\epsilon, \eta]/(\epsilon^2, \eta^2 - 2g_0\epsilon) \simeq \mathbb{F}_p[\eta]/(\eta^4), \quad g_0 = -ab/S \neq 0. \quad (\text{IC21})$$

At M it is

$$\boxed{\mathbb{F}_p[\epsilon, \eta]/(\epsilon^3, \eta^2 - 3\epsilon^2)}. \quad (\text{IC22})$$

These follow from the actual reduced quartics, respectively $-S^{-1}T^2(T-a)(T-b)$ and $T^3(1-T/a)$, using the supplied multiple-root algebra [11, equation (23)]. Their dimensions are four and six. Multiplication by ϵ has blocks (2, 2) or (3, 3). Multiplication by η has blocks (4) or (4, 2): on $1, \epsilon, \epsilon^2, \eta, \epsilon\eta, \epsilon^2\eta$ in (IC22), its successive ranks are 6, 4, 2, 1, 0. Thus (IC22) is not a single length-six chain.

The normalization map modulo p has kernel dimension two on the E collision block and five on the M collision block. For E, T becomes zero in each normal factor modulo p , while η gives its two labelled nonzero nilpotents. For M, both $T = p\zeta$ and $\eta = p\theta$ become zero in the normal special fibre. These are the kernels dictated by (IC20), not a deletion of the original labels.

Theorem 30.1 (The added E point stops at the third digit). *The point $(T, \eta) = (0, 0)$ of (IC21) has exactly p lifts solving both original equations modulo p^2 , and has no lift modulo p^3 .*

Proof. Write $\mathcal{H}(T) = U(T)(T-p)(T-pc)$, with $U(T) = -(T-a)(T-b)/S$ a unit near zero. A putative lift modulo p^2 is $T = pw, \eta = pv$, with $w, v \in \mathbb{F}_p$. The first equation is automatic. The second gives

$$0 \equiv pU(0)(2w - 1 - c) \pmod{p^2},$$

so $w = (1 + c)/2 = 5/8 \pmod{p}$. The p values of v are all possible. For any higher lift of that T ,

$$\frac{\mathcal{H}(T)}{p^2} \equiv -U(0)\frac{(1-c)^2}{4} = \frac{9ab}{64S} \not\equiv 0 \pmod{p}. \quad (\text{IC23})$$

Thus the root equation already fails modulo p^3 , independently of η . $\square$

The finite flat completion is proper, but this boundary point has no integral $\mathfrak{o}$ -section. The ramified generic points remain over their actual quadratic extensions. At $p = 1009$ in the first row of (IC15), the entire generic signed fibre has no K -point despite an original ES solution and a rational special boundary point. No smooth Hensel hypothesis is available at $\eta = 0$.

For M, the collision block instead has the exact integral normalization chart

$$G(Z) = (Z-1)(Z-b)(Z-c), \quad G(Z) = 0, \quad \theta^2 = \frac{a-pZ}{S}G'(Z), \quad (\text{IC24})$$

with $T = pZ, \eta = p\theta$. This is finite etale of rank six, since the reduced roots are distinct and the right side is a unit there. The inverse divides exactly by those powers of p . The character product in (IC14) proves that at least one cluster root has two K -rational signed lifts. Thus the M boundary point does lift to $\mathfrak{o}$, although its full multiplicity and the index p^9 are still retained.

The original physical inverse (IC10) has pole $v_p(\xi) = -1/2$ on the E cluster and $v_p(\xi) = -1$ on the M cluster. Inverting η in (IC21) or (IC22) annihilates that local algebra; it does not turn the added state into a finite original affine inverse. The supplied complex metric limit remains the one in [11, equations (21)–(22)]; no identification of that complex norm with this nonarchimedean valuation is asserted.

31 The generic prime-sign symmetry has an integral defect

The distinguished-root projector is

$$e_p(T) = \frac{\prod_{i=1}^3 (T - \ell_i)}{\prod_{i=1}^3 (p - \ell_i)}. \quad (\text{IC25})$$

Its values are $(1, 0, 0, 0)$. Its coefficients require an exact denominator p in E and p^2 in M , since its numerator is monic and the denominator has those respective valuations. This is also the supplied g_p/d_p projector [11, equations (9)–(11)].

Among the supplied four generic deck sign maps, the prime-only flip is

$$T \mapsto T, \quad \eta \mapsto (1 - 2e_p(T))\eta. \quad (\text{IC26})$$

Uniqueness in the original free basis (IC9) proves that it does not preserve $\hat{\mathcal{S}}$. Nor does its negative, the denominator-only flip. The identity and the simultaneous sign $\eta \mapsto -\eta$ do preserve it. All four extend to the normalization (IC16), where their action is an independent sign on the labelled quadratic factors. This conclusion concerns those four source deck maps, not all automorphisms of a specialized split algebra.

32 Literal channel interpolation and reciprocal inversion

Suppose actual E and M states are both supplied at the same p . Denote their ordered literal roots by x_i and y_i . The exact root-algebra maps are

$$F(T) = \sum_i y_i \frac{\prod_{j \neq i} (T - x_j)}{\prod_{j \neq i} (x_i - x_j)}, \quad G(T) = \sum_i x_i \frac{\prod_{j \neq i} (T - y_j)}{\prod_{j \neq i} (y_i - y_j)}. \quad (\text{IC27})$$

Evaluation proves both inverse compositions on the respective quotient rings. Under their common label coordinates in K^4 , the integral orders obey

$$\mathcal{O}_M \subset \mathcal{O}_E, \quad [\mathcal{O}_E : \mathcal{O}_M] = p^2. \quad (\text{IC28})$$

Indeed the three M cluster values are congruent modulo p , in particular at the E pair of labels $0, 3$. The already proved indices give the ratio. Thus F is integral. In the reverse interpolation, the summand at label 2 has unit output x_2 and a denominator of valuation two; the other cluster outputs are divisible by p . Consequently

$$\boxed{\min_j v_p(G_j) = -2, \quad p^2 G(T) \bmod p = c_0 T^2 (T - a_M), \quad c_0 \neq 0.} \quad (\text{IC29})$$

This proves the exact lost factor, not just an upper denominator estimate. At $p = 1009$, the pair of $(a, u) = (253, 11)$ states in (IC15) gives $c_0 = 306$ and $p^2 G \equiv 275T^2 + 306T^3 \pmod{1009}$.

The full signed generic algebras need not be isomorphic over K : E has ramified field factors and M has none. The surviving comparison is explicit after a specified extension. Adjoin square roots $\delta_i^2 = d_i^M/d_i^E$ and send η_M to $\delta(T)\eta_E$, where δ is the labelled interpolation of the δ_i . The inverse uses their reciprocals. The extension has degree at most four: for odd p , unit classes are square or a fixed nonsquare unit, and the only additional parity is that of p . The original integral images must still be tested by (IC17); they are not made unit lattices by this field isomorphism.

This is applicable to actual source-changing returns, for example the preceding capacity certificate [13, sharp finite repair] at $p = 825241$:

$$E : (a, R, u, h, r, s, \kappa) = (206321, 43, 2240439739, 19, 10859, 1, 208402140),$$

$$M : (a, R, u, h, r, s, \lambda) = (217170, 43439, 25, 1, 5, 43434, 1).$$

The latter is supplied there by the noncanonical original cofactor word 25, not by interpolation in (IC27). The current certificate transports both actual states through (IC18)–(IC29), preserving their different factor inventories. No M state is assumed to exist in order to use (IC27) as an existence proof.

32.1 The reciprocal model changes the lattice and the signed square class

Keep all four roots while applying $z_i = p/\ell_i$. On the literal root algebra,

$$\boxed{I(T) = -\frac{p}{D}(AT^3 + T^2 + BT + C) = p/T.} \quad (\text{IC30})$$

Its exact coefficient denominator is p in E and p^2 in M . The transformed monic polynomial is $Z^4 L(p/Z)/L(0)$, where $L = \prod(T - \ell_i)$. If $N = \prod_i \ell_i$ and the normalized reciprocal quartic is $\tilde{\mathcal{H}} = -\frac{1}{5} \prod(Z - z_i)$, differentiation gives

$$\boxed{\tilde{\mathcal{H}}'(z_i) = \omega \frac{\mathcal{H}'(\ell_i)}{\ell_i^2}, \quad \omega = -\frac{p^3 S}{5N}.} \quad (\text{IC31})$$

The four reciprocal roots sum to five; the coefficient $-1/5$ is therefore fixed, not fitted. The signed comparison needs the recorded quadratic twist $\sqrt{\omega}$. Its valuation is one in E and zero in M .

The reciprocal discriminant valuation is the literal value plus $12 - 6v_p(N)$, hence two in E and zero in M . The reciprocal root order has index p in E and is maximal in M . In E its colliding labels are now 1, 2, not 0, 3. Removing the separately marked root $z_0 = 1$ yields the preceding reciprocal cubic, whose signed derivatives acquire the further factors $-(z_i - 1)/5$ from the augmented quartic. This states the maps and square classes rather than conflating the earlier seven-point and current eight-point fibres.

33 A single missing prime passes any fixed finite integrality window

The supplied real separation estimates [11, §1.2, equations (2)–(4)] are necessary at an integral witness. The next result gives a precise scope test for treating them, together with a finite local completion, as sufficient. It retains both numerical channel gates and the entire allowed numerical range of the proposed integer u . The one failed condition is original square-divisor availability, at one explicitly designated prime.

Theorem 33.1 (A joint, single-prime source obstruction). *For every fixed finite set of rational primes Σ , there are infinitely many hard primes p and integers a, R, u such that*

$$p/4 < a < p/2, \quad R = 4a - p = 3, \quad 0 < u < a^2, \quad R \mid 4u + 1, \quad R \mid u + a, \quad (\text{IC32})$$

yet $u \nmid a^2$. There is just one extra prime occurrence in u , at a specified prime $l \notin \Sigma \cup \{p\}$. In particular

$$a^2/u \in \mathbb{Z}_q \quad (q \in \Sigma \cup \{p\}), \quad v_l(a^2/u) = -1.$$

The literal E and M rational denominator returns from these same numerical coordinates satisfy the ES identity and all the following conditions:

1. *their only nonintegral denominator has reduced denominator exactly l ;*
2. *at p they have the respective E and M local signatures, normalization indices, conductors and invariant factors proved above;*
3. *their four positive literal roots have successive gaps greater than one;*
4. *with $s_p = (p^2 + 3p)/4$ and $B_p = p + s_p(s_p + 1)$, both satisfy*

$$S < B_p, \quad \max(x, y, z) < s_p(s_p + 1)/3, \quad \text{Disc}(\mathcal{H}) \geq 144/B_p^6, \quad 2/S \leq |\mathcal{H}'(\ell_i)| \leq S^2;$$

5. the same prime p has separately supplied original integer E and M states. The failed source in (IC32) is not an ES counterexample.

Proof. Choose a prime $l \equiv 11 \pmod{12}$ outside Σ . Dirichlet's theorem [3, Theorem 18.1] permits this choice, and then supplies infinitely many primes in the reduced CRT progression

$$p \equiv 1 \pmod{840}, \quad p \equiv -1 \pmod{l}.$$

The moduli are coprime. Take $p > 4l$ and set

$$a = (p+3)/4, \quad R = 3, \quad u = al, \quad h = a/l, \quad r = l, \quad s = 1, \quad \kappa = (pl+1)/3, \quad \lambda = (l+1)/3. \quad (\text{IC33})$$

Both quotients are positive integers, and $l < a$ gives $0 < u < a^2$. Modulo three, $p \equiv a \equiv 1$ and $l \equiv 2$, so both gates hold. But $a \equiv 1/2 \pmod{l}$ and $\gcd(a, u) = a$. Thus $h = a/l$ is nonintegral with exact denominator l . Every prime of a occurs in u with its original exponent; the only extra coordinate is $v_l(u) = 1$, while $v_l(a) = 0$. In the centered original box this is exactly $\beta_l = 1$ outside $[0, 0]$. All the equations $a = hrs$, $u = hr^2$, and $\gcd(r, s) = 1$ hold, but the positive rational grade h is not an allowed original integer grade.

The two explicit returns are

$$E_{\text{rat}} : (a, a\kappa/l, pa\kappa), \quad M_{\text{rat}} : (a, pa\lambda/l, pa\lambda). \quad (\text{IC34})$$

Multiplication proves their reciprocal identities. Since p, a, κ, λ are all units at l , the middle displayed denominator has exact reduced denominator l and the other two are integers. Every declared finite place therefore passes the denominator test and the original divisor valuation test.

At the original prime, $p \nmid a, l, \kappa, \lambda$ and $(l/p) = (p/l) = (-1/l) = -1$. Also $a \equiv 3/4 \pmod{p}$ is a square. Thus the same unit and character calculations as Theorem 27.2 apply to (IC34): in the first return $4a\kappa \equiv 1$, while the second has normalized root product $a^2\lambda^2/l$ a nonsquare. The reduction patterns, local field alternatives and all integral normalization formulas follow exactly. This verifies local algebra for rational targets; it does not insert a missing integer factor into the original box.

For both triples write their largest denominator as $z = s_p q$, where $q = \kappa$ or λ and $s_p = pa$. Since $l < a$, $\kappa = (pl+1)/3 < (s_p+1)/3$, and the same bound holds for λ . Hence $z < s_p(s_p+1)/3$. Their other denominator is $z/(pl)$ in E and z/l in M . It is greater than $pa/3 > p+1$; also $p-a > 1$ and $z-y > 1$. Thus the ordered root list a, p, y, z has successive gaps greater than one, and $S = p+a+y+z < p+3z < B_p$. Four such roots have Vandermonde modulus at least twelve. The product of three differences at a selected root is at least two and at most S^3 . The claimed discriminant and derivative bounds follow from $\mathcal{H} = -S^{-1} \prod (T - \ell_i)$.

Finally retain the genuine endpoint data

$$R_E = l, \quad a_E = u_E = (p+l)/4, \quad (h_E, r_E, s_E, \kappa_E) = (a_E, 1, 1, (p+1)/l).$$

They are integral and return $(a_E, a_E\kappa_E, pa_E\kappa_E)$. There is also a genuine middle state: put $j = (l+1)/4$, $c = (p+1)/l$, then

$$(h_M, r_M, s_M, \lambda_M) = (j, 1, c, 1), \quad a_M = jc, \quad R_M = c+1, \quad u_M = j.$$

Its identity is $p = lc - 1 = 4jc - c - 1$. Since $l \geq 11$ and $c > 4$, both first-half inequalities hold. This gives the last assertion without assuming ES. $\square$

For $\Sigma = \{2, 3, 5, 7, 11\}$, choose $l = 23$ and the hard prime $p = 35281$. The numerical source is

$$a = 8821, \quad R = 3, \quad u = 202883, \quad (h, r, s, \lambda, \kappa) = (8821/23, 23, 1, 8, 270488).$$

The two nonintegral denominator triples are

$$E_{\text{rat}} = (8821, 2385974648/23, 84179571556088),$$

$$M_{\text{rat}} = (8821, 2489709608/23, 2489709608). \quad (\text{IC35})$$

The E target has zero rational local signed points and the M target has four. Their local normalization and ramification types are different. The only forbidden original factor is $23 \mid u$, while $23 \nmid a = 8821$. A second complete example at $p = 54601$ is included in the certificate. Trial division proves these displayed primes.

This theorem does not concern all places simultaneously: the omitted place is explicit. Nor does it exclude an adaptive test of every possible denominator prime. It proves that a fixed finite collection of local observations, even with the original numerical ranges, both channel congruences and the stated real bounds, is not original square-divisor membership. For the broader arithmetic-approximation context, compare [4, Theorems 1.1, 1.8–1.9]. The separate companion fragment `crt_control_note.tex` preserves an additional exact CRT construction with three larger-prime control targets; those targets and their full equations are also independently checked.

34 Verification scope, information loss, and continuation

The full replay covers every original first-half box for the 99 primes $p \equiv 1 \pmod{12}$ through 3000: 868359 square divisors and 4505 labelled states, of which 2451 are E and 2054 are oriented M. It compares all derivative valuations, characters, coefficient identities and local point alternatives. The deep lattice calculations expand every original normalization matrix at the four $p = 1009$ examples. The separate checker uses all minors for the invariant factors and independent Lagrange inversion, rather than the first program’s elimination. Both test the direct p^2/p^3 boundary equations, the actual capacity crossing, and every supplied rational control.

The main verifier is standard-library Python. The separate implementation imports none of it or of the preceding work. Its original source enumeration is by primitive coprime r, s and $h = a/(rs)$, rather than by divisors of a^2 . The prime domains, original vectors, ordered states and finite computation ranges are all recorded in JSON. Checks of a finite prefix are not a proof that the original source is nonempty at every prime.

No source coefficient is created by normalization. The exact sequence $0 \rightarrow \hat{\mathcal{S}} \rightarrow \mathcal{N} \rightarrow \mathcal{N}/\hat{\mathcal{S}} \rightarrow 0$ retains all the finite torsion coordinates in (IC18)–(IC20). Taking a quotient, a rational extension, or a reduction modulo p is accompanied by its calculated kernel. A receiving zero is not relabelled as absence of an original divisor word, and an added nilpotent boundary point is not promoted to an original E/M word. The countermodels in Theorem 33.1 make that latter distinction quantitative.

The remaining Turn 7 task is still to force a positive original integer coefficient for each prescribed hard prime. The completed same-grade capacity classification remains available, and its actual source-changing maps are compatible with the lattice comparison here. Finite rank, properness, local normalization and the displayed approximation bounds do not supply the initial arithmetic state by themselves. No Turn 8 witness guarantee, universal ES proof, Lean build or independent mathematical review is claimed.

Defining-prime normalization, gluing and local characters

The general collision calculation below includes the separated exterior and middle orders as exact specializations and then keeps every additional closer-pair valuation. Its translated chart is related to the original chart by a quadratic twist whose square class is retained.

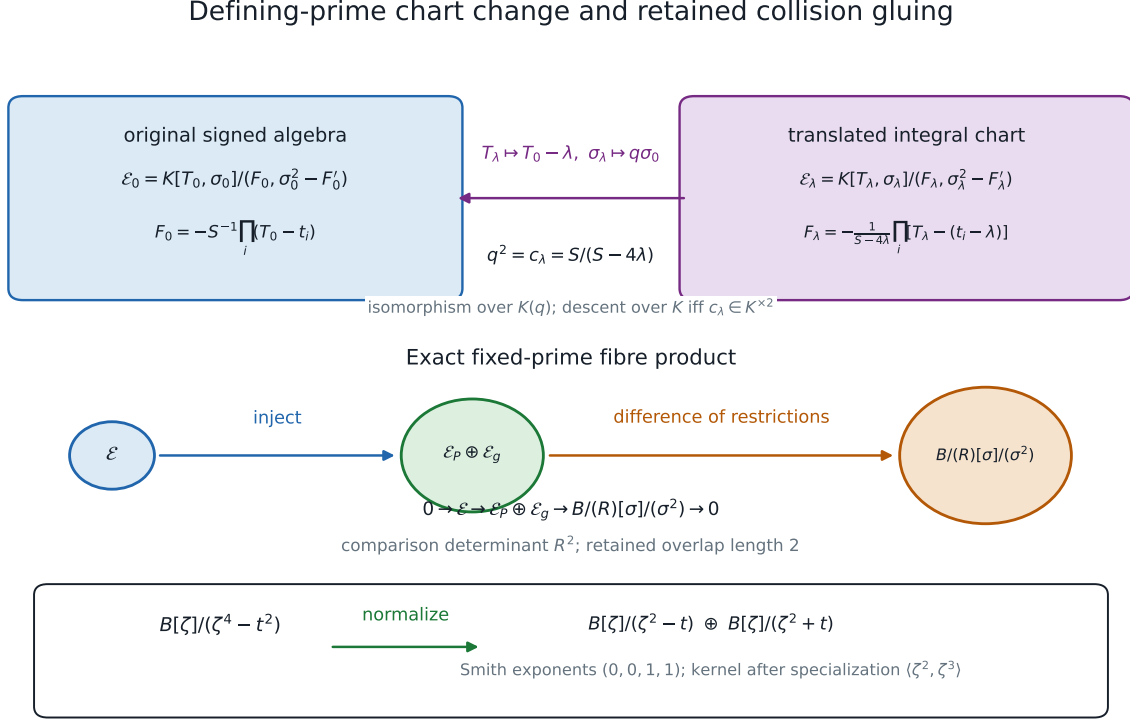


Figure 9: The exact translated-chart quadratic twist, rank-2 + 6 fibre product and collision normalization. The displayed base extension is part of the map and is not discarded during descent.

The ES bridge lets the eight-state completion be tested at the **defining prime of an actual Erdős–Straus witness**. That produces a new arithmetic distinction:

On the separated residue stratum:

one denominator divisible by p : $\tilde{\mathcal{E}}_p/\mathcal{E}_p \cong (\mathbb{Z}/p\mathbb{Z})^2$,

two denominators divisible by p : $\tilde{\mathcal{E}}_p/\mathcal{E}_p \cong (\mathbb{Z}/p\mathbb{Z})^2 \oplus (\mathbb{Z}/p^2\mathbb{Z})^2 \oplus \mathbb{Z}/p^3\mathbb{Z}$.

(1)

Here $\mathcal{E}_p$ is the **original signed quartic algebra**, and $\tilde{\mathcal{E}}_p$ is its integral normalization. Both retain rank eight. I also calculate the complete answer when additional residue collisions occur.

The second case is particularly informative: **the integral signed fibre is nonreduced, although its normalized field factors are all unramified or split**. Thus ramification of a field extension, failure of an integral observation to be invertible, and loss of geometric point labels are three different phenomena, with explicit maps between them.

There is a corresponding advance in the complex metric calculation. On the ES prime–denominator collision, the algebraic splitting has a divergent Gram determinant, but the complete minimum over its original evaluation kernel cancels that divergence exactly. The four unregularized signed evaluations have covariance eigenvalues of orders

$1, \quad |t|, \quad |t|^2, \quad |t|^3,$

(2)

with all four leading constants evaluated below in the fixed original Gamma-conductor metric.

I used the ES crosswalk at commit `2b5ab3b63471092624b4fcfac2ab2a0fbafa23a3`, the zeta workbench at `84d8f514c7df50b353a2aa1d039c485234c0a547`, and the complete preceding finite-completion manuscript. The ES source supplies the normalized prime-denominator quartic; the finite-completion source supplies the signed algebra. The normalization, integral invariant factors, ES-restricted gluing, and metric calculations below are the continuation.

34.1 1. Normalize the ES coefficient hypersurface without losing the prime at its boundary

For an original ES solution, retain

$$\frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}, \quad S = p + x + y + z,$$

and

$$F(T) = -\frac{1}{S}(T-p)(T-x)(T-y)(T-z) = AT^4 + T^3 + BT^2 + CT + D.$$

With

$$s_1 = x + y + z, \quad s_2 = xy + xz + yz, \quad s_3 = xyz,$$

the exact coefficients are

$$A = -S^{-1}, \quad B = A(ps_1 + s_2), \quad C = -5As_3, \quad D = Aps_3,$$

and

$$ps_2 = 4s_3, \quad p = -\frac{5D}{C}.$$

The ES workbench proves the coefficient equation

$$625AD^3 - 125CD^2 + 25BC^2D - 4C^4 = 0. \quad (3)$$

Its earlier inverse uses $C \neq 0$. At the defining prime of an integral witness, however, C and D both reduce to zero. The inverse therefore needs an actual extension rather than cancellation of the displayed denominator.

Let

$$\mathcal{S} = \mathbb{C}[A, A^{-1}, B, C, D] / (625AD^3 - 125CD^2 + 25BC^2D - 4C^4).$$

Use P for the retained distinguished-root parameter, so that its arithmetic value will be p . The normalization is

$$\boxed{\mathcal{N} = \mathbb{C}[A, A^{-1}, P, W],}$$

with the exact map

$$\boxed{B = AW - P - AP^2, \quad C = -\frac{5}{4}APW, \quad D = \frac{1}{4}AP^2W.} \quad (4)$$

The original symmetric data are recovered by

$$s_1 = -A^{-1} - P, \quad s_2 = W, \quad s_3 = PW/4.$$

Thus the original reciprocal relation remains $Ps_2 = 4s_3$.

To prove normalization, put

$$R_0 = \mathbb{C}[A, A^{-1}, B, C].$$

The parameter P satisfies the monic equation

$$P^3 + A^{-1}P^2 + (B/A)P + \frac{4C}{5A} = 0. \quad (5)$$

Both $\mathcal{S}$ and $\mathcal{N}$ are free of rank three over R_0 . In their respective bases

$$(1, D, D^2), \quad (1, P, P^2),$$

the inclusion is exactly

$$\boxed{\text{diag} \left(1, -\frac{C}{5}, \frac{C^2}{25} \right)}. \quad (6)$$

It is injective, finite, and becomes an isomorphism after inverting C . The ring $\mathcal{N}$ is a Laurent polynomial ring and hence normal. This proves the assertion.

More precisely,

$$\boxed{\mathcal{N}/\mathcal{S} \cong R_0/(C) \oplus R_0/(C^2) \quad \text{as an } R_0\text{-module},} \quad (7)$$

and its exact conductor ideals are

$$\boxed{\mathfrak{c}_{\mathcal{S}} = (C, D)^2, \quad \mathfrak{c}_{\mathcal{N}} = C^2\mathcal{N}.} \quad (8)$$

For the last statement, write an element of $\mathcal{S}$ as

$$a + CbP + C^2cP^2.$$

Requiring its products with P and P^2 to remain in $\mathcal{S}$, using (5), forces a to be divisible by C^2 and b by C . Conversely, every element of $C^2\mathcal{N}$ and every product of it with $\mathcal{N}$ belongs to $\mathcal{S}$.

This calculation identifies the exact information missing from the coefficient formula $P = -5D/C$. At $C = D = 0$, the distinguished parameter is retained in the normalization and satisfies

$$P(AP^2 + P + B) = 0.$$

It is not recovered by assigning a value to $0/0$.

The same formulas work over $\mathbb{Z}[1/10, A^{-1}]$, so they are available at every hard prime considered below.

34.2 2. Every hard-prime witness has four distinct complex roots, but collides at its own prime

The six hard classes in the ES workbench all satisfy $p \equiv 1 \pmod{12}$. The following argument applies to **every positive integral ES solution at such a prime**, rather than only the exterior states covered by the preceding crosswalk.

First, the denominators are distinct. If $y = z$, set $d = 4x - p > 0$. The ES equation gives

$$dy = 2px, \quad 2dy = p(d + p),$$

so $d \mid p^2$. But $d \equiv -p \equiv 3 \pmod{4}$, whereas every positive divisor of p^2 is $1 \pmod{4}$, a contradiction.

No denominator equals p . If $x = p$, then

$$(3y - p)(3z - p) = p^2.$$

Both factors are positive and $2 \pmod{3}$, whereas every positive divisor of p^2 is $1 \pmod{3}$. Therefore

$$\boxed{p, x, y, z \text{ are pairwise distinct.}} \quad (9)$$

Now reduce the original equation

$$4xyz = p(xy + xz + yz) \quad (10)$$

at p .

At least one denominator is divisible by p . All three cannot be divisible by p , since division by p would give

$$4 = \frac{1}{X} + \frac{1}{Y} + \frac{1}{Z} \leq 3.$$

Moreover, each denominator divisible by p has valuation exactly one.

If only z is divisible, the right side of (10) has valuation one, so

$$z = pZ, \quad p \nmid Z.$$

The divided equation gives

$$\boxed{Z \equiv \frac{1}{4} \pmod{p}.} \quad (11)$$

In particular, $\nu_p(p - z) = 1$.

If two denominators have valuations $a < b$, valuation comparison in (10) gives $b = 1$, contradicting $b > a \geq 1$. Their valuations must therefore agree. If both were at least two, then $y, z \geq p^2$, and

$$\frac{p}{4} < x \leq \frac{p^2}{4p - 2} < \frac{p + 1}{4},$$

which contains no integer. Hence

$$\boxed{\text{Exactly one or two denominators are divisible by } p, \text{ and every such valuation is } 1.} \quad (12)$$

Thus every actual hard-prime witness lies in the eight-point complex étale locus, while its defining-prime reduction necessarily has a repeated root. That repeated root is supplied by the arithmetic equation itself.

34.3 3. The complete integral normalization matrix

Assume first that $p \nmid S$, so the literal coefficient $A = -1/S$ is a p -adic unit. Write

$$t_0 = p, \quad t_1 = x, \quad t_2 = y, \quad t_3 = z, \quad f(T) = \prod_{i=0}^3 (T - t_i).$$

The integral signed algebra is

$$\boxed{\mathcal{E}_p = \mathbb{Z}_p[T, \sigma] / \left(f(T), \sigma^2 - Af'(T) \right).} \quad (13)$$

It is free of rank eight, with basis

$$1, T, T^2, T^3, \quad \sigma, \sigma T, \sigma T^2, \sigma T^3.$$

This is the A -invertible chart of the preceding finite completion, not the original affine factor chart with $a = 1/\sigma$. The completion's multiplication-trace convention is retained.

Define

$$n_{ij} = \nu_p(t_i - t_j), \quad b_i = \sum_{j \neq i} n_{ij},$$

$$m_i = \left\lfloor \frac{b_i}{2} \right\rfloor, \quad \epsilon_i = b_i - 2m_i \in \{0, 1\},$$

and retain the full unit

$$u_i = p^{-b_i} A f'(t_i) \in \mathbb{Z}_p^\times.$$

In the generic fibre,

$$\mathcal{E}_p \otimes \mathbb{Q}_p = \prod_{i=0}^3 \mathbb{Q}_p[\sigma_i]/(\sigma_i^2 - A f'(t_i)).$$

Its integral normalization is

$$\boxed{\tilde{\mathcal{E}}_p = \prod_{i=0}^3 \mathbb{Z}_p[\tau_i], \quad \tau_i = \sigma_i/p^{m_i}, \quad \tau_i^2 = p^{\epsilon_i} u_i.} \quad (14)$$

Each factor is interpreted as a quadratic field algebra or a split quadratic algebra, as appropriate.

For completeness, no integral-basis theorem is being used without its local check. If $x + y\tau_i$ is integral, its trace makes $2x$ integral, hence $x \in \mathbb{Z}_p$. Its square after subtracting x gives

$$2\nu_p(y) + \epsilon_i \geq 0,$$

so $y \in \mathbb{Z}_p$. Thus (14) is the full integral closure. Unit square roots and the decomposition into separated residue clusters use the ordinary Henselian lifting statements with $p \neq 2$ (The Stacks Project Authors, 2026, §10.153). (The Stacks Project)

Let

$$V = (t_i^j)_{0 \leq i, j \leq 3}.$$

In the ordered normalization basis consisting first of its four constants and then its four τ_i , the inclusion is exactly

$$\boxed{\mathbf{C}_p = \begin{pmatrix} V & 0 \\ 0 & \text{diag}(p^{m_i})V \end{pmatrix}.} \quad (15)$$

This evaluates the entire normalization map. The inverse over $\mathbb{Q}_p$ is the corresponding labelled Lagrange interpolation, with the displayed powers of p retained.

Consequently,

$$\boxed{\begin{aligned} \text{length}_{\mathbb{Z}_p}(\tilde{\mathcal{E}}_p/\mathcal{E}_p) &= 2 \sum_{i < j} n_{ij} + \sum_i m_i \\ &= 3 \sum_{i < j} n_{ij} - \frac{1}{2} \sum_i \epsilon_i. \end{aligned}} \quad (16)$$

The exact trace-discriminant check is

$$\nu_p(\det T_{\mathcal{E}_p}) = 6 \sum_{i < j} n_{ij}, \quad \nu_p(\det T_{\tilde{\mathcal{E}}_p}) = \sum_i \epsilon_i.$$

Their difference is twice (16), as required by the squared determinant of (15). The trace here is multiplication trace, with the discriminant convention of The Stacks Project, §49.3. (The Stacks Project)

34.3.1 One p -divisible denominator: every invariant factor

Let

$$z = pZ, \quad p \nmid xyZ, \quad n = \nu_p(x - y).$$

The pair p, z has difference valuation one by (11), and its differences from x, y are units. Thus the complete Smith exponents of (15) are the multiset

$$\boxed{\{0, 0, 1, 1\} \cup \left\{0, \left\lfloor \frac{n}{2} \right\rfloor, n, n + \left\lfloor \frac{n}{2} \right\rfloor\right\}}. \quad (17)$$

An exponent zero denotes a unit invariant factor.

To prove (17), separate the two residue clusters by their unit resultant. For a two-root cluster whose gap has valuation n , the ordinary evaluation matrix has Smith exponents $0, n$. Its two derivative valuations both equal n , so the odd evaluation block is multiplied by $p^{\lfloor n/2 \rfloor}$. This gives the second set in (17). The first set is the same calculation for $n = 1$.

In particular,

$$\boxed{\text{length}(\tilde{\mathcal{E}}_p/\mathcal{E}_p) = 2 + 3n - (n \bmod 2)}. \quad (18)$$

When $x \not\equiv y \pmod{p}$, this is the first case of (1).

34.3.2 Two p -divisible denominators: every invariant factor

Write

$$y = pY, \quad z = pZ, \quad p \nmid xYZ.$$

The exact reduced relation is

$$Y + Z \equiv 4YZ \pmod{p}. \quad (19)$$

The three scaled roots $1, Y, Z$ cannot all coincide. If two coincide, there is only one closer pair: $Y = Z$ forces $Y = Z = 1/2$, whereas $Y = 1$ forces $Z = 1/3$, and conversely.

Let

$$n = \max\{\nu_p(p - y), \nu_p(p - z), \nu_p(y - z)\} \geq 1, \quad m = \left\lfloor \frac{n+1}{2} \right\rfloor.$$

The other two differences within this three-root cluster have valuation one. The full Smith exponents are

$$\boxed{0, 0, 0, 1, 1, m+1, n+1, n+m+1}. \quad (20)$$

For the proof, the three-root Vandermonde has exponents

$$0, 1, n+1.$$

The normalization multipliers on its rows have exponents $1, m, m$. The least valuations of its 1×1 , 2×2 , and 3×3 minors are respectively

$$1, \quad m + 2, \quad n + 2m + 3.$$

Its odd-block Smith exponents are therefore

$$1, \quad m + 1, \quad n + m + 1.$$

The isolated unit root x contributes two further unit factors. This proves (20), including all complementary directions.

Its length is

$$\boxed{3n + 6 - ((n + 1) \bmod 2)}. \quad (21)$$

When $1, Y, Z$ are distinct modulo p , $n = 1$, giving the second case of (1).

In that case all three clustered derivative valuations are two. Thus all four normalized quadratic factors in (14) are unramified or split, even though the original integral fibre has a length-six nonreduced block.

34.3.3 The $p \mid S$ case has an integral translated chart, related by an exact quadratic twist

When $p \mid S$, the literal normalized coefficients are not all integral. Put

$$F_0(T) = -\frac{1}{S} \prod_i (T - t_i).$$

Take the recorded shift $\lambda = 1$ and define

$$F_\lambda(T) = -\frac{1}{S - 4\lambda} \prod_i [T - (t_i - \lambda)].$$

Since $p > 3$, $S - 4\lambda$ is a unit. The root differences are unchanged, so (15)–(21) apply in this integral chart.

Its return to the original coefficient chart is

$$\boxed{F_\lambda(T) = \frac{S}{S - 4\lambda} F_0(T + \lambda), \quad \sigma_\lambda^2 = \frac{S}{S - 4\lambda} \sigma_0^2.} \quad (22)$$

Write

$$c_\lambda = \frac{S}{S - 4\lambda}.$$

The translated signed algebra is the quadratic twist of the original signed algebra by the square class of c_λ . More exactly, after adjoining an element q with

$$q^2 = c_\lambda,$$

the assignment

$$\boxed{T_\lambda \mapsto T_0 - \lambda, \quad \sigma_\lambda \mapsto q\sigma_0}$$

is an isomorphism of signed algebras over $\mathbb{Q}_p(q)$: translation carries the root polynomial to $F_0(T_0)$, and

$$(q\sigma_0)^2 = c_\lambda F'_0(T_0) = F'_\lambda(T_0 - \lambda).$$

It descends to an isomorphism over $\mathbb{Q}_p$ exactly when the square class of c_λ is trivial. Therefore the Smith data obtained from this translated integral model are data for that explicit quadratic twist unless such a descent, together with its effect on the integral lattice, is proved. Both square-root choices and the translation are retained. No original-algebra conclusion below uses the translated chart without this qualification.

34.4 4. Actual hard-prime examples, including the boundary points

At the hard prime $p = 1201$, two exact positive solutions are

$$(x, y, z) = (306, 16218, 1082101) \quad (23)$$

and

$$(x, y, z) = (306, 21618, 61251). \quad (24)$$

The first is an original exterior state with

$$h = 17, \quad r = 1, \quad s = 18, \quad \kappa = 53,$$

$$R = 23, \quad u = 17, \quad D_E = 3, \quad 23 \cdot 53 = 1201 + 18.$$

Its denominator formulas are precisely

$$x = hrs, \quad y = hsk, \quad z = phrk.$$

These are the retained Turn 7 coordinates, not another Egyptian-fraction parametrization. The identities can be checked without numerical approximation:

$$\frac{1}{16218} + \frac{1}{1082101} = \frac{23}{306 \cdot 1201},$$

and

$$\frac{1}{18} + \frac{1}{51} = \frac{23}{306}.$$

For (23), the normalized coefficient vector reduces to

$$\boxed{(A, B, C, D) \equiv (410, 100, 0, 0) \pmod{1201}.} \quad (25)$$

The colliding prime and divisible-denominator labels give the local algebra

$$\boxed{\mathbb{F}_{1201}[r, \sigma]/(r^2, \sigma^2 - 200r) \cong \mathbb{F}_{1201}[\sigma]/(\sigma^4).} \quad (26)$$

The normalization inclusion has six unit Smith factors and two factors 1201. For (24),

$$\boxed{(A, B, C, D) \equiv (522, 0, 0, 0) \pmod{1201}.} \quad (27)$$

Here the prime and both divisible-denominator labels give

$$\boxed{\mathbb{F}_{1201}[r, \sigma]/(r^3, \sigma^2 - 3r^2),} \quad (28)$$

of length six. Its normalization inclusion has exponents

$$0, 0, 0, 1, 1, 2, 2, 3.$$

These are explicit critical boundary points of the finite completion: $r = \sigma = 0$, where its Jacobian $2\sigma^3$ vanishes. The original affine factor map still has Jacobian -2 , and its missing integral lifts have

$$a = 1/\sigma.$$

In the first example their p -adic valuation is $-1/2$; in the second it is -1 .

The nilpotent actions are also different. In (26), multiplication by σ has one block of length four, while multiplication by r has two blocks of length two. In (28), multiplication by σ has blocks of lengths four and two, whereas multiplication by r has two blocks of length three.

More generally, for $m \geq 2$, in a multiplicity- m local block

$$k[r, \sigma]/(r^m, \sigma^2 - r^{m-1}u(r)), \quad u(0) \neq 0,$$

the ranks of multiplication by $\sigma, \sigma^2, \sigma^3$ are

$$m+1, \quad 2, \quad 1.$$

Hence its complete Jordan list is

$$\boxed{4, \underbrace{2, \dots, 2}_{m-2 \text{ blocks}},} \quad (29)$$

while multiplication by r retains two length- m blocks.

The restriction $m \geq 2$ is necessary. When $m = 1$, the relation is $\sigma^2 = u(0) \in k^\times$, so multiplication by σ is invertible and is not nilpotent; the Jordan list in (29) does not apply.

34.4.1 The original exterior cofactor determines a local quadratic field

There is a further arithmetic identification. For an original exterior state, put $z = pZ$ and retain the exact exterior cofactor

$$\boxed{D_E = \frac{4u+1}{R} = \frac{r+\kappa}{s}.}$$

When $p \nmid S$,

$$\frac{xy}{x+y} = \frac{Z}{D_E}$$

holds exactly in the original h, r, s, κ coordinates. Together with $Z \equiv 1/4 \pmod{p}$, it gives

$$\boxed{\frac{F'(p)}{p} \equiv -\frac{3}{16D_E} \pmod{p}.} \quad (30)$$

For $p \equiv 1 \pmod{12}$, -3 is a square modulo p : if ζ is a primitive cube root in $\mathbb{F}_p$, then $(2\zeta + 1)^2 = -3$. Hensel lifting therefore proves

$$\boxed{\mathbb{Q}_p(\sqrt{F'(p)}) \cong \mathbb{Q}_p(\sqrt{pD_E}).} \quad (31)$$

The isomorphism multiplies by the actual lifted unit square root; it does not set that unit equal to one. The two factor signs remain distinct. (The Stacks Project)

Thus the finite signed cover records the quadratic class of the **original exterior cofactor**. For (23), $D_E = 3$, and the prime-marked factor is a ramified quadratic field. For the separated two-divisible case, the normalized factors instead have even derivative valuations and are unramified or split.

34.5 5. Restricting the cover to ES gives a $2 + 6$ decomposition with a calculable gluing defect

Over the smooth normalized ES parameter space, retain

$$g(T) = T^3 - s_1 T^2 + \frac{4s_3}{P} T - s_3, \quad F(T) = A(T - P)g(T),$$

and put

$$R = g(P) = P^3 - P^2 s_1 + 3s_3.$$

The signed algebra has two natural quotients:

$$\mathcal{E}_P = B[\sigma]/(\sigma^2 - AR),$$

of rank two, and

$$\mathcal{E}_g = B[T, \sigma]/(g(T), \sigma^2 - A(T - P)g'(T)),$$

of rank six.

Their complete relation is

$$\boxed{0 \longrightarrow \mathcal{E} \longrightarrow \mathcal{E}_P \oplus \mathcal{E}_g \longrightarrow B/(R)[\sigma]/(\sigma^2) \longrightarrow 0.} \quad (32)$$

The final arrow is the difference of the two restrictions. Equivalently, $\mathcal{E}$ is their fibre product over the displayed length-two overlap.

This follows from the exact ideals

$$(T - P) \cap (g) = ((T - P)g), \quad (T - P, g) = (T - P, R),$$

applied separately to the even and odd polynomial parts. Thus the prime-marked and denominator-marked sectors split after R is inverted, but they have a retained gluing module at a prime-denominator collision.

The ordinary root Chinese-remainder matrix is

$$\mathbf{C}_R = \begin{pmatrix} 1 & P & P^2 & P^3 \\ 1 & 0 & 0 & s_3 \\ 0 & 1 & 0 & -4s_3/P \\ 0 & 0 & 1 & s_1 \end{pmatrix}, \quad \det \mathbf{C}_R = -R. \quad (33)$$

The signed comparison consists of two such blocks, with determinant R^2 .

If $\Delta_g = \text{disc}(g)$, the two trace determinants are exactly

$$\boxed{\det T_{\mathcal{E}} = 256A^4 R^6 \Delta_g^3, \quad \det T_{\mathcal{E}_P \oplus \mathcal{E}_g} = 256A^4 R^2 \Delta_g^3.} \quad (34)$$

The factor R^4 is the square of the comparison determinant. On $\Delta_g \neq 0$, the direct sum is the normalization near $R = 0$.

34.5.1 The ES collision has a different extension from a transverse ambient collision

Suppose one denominator is $P + t$, with the other two roots remaining distinct from P . Locally,

$$F = Ax(x - t)h(P + x), \quad x = T - P, \quad h(P) \neq 0.$$

Retain an analytic square root of the nonzero unit and set

$$\zeta = \sigma / \sqrt{Ah(P + x)}.$$

Then

$$\zeta^2 = 2x - t,$$

and the complete local algebra becomes

$$\boxed{B[\zeta]/(\zeta^4 - t^2), \quad x = \frac{\zeta^2 + t}{2}.} \quad (35)$$

Its normalization is the sum of

$$\zeta^2 = t, \quad \zeta^2 = -t.$$

In the bases $(1, \zeta, \zeta^2, \zeta^3)$ and the two copies of $(1, \zeta)$, the comparison has determinant $4t^2$ and Smith exponents

$$0, 0, 1, 1.$$

Its cokernel is $(B/(t))^2$.

At $t = 0$, the original fibre is $\mathbb{C}[\zeta]/(\zeta^4)$. Normalizing first and then specializing gives two copies of $\mathbb{C}[\zeta]/(\zeta^2)$. The specialization map kills precisely

$$\text{span}\{\zeta^2, \zeta^3\}.$$

The associated Tor_1 term retains those two directions in the specialized exact sequence. The connection residue in the original four-dimensional extension is

$$\boxed{0, \quad \frac{1}{2}, \quad 1, \quad \frac{3}{2},} \quad (36)$$

whereas the normalized extension has

$$0, \quad 0, \quad \frac{1}{2}, \quad \frac{1}{2}.$$

Their trace difference is two, exactly

$$t \partial_t \log \det \mathbf{C} = 2.$$

The integer parts in (36) therefore have a measured lattice meaning; discarding them changes the extension.

The ES-restricted geometric monodromy is correspondingly

$$\boxed{\left\{ (\epsilon, \pi) : \pi \in S_3 \text{ fixes } P, \quad \prod_{\alpha=0}^3 \epsilon_\alpha = \text{sgn } \pi \right\}, \quad \text{order } 48.} \quad (37)$$

The intrinsic prime prevents arbitrary permutation of all four root labels. Denominator collisions generate S_3 ; a loop around $R = 0$ flips the signs at P and the colliding denominator without exchanging their labels. Those three pair flips generate the full eight-element even-sign subgroup. This proves (37), rather than simply restricting the number 192 by a dimension count.

34.6 6. Carry the gluing defect through the fixed original Gamma metric

The preceding ES bridge gives the fixed maps

$$\Phi_* = (V_*^\top)^{-1} K^{-1}, \quad \Psi_* = B_{v,*} (V_*^\top)^{-1} K^{-1},$$

with

$$T_{\mathcal{A},*} \Phi_* = \Psi_*.$$

The period, quartet, conductor moments, and reference matrices remain fixed while the ES coefficients vary.

Use two copies of this actual four-dimensional receiving map:

$$f_0(T) + \sigma f_1(T) \mapsto (\Psi_*[f_0], \Psi_*[f_1]), \quad \deg f_i \leq 3. \quad (38)$$

Every algebra multiplication is transported by its full matrix under (38). It is not identified with multiplication by S' in the conductor target.

Let $\mathcal{H}_\Gamma$ be the original monomial Gamma Gram on $\mathcal{P}_3$, and put

$$H_* = \Psi_*^* \mathcal{H}_\Gamma \Psi_*, \quad v(r) = (1, r, r^2, r^3)^\top.$$

Define its exact evaluation covariance

$$K_*(r, s) = v(r)^\top H_*^{-1} \overline{v(s)}. \quad (39)$$

This is a fixed four-dimensional expression, and it can be evaluated without an unnamed inverse. Put $b = s/2$, $M = (2\pi)^{s/2}$, and retain the original target centre c_* . Its four orthonormal polynomials are

$$\begin{aligned} \phi_0(S') &= M^{-1/2}, \\ \phi_1(S') &= \frac{-i(S' - c_*)}{\sqrt{Mb}}, \\ \phi_2(S') &= -\frac{(S' - c_*)^2 + b}{\sqrt{2Mb(b+1)}}, \\ \phi_3(S') &= \frac{i[(S' - c_*)^3 + (3b+2)(S' - c_*)]}{\sqrt{6Mb(b+1)(b+2)}}. \end{aligned}$$

If

$$q_j(r) = v(r)^\top \Psi_*^{-1}[\phi_j],$$

then

$$\boxed{K_*(r, s) = \sum_{j=0}^3 q_j(r) \overline{q_j(s)}}. \quad (40)$$

All source masses, centres, and conductor coefficients remain in these finite polynomials.

34.6.1 The complete minimum cancels the splitting pole

Observe the prime coefficient pair

$$f_0(P) = u, \quad f_1(P) = v.$$

The exact minimum in the metric (38) is

$$\boxed{\min \|f_0 + \sigma f_1\|^2 = \frac{|u|^2 + |v|^2}{K_*(P, P)}}. \quad (41)$$

Its minimum representatives are

$$f_0(r) = u \frac{K_*(r, P)}{K_*(P, P)}, \quad f_1(r) = v \frac{K_*(r, P)}{K_*(P, P)}.$$

The algebraic Chinese-remainder section instead uses

$$e_P(r) = g(r)/R.$$

Its exact excess is

$$\boxed{\left\| \Psi_* \frac{g}{R} \right\|_\Gamma^2 = \frac{1}{K_*(P, P)} + \left\| \Psi_* \left(\frac{g}{R} - \frac{K_*(\cdot, P)}{K_*(P, P)} \right) \right\|_\Gamma^2}. \quad (42)$$

The second term lies in the whole original evaluation kernel and is orthogonal to the minimum section.

For the two-column signed section, the determinant ratio is consequently

$$\boxed{\frac{\det G_{\text{CRT}}}{\det G_{\text{min}}} = \left[\frac{K_*(P, P) \|\Psi_* g\|_\Gamma^2}{|R|^2} \right]^2}. \quad (43)$$

Thus the algebraic section has a fourth-order Gram-determinant pole as $R \rightarrow 0$, while the attained coefficient quotient (41) remains positive and finite. The pole is carried by the explicitly displayed kernel correction, rather than removed by a normalization convention.

34.6.2 All four raw signed-evaluation eigenvalues

The raw point values are different observations:

$$f_0(P) \pm \sigma_P f_1(P), \quad f_0(P+t) \pm \sigma_{P+t} f_1(P+t).$$

A unitary sum/difference transformation splits their covariance into

$$2K_2, \quad 2D_\sigma K_2 D_\sigma^*,$$

where

$$K_2 = \begin{pmatrix} K_*(P, P) & K_*(P, P+t) \\ K_*(P+t, P) & K_*(P+t, P+t) \end{pmatrix}.$$

Set

$$K = K_*(P, P),$$

$$L = \sum_j |q'_j(P)|^2 - \frac{|\sum_j q'_j(P) \overline{q_j(P)}|^2}{K} > 0,$$

and

$$H_0 = |A_0 h(P)|.$$

The strict positivity of L follows from independence of evaluation and derivative on $\mathcal{P}_3$; explicitly,

$$KL = \sum_{i < j} |q_i(P)q'_j(P) - q_j(P)q'_i(P)|^2.$$

Since

$$\sigma_P^2 = -tA_0h(P)(1 + O(t)), \quad \sigma_{P+t}^2 = tA_0h(P)(1 + O(t)),$$

the four covariance eigenvalues are

$$\begin{aligned} \lambda_1 &= 4K(1 + o(1)), \\ \lambda_2 &= 4H_0K |t|(1 + o(1)), \\ \lambda_3 &= L |t|^2(1 + o(1)), \\ \lambda_4 &= H_0L |t|^3(1 + o(1)). \end{aligned} \tag{44}$$

The corresponding attained raw-value metric has their reciprocals. This calculation can be realized on an explicit ES coefficient family:

$$\left(P; P+t, 2P, \frac{2P(P+t)}{5P+7t} \right). \tag{45}$$

Its reciprocal sum is identically $4/P$. At $t = 0$,

$$H_0 = \left| \frac{3P}{22} \right|.$$

Thus even the local factor in (44) is explicit. The family is an algebraic ES family; it is not claimed to consist of integral witnesses for arbitrary t .

The exact passage to retained jets is

$$a_r = \frac{u_{r,+} + u_{r,-}}{2}, \quad b_r = \frac{u_{r,+} - u_{r,-}}{2\sigma_r},$$

followed by

$$\left(a_P, \frac{a_{P+t} - a_P}{t}, b_P, \frac{b_{P+t} - b_P}{t} \right). \tag{46}$$

Its determinant has modulus

$$\frac{1}{4|\sigma_P \sigma_{P+t} t^2|}.$$

The limiting jet covariance is two copies of the positive evaluation-derivative Gram. Its singular coordinate factors account exactly for the $|t|^6$ covariance determinant in (44).

This completes the metric comparison between the coalescing signed points, their retained jet algebra, and the specified original Gamma-conductor receiver.

34.7 7. How this returns to the zeta workbench

The common arithmetic object is the rational signed algebra

$$\mathcal{E}_{\mathbb{Q}} = \mathbb{Q}[T, \sigma]/(f, \sigma^2 - F').$$

It has two explicit base changes:

$$\mathcal{E}_{\mathbb{Q}} \otimes_{\mathbb{Q}} \mathbb{Q}_p,$$

with the integral normalization matrix (15), and

$$\mathcal{E}_{\mathbb{Q}} \otimes_{\mathbb{Q}} \mathbb{C} \xrightarrow{\Psi_*^{\oplus 2}} \mathcal{P}_3^{\oplus 2},$$

with the original Gamma metric and minimum calculations (41)–(46).

No p -adic valuation has been assigned to an unspecified transcendental period. The fixed complex conductor retains its nonzero moment μ_v , and its determinant remains the original one.

The newest zeta workbench also supplies a finite-prime theta transfer. At the same rational prime p , its original multiplier is

$$b_p(s) = 1 - p^{s-1}.$$

At a stipulated zeta root $\rho = 1/2 + \delta + i\gamma$,

$$\boxed{|b_p(\rho)| \geq 1 - p^{-1/2+\delta} > 0.} \quad (47)$$

For a full primary of order m , its jet matrix retains the coefficients

$$\frac{b_p^{(j)}(\rho)}{j!} = -\frac{p^{\rho-1}(\log p)^j}{j!}, \quad j \geq 1,$$

and determinant $b_p(\rho)^m \neq 0$. These are exactly the source's finite-prime multiplier and full-jet conventions.

Therefore the defining-prime degeneration calculated here does not manufacture a zero of the theta-transfer multiplier or of the weighted conductor. It instead supplies the missing arithmetic and metric accounting for the ES–Fable map: the complete integral cokernel, the retained collision algebra, the changed extension lattice, and the actual minimum that removes an algebraic-section pole.

The substantive new conclusions are equations (6)–(8), (15)–(21), (25)–(31), (32)–(37), and (41)–(46). They go beyond the previous eight-point count and generic discriminant calculation. In particular, they exhibit explicit singular integral points coming from an actual hard-prime witness and evaluate what survives after normalization and after the specified Gamma minimum.

These are written derivations. The repository checker `verify_defining_prime.py` now independently replays 80 exact identities covering the coefficient map, representative Smith strata, both $p = 1201$ witnesses, local nilpotent actions, the chart twist, collision comparison, explicit ES family, and local character example; its machine-readable receipt is `results/verification.json`. The checker is a certificate for those displayed calculations, not a replacement for the quantified arguments. The growing- k RH projected-current phase and the independent proper-source target minimum are not assigned values from this fixed finite receiver.

35 Exact crosswalk to the original E/M integral completion

35.1 Objects and bases

Let $p \equiv 1 \pmod{12}$ be prime and let

$$\frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$$

be a positive integral witness with $p \nmid S = p + x + y + z$. Retain the literal roots

$$t_0 = p, \quad t_1 = x, \quad t_2 = y, \quad t_3 = z, \quad f(T) = \prod_{i=0}^3 (T - t_i), \quad A = -S^{-1}.$$

The two packets use different letters for the same rank-eight order:

$$\mathcal{E}_p = \mathbb{Z}_p[T, \sigma] / (f, \sigma^2 - Af') = \mathbb{Z}_p[T, \eta] / (f, \eta^2 - Af').$$

The identification is the identity on T and sends $\sigma \mapsto \eta$. In both sources the ordered original basis is

$$(1, T, T^2, T^3, \sigma, \sigma T, \sigma T^2, \sigma T^3).$$

For

$$b_i = \sum_{j \neq i} \nu_p(t_i - t_j), \quad m_i = \left\lfloor \frac{b_i}{2} \right\rfloor, \quad \epsilon_i = b_i - 2m_i,$$

put

$$u_i = p^{-b_i} Af'(t_i), \quad \tau_i = \sigma_i / p^{m_i}, \quad \tau_i^2 = p^{\epsilon_i} u_i.$$

The integral normalization in both presentations is

$$\tilde{\mathcal{E}}_p = \prod_{i=0}^3 \mathbb{Z}_p[\tau_i].$$

The complete inclusion matrix, from the displayed original basis to the normalization basis consisting of the four constants followed by the four τ_i , is

$$C_p = \begin{pmatrix} V & 0 \\ 0 & \text{diag}(p^{m_i})V \end{pmatrix}, \quad V = (t_i^j)_{0 \leq i, j \leq 3}.$$

Thus the comparison below is an equality of integral maps, not a comparison of dimensions or a relabelling of two unrelated orders.

35.2 Exterior channel: the general formula gives index p^2

In the separated exterior channel, exactly one denominator is divisible by p . Relabel it as $z = pZ$. The prime equation gives $Z \equiv 1/4 \pmod{p}$, so

$$\nu_p(p - z) = 1.$$

The other two denominators x, y are units. In the original E state they are distinct modulo p , so

$$n = \nu_p(x - y) = 0.$$

The general two-cluster Smith formula is

$$\{0, 0, 1, 1\} \cup \left\{0, \left\lfloor \frac{n}{2} \right\rfloor, n, n + \left\lfloor \frac{n}{2} \right\rfloor\right\}.$$

Substitution of $n = 0$, with no deletion of unit factors, gives

$$\boxed{(0, 0, 0, 0, 0, 0, 1, 1)}.$$

Consequently

$$\text{length}_{\mathbb{Z}_p}(\tilde{\mathcal{E}}_p/\mathcal{E}_p) = 2, \quad [\tilde{\mathcal{E}}_p : \mathcal{E}_p] = p^2,$$

and the torsion module is exactly

$$\tilde{\mathcal{E}}_p/\mathcal{E}_p \cong (\mathbb{Z}_p/p)^2.$$

This is the exterior statement labelled (IC18)–(IC20) in the integral completion.

35.3 Middle channel: the general formula gives index p^9

In the separated middle channel, write

$$y = pY, \quad z = pZ, \quad Y + Z \equiv 4YZ \pmod{p}.$$

The three scaled residues $1, Y, Z$ are pairwise distinct in the original M state. Hence every difference inside the cluster $\{p, y, z\}$ has valuation one. In the notation of the general three-root formula,

$$n = 1, \quad m = \left\lfloor \frac{n+1}{2} \right\rfloor = 1.$$

Substitution in

$$0, 0, 0, 1, 1, m+1, n+1, n+m+1$$

gives

$$\boxed{(0, 0, 0, 1, 1, 2, 2, 3)}.$$

Therefore

$$\text{length}_{\mathbb{Z}_p}(\tilde{\mathcal{E}}_p/\mathcal{E}_p) = 9, \quad [\tilde{\mathcal{E}}_p : \mathcal{E}_p] = p^9,$$

and

$$\tilde{\mathcal{E}}_p/\mathcal{E}_p \cong (\mathbb{Z}_p/p)^2 \oplus (\mathbb{Z}_p/p^2)^2 \oplus \mathbb{Z}_p/p^3.$$

This is exactly the middle statement (IC19)–(IC20). The substantial index does not imply ramification: here all three clustered derivative valuations are two, so the normalized quadratic factors are unramified or split.

35.4 Special fibres and the retained maps

For the exterior pair, translate the colliding root to $r = 0$. The reduced local algebra has the exact form

$$k[r, \sigma]/(r^2, \sigma^2 - cr), \quad c \in k^\times.$$

The map

$$k[r, \sigma]/(r^2, \sigma^2 - cr) \longrightarrow k[\sigma]/(\sigma^4), \quad r \longmapsto c^{-1}\sigma^2, \quad \sigma \longmapsto \sigma$$

is an isomorphism. Its inverse sends the class of σ to σ ; the defining relations give $r = c^{-1}\sigma^2$ and $\sigma^4 = c^2r^2 = 0$. This is the integral completion's E fibre, with its original unit $c = 2g_0$ retained.

For the middle triple, the reduced local algebra is

$$k[r, \sigma]/(r^3, \sigma^2 - 3r^2),$$

which is literally the integral completion's M fibre after $\sigma \mapsto \eta$. Multiplication by r has two length-three blocks; multiplication by σ has blocks of lengths four and two. These block statements describe different endomorphisms of the same six-dimensional algebra and are not substitutions for the Smith lattice calculation.

35.5 Scope

The crosswalk proves that the p^2 exterior defect and p^9 middle defect are the separated $n = 0$ and $n = 1$ instances of the full defining-prime Smith classification. It does not turn a nonreduced special fibre into an Erdős–Straus counterexample, and it does not identify an integral conductor ideal with the fixed complex weighted conductor.

36 The exact local signed-label representation

36.1 Normalized factors and their characters

Let $p > 2$, let $K = \mathbb{Q}_p$, and retain the integral signed order

$$\mathcal{E}_p = \mathbb{Z}_p[T, \sigma]/(f(T), \sigma^2 - Af'(T))$$

from the defining-prime calculation. For the distinct labelled roots $t_0, \dots, t_3$, put

$$b_i = \sum_{j \neq i} \nu_p(t_i - t_j), \quad m_i = \left\lfloor \frac{b_i}{2} \right\rfloor, \quad \epsilon_i = b_i - 2m_i \in \{0, 1\},$$

$$u_i = p^{-b_i} Af'(t_i) \in \mathbb{Z}_p^\times, \quad \tau_i = \sigma_i/p^{m_i}, \quad \tau_i^2 = p^{\epsilon_i} u_i.$$

Fix a coefficient field of characteristic zero. Let χ_i be the quadratic character of $G_K = \text{Gal}(\overline{K}/K)$ attached to the quadratic K -algebra $K[\tau_i]/(\tau_i^2 - p^{\epsilon_i} u_i)$; it is trivial when that algebra splits. The permutation representation on the two geometric signed labels of this factor is

$$W_i \cong \mathbf{1} \oplus \chi_i.$$

This is an equality of G_K -representations. It retains a split factor as two fixed labels and a field factor as the two embeddings exchanged by its quadratic character.

36.2 The cluster quotient

Let $\mathcal{C}$ be a residue cluster of $m \geq 2$ root labels. Its nonreduced special fibre has one reduced geometric point. The common reduced point supplies the diagonal invariant line

$$\mathbf{1}_\Delta \longrightarrow \bigoplus_{i \in \mathcal{C}} W_i, \quad 1 \longmapsto ((1, 1))_{i \in \mathcal{C}}.$$

Define the signed-label defect representation by the exact quotient

$$0 \longrightarrow \mathbf{1}_\Delta \longrightarrow \bigoplus_{i \in \mathcal{C}} (\mathbf{1} \oplus \chi_i) \longrightarrow V_{\mathcal{C}} \longrightarrow 0.$$

Representations of the finite elementary 2-group generated by the χ_i are semisimple in characteristic zero. Removing the displayed diagonal trivial line therefore gives the exact decomposition

$$\boxed{V_{\mathcal{C}} \cong \mathbf{1}^{m-1} \oplus \bigoplus_{i \in \mathcal{C}} \chi_i.} \quad (\text{LG1})$$

In particular, $\dim V_{\mathcal{C}} = 2m - 1$. This construction records the loss from $2m$ generic signed labels to one reduced special-fibre label. It does not identify a nonreduced scheme with its normalization.

Every quadratic extension of $\mathbb{Q}_p$ for odd p is tame. Moreover,

- if $\epsilon_i = 0$, the factor is split or unramified and $a(\chi_i) = 0$;
- if $\epsilon_i = 1$, the factor is ramified quadratic and $a(\chi_i) = 1$.

Thus the Swan conductor is zero and

$$\boxed{a(V_{\mathcal{C}}) = \sum_{i \in \mathcal{C}} \epsilon_i.} \quad (\text{LG2})$$

The same conductor is obtained before quotienting by $\mathbf{1}_\Delta$, because a trivial representation has conductor zero.

36.3 Equality with the integral defect formula

For all four labels, let $n_{ij} = \nu_p(t_i - t_j)$. The normalization calculation proves

$$L := \text{length}_{\mathbb{Z}_p}(\tilde{\mathcal{E}}_p/\mathcal{E}_p) = 3 \sum_{i < j} n_{ij} - \frac{1}{2} \sum_i \epsilon_i.$$

Let $W = \bigoplus_i (\mathbf{1} \oplus \chi_i)$. Since $a(W) = \sum_i \epsilon_i$, multiplication by two and rearrangement give the exact identity

$$\boxed{2L + a(W) = 6 \sum_{i < j} \nu_p(t_i - t_j).} \quad (\text{LG3})$$

This relates two different invariants of the same normalized factors: the left term L measures the integral lattice defect, while $a(W)$ measures ramification of the generic signed-label representation. Neither determines the other.

36.4 Exterior pair

For a separated one-divisible-denominator state, write $z = pZ$. The collision cluster is $\{p, z\}$, with $\nu_p(p - z) = 1$. The normalized derivative units at the two labels have ratio

$$\frac{p^{-1}Af'(p)}{p^{-1}Af'(z)} = -\frac{(p-x)(p-y)}{(z-x)(z-y)} \equiv -1 \pmod{p}.$$

Because $p \equiv 1 \pmod{4}$, -1 is a square in $\mathbb{F}_p^\times$, so the two ramified quadratic characters are equal. The exterior-cofactor calculation identifies their common field as

$$\mathbb{Q}_p(\sqrt{pD_E}), \quad D_E = \frac{4u+1}{R} = \frac{r+\kappa}{s}.$$

Hence

$$\boxed{V_{\{p,z\}} \cong \mathbf{1} \oplus \chi_{pD_E}^{\oplus 2}, \quad a(V_{\{p,z\}}) = 2, \quad \dim V_{\{p,z\}}^{I_p} = 1.} \quad (\text{LG4})$$

The invariant line is the remaining trivial summand; each ramified character has no inertia invariants.

36.5 Middle triple

For a separated two-divisible-denominator state, write

$$y = pY, \quad z = pZ, \quad Y + Z \equiv 4YZ \pmod{p},$$

with $1, Y, Z$ pairwise distinct. Each difference in the cluster $\{p, y, z\}$ has valuation one, so every clustered derivative valuation is two. All three χ_i are therefore unramified or trivial.

For $\{\alpha, \beta, \gamma\} = \{1, Y, Z\}$, the normalized derivative unit is

$$u_\alpha \equiv (\alpha - \beta)(\alpha - \gamma) \pmod{p}. \quad (\text{LG5})$$

Indeed the omitted common factor is $-Ax = x/S \equiv 1 \pmod{p}$, so LG5 retains the complete square class. The product of the three units is

$$-[(1-Y)(1-Z)(Y-Z)]^2.$$

At $p \equiv 1 \pmod{4}$, this is a square. Consequently either all three unramified characters are trivial, or exactly two are the nontrivial unramified quadratic character. Since LG1 contributes

two further trivial summands, the Frobenius characteristic polynomial on the rank-five defect representation is exactly one of

$$\boxed{(1 - T)^5, \quad (1 - T)^3(1 + T)^2.} \quad (\text{LG6})$$

36.6 The $p = 1201$ middle witness

For

$$(p; x, y, z) = (1201; 306, 21618, 61251)$$

one has $Y = 18$ and $Z = 51$. Formula LG5 gives the three retained units

$$850, \quad 640, \quad 449 \pmod{1201}.$$

Euler's criterion gives their characters

$$-1, \quad +1, \quad -1.$$

The isolated root $x = 306$ has derivative unit 42, which is a square modulo 1201. Thus the normalized signed fibre has four $\mathbb{F}_{1201}$ -points: two from the isolated split factor and two from the one split clustered factor. The original special fibre has those two isolated points plus its one reduced cluster point, hence three. The rank-five defect representation has

$$\boxed{\det(1 - T \text{Frob} \mid V) = (1 - T)^3(1 + T)^2}$$

and Frobenius trace $3 - 2 = 1$, exactly the point-count difference $4 - 3$.

The integral defect remains independent data:

$$\text{length}(\tilde{\mathcal{E}}_{1201}/\mathcal{E}_{1201}) = 9, \quad \text{Smith}(\mathbb{C}_{1201}) = (0, 0, 0, 1, 1, 2, 2, 3).$$

This example proves that an unramified normalized representation can coexist with a large integral gluing defect. It does not identify local Frobenius with any complex zeta operator.

37 A root-algebra bridge and its exact projective obstruction

After choosing a bijection $t_j \leftrightarrow \rho_{*,j}$ between the four distinct ES roots and the four fixed reference roots, let $\ell_j^\rho(S)$ be the Lagrange polynomial satisfying $\ell_j^\rho(\rho_{*,k}) = \delta_{jk}$. Then

$$\Theta : \frac{\mathbb{C}[T]}{\prod_j (T - t_j)} \longrightarrow \frac{\mathbb{C}[S]}{\prod_j (S - \rho_{*,j})}, \quad [f] \longmapsto \sum_{j=1}^4 f(t_j) [\ell_j^\rho(S)] \quad (\text{EZ36})$$

is a unital algebra isomorphism. Its inverse evaluates at the reference roots and interpolates at the ES roots; hence both kernels are zero. It depends on the chosen bijection. It is generally not induced by one fractional-linear map. Such a map exists exactly when the corresponding ordered cross ratios agree, up to the six values produced by relabelling. This is the exact projective obstruction; failure of that stronger map does not erase EZ31–36.

38 Scope of the quartic-to-conductor core

The new composite is

$$\begin{array}{c}
\text{actual ordered ES exterior state} \\
\downarrow \text{EZ18-23} \\
(p; x, y, z) \\
\downarrow \text{normalized product EZ6} \\
J_4 \in \mathcal{B}, \quad p = -5u_4/u_3, \quad G(J_4) = 0 \\
\downarrow P^{-1} \\
\{q_{p,\pm}, q_{x,\pm}, q_{y,\pm}, q_{z,\pm}\} \\
\downarrow (\Phi_*, \Psi_*, T_{A,*}) \\
\text{eight states in the fixed weighted-conductor square.}
\end{array}$$

The map begins with an actual ES witness and therefore is not a proof that a witness exists for every p . The hypersurface $G = 0$ is not by itself an arithmetic occupancy theorem. The fixed quartet and fixed conductor data do not vary with p, x, y, z , and the construction proves no zero of the Riemann zeta function. The signed-root monodromy group of order 192 belongs to the generic eight-sheet cover over $\mathcal{B}$. After restricting to a fixed intrinsic prime and to $\mathcal{B}_p$, Theorem 23.2 gives the different order-48 group and its rank-2 + 6 sheet decomposition. Neither group is attributed to the seven-state boundary suspension. The finite completion EZ67 adds nonreduced collision fibres but leaves the original meromorphic inverse pole visible; its local paired-root map does not identify the global ES and zeta constructions.

Illustrated guide to the new five-track continuation

The following three figures display the proved maps without replacing their coordinate formulas. Tracks I–III refine the ES/Fable interfaces. Tracks IV–V are RH-facing conditioning and certification interfaces and are not claims of progress toward ES occupancy.

Positive-real Erdős–Straus solutions cross the extra odd-frame divisor

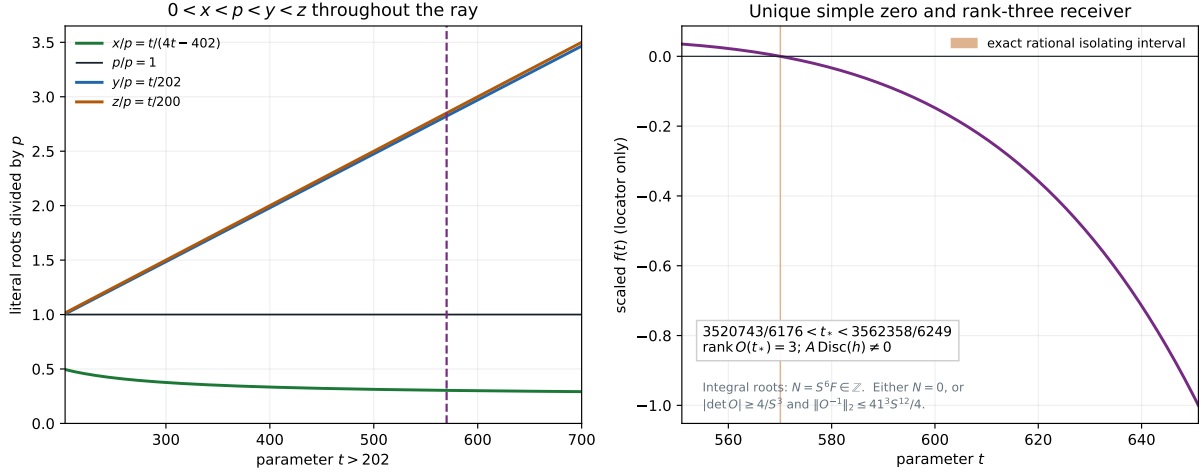


Figure 10: The exact positive-real solution ray and its unique rank-three odd receiver. The plotted polynomial is a locator; the rational isolating interval, Sturm count, nonzero minor and integral dichotomy are proved in Track I, Sections 3–4.

Trace degeneration is multiplication by $2\eta^3$

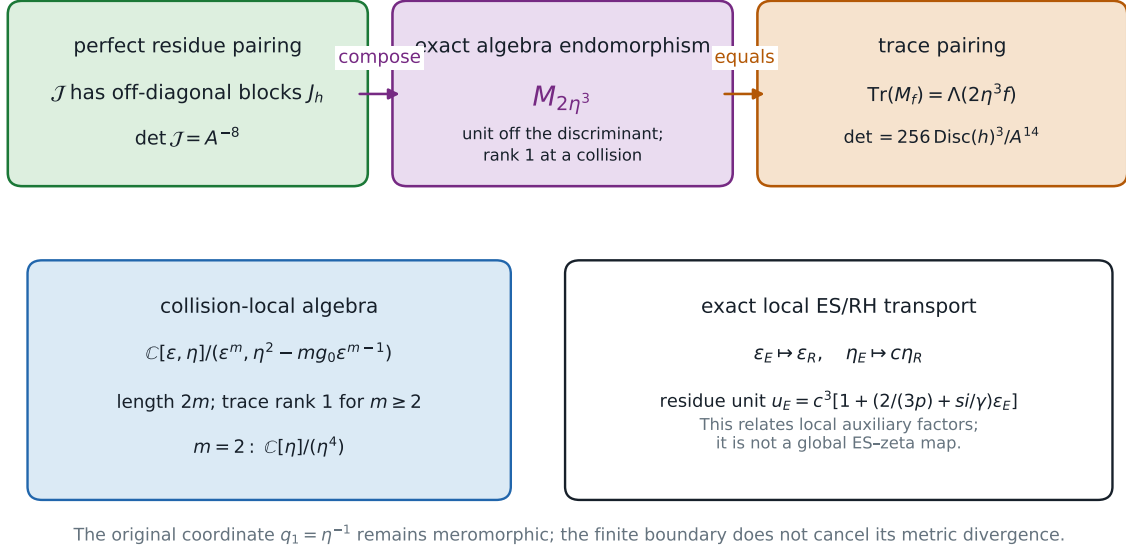


Figure 11: The perfect residue pairing reaches the trace pairing through the specific endomorphism $M_{2\eta^3}$. Track II, Sections 3–5 prove the determinants, collision ranks, local algebra and nonconstant residue unit.

Residue duality reaches a native moment Gram through a typed multiplier

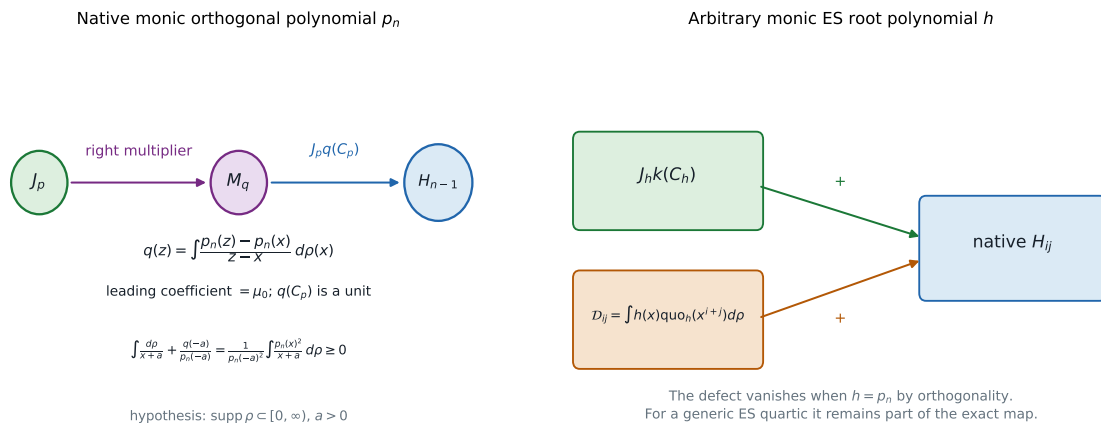


Figure 12: The native orthogonal-polynomial multiplier and the stronger arbitrary-polynomial identity retaining its relation defect. Track III proves the multiplier, support hypothesis, resolvent error and defect term.

39 Combined ES–Fable–RH research continuation

20 September 2026. Five connected derivation tracks.

The source ledger at the end identifies the pinned snapshots and reading scope. The ZIP contains the executable exact checks and complete receipts.

Verification: 541 exact checks and eleven deliberately false-inference controls per interpreter mode; normal and optimized outputs are byte-identical. This is not Lean verification and does not evaluate unknown native arithmetic moments.

40 Track I — The odd receiving divisor on the positive-real ES family

Status. New derivations relative to the pinned ES/RH sources; the generic odd-frame factorization and the earlier singular example are independently rechecked. This note proves neither universal ES occupancy nor nonvanishing of the odd determinant at every integral ES witness. Sources and their exact versions are in `SOURCE_LEDGER.md`.

40.1 1. Original source and receiving objects

For an ES solution, retain

$$4/p = 1/x + 1/y + 1/z, \quad S = p + x + y + z, \\ h(r) = -S^{-1}(r - p)(r - x)(r - y)(r - z) = Ar^4 + r^3 + Br^2 + Cr + D.$$

The ES foundation’s EZ6–EZ10 gives

$$A = -S^{-1}, \quad B = -[p(x + y + z) + xy + xz + yz]/S, \quad C = 5xyz/S, \quad D = -pxyz/S,$$

and recovers $p = -5D/C$. The coefficient equation is $625AD^3 - 125CD^2 + 25BC^2D - 4C^4 = 0$. That equation alone has no integrality or positivity condition.

For distinct roots r_j , put $d_j = h'(r_j)$ and choose both signs of $\xi_j^2 d_j = 1$. The original sign-odd columns are

$$o_j = \xi_j(1, -id_j, Ad_j^2 + 2r_j d_j, i(7r_j^2 d_j - 13d_j^2))^T.$$

Let $O = [o_1 \ o_2 \ o_3 \ o_4]$, and let V_h be the coefficient matrix of the remainders of the four polynomials in this formula, in the unmodified basis $1, r, r^2, r^3$. If

$$U = (r_j^n)_{0 \leq n \leq 3, 1 \leq j \leq 4} \text{diag}(\xi_j),$$

then $O = V_h U$. The product of all derivative factors gives

$$\det U = \chi/A^2, \quad \chi \in \{1, -1\}.$$

The sign changes with the chosen ordering and square-root branches; the following divisor does not.

40.2 2. The complete extra determinant factor

Define

$$\begin{aligned} F(A, B, C, D) = & 20(3C - B^2) + A(87B^3 - 282BC - 51D) \\ & + A^2(-28B^4 + 22B^2C + 431C^2 + 204BD) \\ & + A^3(224B^2D - 168BC^2 - 856CD) - 448A^4D^2. \end{aligned}$$

Polynomial division gives

$$\det V_h = 4F/A, \quad \boxed{\det O = 4\chi F/A^3.}$$

This is checked over the rational function field, not fitted from samples. For an independently inspectable determinant, after the constant first row and the cancelling factors $(-i)i = 1$, the remaining coefficient matrix is

$$\begin{pmatrix} 2B & 3 & 4A \\ 4ABC - 8AD - 7C & 4AB^2 - 2AC - 16A^2D - 5B & 4AB - 8A^2C - 3 \\ 20C/A + 76D - 52BC & 20B/A + 5C - 52B^2 + 208AD & 20/A - 66B + 104AC \end{pmatrix}.$$

Its determinant is exactly $4F/A$. In particular this is not the discriminant of h , the Jacobian of the nonlinear Fable map, or the determinant of the original linear conductor.

40.3 3. A positive-real ES family crossing this divisor

For every real $p > 0$, define

$$(p; x, y, z) = p \left(1, \frac{t}{4t - 402}, \frac{t}{202}, \frac{t}{200} \right), \quad t > 202.$$

The reciprocal identity follows by adding $(4t - 402)/(pt)$, $202/(pt)$, and $200/(pt)$. The strict ordering is

$$0 < x < p < y < z.$$

Thus the four literal roots are distinct throughout this interval. The normalization retains $A = -1/S$.

At $p = 1$, direct substitution in the preceding polynomial gives

$$F(t) = -\frac{5t^2 f(t)}{(402t^2 + 10099t - 4060200)^6},$$

where the exact integer polynomial is

$$\begin{aligned} f(t) = & 2359424t^{12} \\ & -2380972032t^{11} \\ & -125022004322048t^{10} \\ & +199766203719977640t^9 \\ & -126442260595802541476t^8 \\ & +41598937953326627478150t^7 \\ & -7385542427424681354996253t^6 \\ & +544226493322473956433710982t^5 \\ & +40187872427398803311778426564t^4 \\ & -12845040782988159530278491608280t^3 \\ & +1219651834158063698018605839792000t^2 \\ & -55076685665576109738730352971200000t \\ & +998076020859413480649547526400000000. \end{aligned}$$

The coefficients scale with p by $(A, B, C, D) \mapsto (p^{-1}A, pB, p^2C, p^3D)$. Every term of F has weight two, so the result for any $p > 0$ is $p^2F(t)$.

The following are exact, reproducible polynomial facts, proved by the Euclidean/Sturm procedures implemented in `check_es_slice.py`:

$$\gcd(f, f') = 1, \quad \#\{t \in (551, 651) : f(t) = 0\} = 1,$$

and its unique root t_* lies in

$$\boxed{\frac{3520743}{6176} < t_* < \frac{3562358}{6249}}.$$

The decimal $t_* \approx 570.0684909$ is only a locator. No floating-point value is used to establish the root or the rank. The determinant scalar is negative at 551 and positive at 651. There is no denominator zero on that interval. A displayed 3-by-3 coefficient minor, stored exactly in `results/es_frame_divisor.json`, has numerator coprime to f , so it is nonzero at t_* . Consequently

$$\boxed{\text{rank } O(t_*) = 3}$$

while $A \text{Disc}(h) \neq 0$. The nonlinear inverse still has eight separate simple branches. The receiving frame alone has lost one direction.

Under the literal scaling by p , the odd matrix satisfies

$$O_p = \text{diag}(p^{-1}, p, p^2, p^3)O_1$$

for coherent sign choices. This proves rank preservation for every fixed positive p , without declaring the diagonal map an isometry. The polynomial has no rational root. This is a positive-real ES solution family, including a fixed arbitrary prime value of p , but its singular point is not an integral ES witness. No implication about occupancy or frame degeneracy at integral denominators follows from this example. It proves that positivity and the exact reciprocal equation, by themselves, do not remove the extra receiving divisor.

40.4 4. Integer separation when the extra factor is nonzero

Suppose the four literal roots are distinct positive integers. Their elementary symmetric sums $e_1 = S, e_2, e_3, e_4$ are integers, and

$$(A, B, C, D) = (-S^{-1}, -e_2/S, e_3/S, -e_4/S).$$

Inspection of every denominator in F shows

$$N = S^6 F \in \mathbb{Z}.$$

Therefore, **conditional on** $N \neq 0$,

$$\boxed{|\det O| = 4|N|/S^3 \geq 4/S^3}.$$

For a selected root r_j , the three differences from the remaining roots are distinct nonzero integers. The product of their absolute values is at least two and at most S^3 , so

$$2/S \leq |d_j| \leq S^2, \quad S^{-1} \leq |\xi_j| \leq \sqrt{S/2}.$$

The four entries of o_j consequently have absolute bounds

$$\sqrt{S/2}, \quad S, \quad 3S^2, \quad 20S^3.$$

Here $|\xi_j d_j| = |d_j|^{1/2}$ and $|\xi_j d_j^2| = |d_j|^{3/2}$; no branch phase has been replaced in the exact formulas. Summing the sixteen squared bounds gives $\|O\|_2 \leq 41S^3$ for $S \geq 1$. The exterior identity then proves

$$\|O^{-1}\|_2 \leq \frac{41^3}{4} S^{12} \quad (N \neq 0).$$

For the actual receiving Hermitian form $G > 0$, the map from $(\mathbb{C}^4, G)$ to Euclidean label coordinates has the additional factor $\|G^{-1/2}\|_2$. A different original source metric contributes its own $\|R^{1/2}\|_2$. No coefficient-frame norm is silently identified with the physical norm.

This estimate separates two tasks: arithmetic nonvanishing of the integer N , which is not proved here, and quantitative inverse control once that nonvanishing is known.

40.5 5. Exact return to the original conductor

Let L be the original invertible finite conductor on the appropriate receiving four-plane, with the quotient and target norms in WCF4. Then

$$\ker(LO) = \ker O, \quad \det(LO) = \det L \det O.$$

Thus neither the extra frame divisor nor the missing direction is removed by the original conductor. Its inverse-exterior bound controls the L^{-1} factor where an inverse exists. In the actual-observation construction, replace the older cubic receiver by the source's $F_B = \Lambda_k J_+ K^{-1}$, retaining $\Lambda_k F_+ = F_B$ and $\bar{C}F_B = F_-$. These are maps on the indicated images, not identifications of ES roots with zeta zeros. The full original arithmetic Gram and both residual minimum energies remain those of MO5–MO10.

The next arithmetic question is whether N can vanish at an admissible integral hard-prime witness, or whether an exact congruence/positivity argument excludes it. A real-parameter frame singularity is not an answer to that arithmetic question.

41 Track II — Residue duality, the exact non-unit element, and boundary transport

Status. The finite rank-eight extension was constructed in the preceding continuation. This note calculates its complete residue pairing, trace pairing, determinant and local ES/RH comparison. It does not identify a bilinear residue pairing with the original positive Hermitian metric. A finite flat complete intersection is Gorenstein in the standard sense (Stacks Project, Tag 0C15); the explicit matrices below prove the required duality here directly.

41.1 1. Root algebra and an everywhere nondegenerate pairing

Work over

$$R_0 = \mathbb{C}[A, A^{-1}, B, C, D], \quad h(r) = Ar^4 + r^3 + Br^2 + Cr + D, \quad E = R_0[r]/(h).$$

This is free of rank four in the original basis $1, r, r^2, r^3$. Define the functional

$$\lambda_E(f) = A^{-1}[r^3] \operatorname{rem}_h f.$$

Let C_h be multiplication by r in that basis. Reduction of the powers through degree six gives its full bilinear pairing matrix

$$J_h = \begin{pmatrix} 0 & 0 & 0 & A^{-1} \\ 0 & 0 & A^{-1} & -A^{-2} \\ 0 & A^{-1} & -A^{-2} & A^{-3} - BA^{-2} \\ A^{-1} & -A^{-2} & A^{-3} - BA^{-2} & -A^{-4} + 2BA^{-3} - CA^{-2} \end{pmatrix}.$$

Its anti-triangular pattern gives

$$\boxed{\det J_h = A^{-4}, \quad J_h C_h = C_h^T J_h.}$$

These hold without inverting the discriminant. The pairing is therefore perfect also on nonreduced root fibres. All transposes in this section are ordinary transposes, not conjugate transposes.

At a squarefree target, Lagrange interpolation gives

$$\lambda_E(f) = \sum_j \frac{f(r_j)}{h'(r_j)}.$$

For the original signed moment matrix $U = (r_j^n) \text{diag}(\xi_j)$, with $\xi_j^2 h'(r_j) = 1$, this proves

$$\boxed{UU^T = J_h.}$$

The identity uses the squares ξ_j^2 , not their squared moduli. Neither positive definiteness nor an isometry of the original Gamma or arithmetic norm follows. At collisions the individual entries of the inverse states may diverge while this bilinear product has a finite algebraic continuation.

41.2 2. A finite extension that keeps the original pole

Define

$$\widehat{S} = E[\eta]/(\eta^2 - h'(r)).$$

It is free of rank eight with basis

$$1, r, r^2, r^3, \eta, \eta r, \eta r^2, \eta r^3.$$

The original signed inverse algebra is the exact open subalgebra

$$E[\xi]/(\xi^2 h' - 1) \simeq \widehat{S}[\eta^{-1}], \quad \eta = \xi h', \quad \xi = \eta^{-1}.$$

Substitution proves both compositions. This is a finite flat extension over $A \neq 0$; it is not asserted to be a normalization, nor to repair the separate boundary at $A = 0$.

In these coordinates the original Fable point is exactly

$$q(r, \eta) = \left(\eta^{-1}, -r - i\eta, A\eta^3 + 2r\eta + 3i\eta^2, 7ir^2\eta + (B - 17r + Ar^2)\eta^2 - 13i\eta^3 - 2A\eta^4 \right)^T.$$

The first coordinate is a genuine meromorphic pole. For a fixed injective receiving map F with its actual positive metric Q , put $G = F^* Q F$. If the approach to $\eta = 0$ stays over a finite coefficient-base point, then r, A, B remain bounded; boundedness of r, A, B is the sufficient hypothesis used in the following coordinate calculation. Direct multiplication of every displayed coordinate by η gives

$$\eta q \longrightarrow e_1, \quad \boxed{|\eta| \|Fq\|_Q \longrightarrow \sqrt{G_{11}} > 0.}$$

Bounded literal roots alone are not enough for this limit, because the leading coefficient A may diverge while roots remain bounded. The noncancellation conclusion itself needs no

coefficient-boundedness assumption: the first coordinate of ηq is identically one, so positive definiteness of G gives the uniform estimate

$$|\eta| \|Fq\|_Q = \|F(\eta q)\|_Q \geq \sqrt{\lambda_{\min}(G)} \|\eta q\|_2 \geq \sqrt{\lambda_{\min}(G)} > 0.$$

Adding the finite algebraic boundary therefore does not cancel an original metric infinity.

41.3 3. The exact non-invertible element is multiplication by $2\eta^3$

Write every element as $f + \eta g$, $f, g \in E$, and define

$$\Lambda(f + \eta g) = \lambda_E(g).$$

Products show that its full residue Gram is

$$\mathcal{J} = \begin{pmatrix} 0 & J_h \\ J_h & 0 \end{pmatrix}, \quad \boxed{\det \mathcal{J} = A^{-8}}.$$

Indeed the even-even and odd-odd products have no odd component; each mixed product contributes $\lambda_E(fg)$. Thus this pairing stays perfect everywhere on the base.

Let M_f denote multiplication by an element f in $\widehat{S}$. Then

$$\boxed{\mathrm{Tr}(M_f) = \Lambda(2\eta^3 f)}.$$

Proof: on the squarefree open set, the eight states are $(r_j, \pm \eta_j)$, with $\eta_j^2 = h'(r_j)$. Their trace on $f_0 + \eta f_1$ is $2 \sum_j f_0(r_j)$. On the other hand,

$$\Lambda(2\eta^3(f_0 + \eta f_1)) = 2\lambda_E(h' f_0) = 2 \sum_j f_0(r_j).$$

Both sides are regular functions over the integral base ring R_0 , so equality on the dense squarefree open set proves it globally. Applying this identity to every product proves the pairing factorization

$$\boxed{\text{trace pairing} = \text{residue pairing} \circ M_{2\eta^3}}.$$

The Jacobian determinant of the two defining relations $(h, \eta^2 - h')$, in variables (r, η) , is also $2\eta h' = 2\eta^3$, agreeing with the explicit trace calculation. No general residue theorem is required to verify the displayed formula.

Let $\Delta_h = \mathrm{Disc}(h)$. Since

$$\mathrm{Norm}_{E/R_0}(h') = \Delta_h/A^2,$$

and multiplication by η has determinant Δ_h/A^2 on $\widehat{S}$,

$$\det M_{2\eta^3} = 2^8 \Delta_h^3/A^6.$$

Consequently

$$\boxed{\det(\text{trace Gram}) = 256 \Delta_h^3/A^{14}}.$$

This locates the loss exactly. The residue map to the dual is invertible. Multiplication by $2\eta^3$ is a unit precisely on the squarefree open set; it becomes singular on the boundary. The original linear conductor is a separate map and remains governed by its own nonzero leading moment and WCF quotient.

41.4 4. Local fibre and rank at an arbitrary multiple root

Fix a target with an m -fold root r_0 , and write

$$h(r) = (r - r_0)^m g(r), \quad g_0 = g(r_0) \neq 0, \quad \varepsilon = r - r_0.$$

In the local root algebra $\mathbb{C}[\varepsilon]/(\varepsilon^m)$,

$$h' = mg_0\varepsilon^{m-1}.$$

Therefore the local completed signed fibre is

$$\boxed{\mathbb{C}[\varepsilon, \eta]/(\varepsilon^m, \eta^2 - mg_0\varepsilon^{m-1})}, \quad \text{length} = 2m.$$

Multiplication by ε has two Jordan blocks of size m , since the algebra is free with basis $1, \eta$ over the truncated root algebra. For $m \geq 2$,

$$\eta^3 = mg_0\eta\varepsilon^{m-1} \neq 0, \quad \eta^4 = 0.$$

Every nonconstant monomial annihilates η^3 , while multiplication by 1 does not. Thus multiplication by $2\eta^3$ has rank one on this length- $2m$ local fibre, and the trace pairing has rank one. The residue pairing stays nondegenerate because it is the restriction of the perfect pairing to the corresponding idempotent summand.

For the quartic h , only $m = 2, 3, 4$ can occur. The certificate's additional $m = 5, 6, 7$ checks test the same local formula in the generalized degree- m model; they are not multiplicities of this quartic.

For $m = 2$, $\varepsilon = \eta^2/(2g_0)$, so the algebra is exactly $\mathbb{C}[\eta]/(\eta^4)$. This is one length-four nonreduced state, not four finite distinct points. It retains the limiting algebraic multiplicity of four nearby signed states.

41.5 5. An explicit ES/RH local algebra map, including its residue correction

The ES family

$$(p; x, y, z) = (p; p, 2p/5, 2p)$$

satisfies the original reciprocal equation. Its normalized quartic is

$$h_E(r) = -\frac{5}{22p}(r - p)^2(r - 2p/5)(r - 2p).$$

At the double root p ,

$$g_{0,E} = 3p/22, \quad g_{1,E} = 1/11, \quad \eta_E^2 = (3p/11)\varepsilon_E.$$

For $p = 5$, this is the actual integral example $(5; 5, 2, 10)$. It is not in the hard-prime domain from the preceding continuation.

Fix the source parameter $\gamma > 0$. The RH source's *auxiliary* paired-root polynomial is

$$h_R(r) = -\frac{1}{2}((r - \frac{1}{2})^2 + \gamma^2)^2.$$

At $r_s = 1/2 + si\gamma$, $s = \pm 1$,

$$g_{0,R} = 2\gamma^2, \quad g_{1,R} = -2si\gamma, \quad \eta_R^2 = 4\gamma^2\varepsilon_R.$$

Choose either square root $c^2 = 3p/(44\gamma^2)$. Then

$$\phi : \eta_E \mapsto c\eta_R, \quad \varepsilon_E \mapsto \varepsilon_R$$

is an isomorphism of the two length-four local algebras. It intertwines centred multiplication exactly; the uncentred variable acquires the explicit scalar correction $p - r_s$.

The full residue functional contains the derivative of the cofactor. At a double root,

$$\lambda_E(f_0 + f_1\varepsilon) = f_1/g_0 - f_0g_1/g_0^2.$$

Therefore the residue functionals under ϕ are **not** in general related by just a constant. The exact relation is

$$\Lambda_R(\phi f) = \Lambda_E(u_E f), \quad u_E = c^3 \left[1 + \left(\frac{2}{3p} + \frac{si}{\gamma} \right) \varepsilon_E \right].$$

To check it, write the odd part of f as $\eta_E(b_0 + b_1\varepsilon_E)$. The two sides are respectively

$$c \left(b_1/g_{0,R} - b_0g_{1,R}/g_{0,R}^2 \right)$$

and the same expression after using $g_{0,E} = c^2 g_{0,R}$. The even part contributes zero. The bracketed element is a unit because $\varepsilon_E^2 = 0$; its inverse is obtained by changing the sign of its nilpotent term, followed by c^{-3} . This is an explicit correction, not an assumed residue isometry.

In the basis $1, \eta, \eta^2, \eta^3$, the algebra map has matrix $\text{diag}(1, c, c^2, c^3)$. A positive Hermitian metric transforms by its complete conjugate congruence, with determinant factor $|c|^{12}$. The algebra map does not supply numerical values for either original native Gram.

41.6 6. Why this does not produce purity by itself

For a nonzero nilpotent N with $N^2 = 0$, every positive Hermitian metric G has an indefinite defect $N^*G + GN$. Choose $Nv = w \neq 0$, $Nw = 0$. On the two vectors (w, v) , the upper-left entry is zero and the off-diagonal entry is $\|w\|_G^2$. The determinant of that compression is $-\|w\|_G^4 < 0$. Thus both signs occur.

The completed local algebra and its perfect bilinear pairing are useful exact structures. Neither removes the native metric pole nor proves an isometric arithmetic action or RH. The separate analytic metric return is addressed in Tracks III–V.

41.7 7. Transport through the fixed original conductor

Let the original fixed four-plane maps be $\Phi_* : \mathbb{C}^4 \rightarrow U_*$, $\Psi_* : \mathbb{C}^4 \rightarrow W_*$, with $L\Phi_* = \Psi_*$, as in the source. Realize the eight coefficient coordinates through

$$F_U = \text{diag}(\Phi_*, \Phi_*), \quad F_W = \text{diag}(\Psi_*, \Psi_*), \quad (L \oplus L)F_U = F_W.$$

For every displayed algebra multiplication matrix M_a , define $M_a^U = F_U M_a F_U^{-1}$ and $M_a^W = F_W M_a F_W^{-1}$. Then

$$(L \oplus L)M_a^U = M_a^W(L \oplus L).$$

This includes the non-unit multiplication $M_{2\eta^3}$, at every allowed auxiliary parameter. The residue and trace matrices transport by $F^{-T} \mathcal{J} F^{-1}$, and their factorization persists. The original Hermitian forms transport separately by conjugate transpose. Every map here is on its specified image, with the reference conductor and norms fixed. The inverse-exterior factors of $L \oplus L$ remain exactly those of the source's direct-sum conductor; no boundary singularity is reclassified as a zero of its original triangular determinant.

42 Track III — The exact map from residue duality to native moment Grams

Status. A finite algebraic identity for the original positive measures. It explains exactly which multiplier converts a residue pairing into a native Gram, and exactly which correction survives for an arbitrary ES root polynomial. The classical Gauss/Stieltjes context is not claimed as new; the explicit receiving factorization and residual are the objects used here. The finite diagnostic uses six rational atoms and is not an evaluation of the arithmetic density.

42.1 1. Original positive moment problem

Let $d\rho$ be one of the programme's actual native parity measures or its positive tilt, with $\text{supp } \rho \subset [0, \infty)$. Retain its full mass μ_0 , moments $\mu_j = \int x^j d\rho$, and original real coordinate x . Assume its polynomial Grams through the degree in use are positive definite. For the native measures this follows from their positive-almost-everywhere density and finite moments.

For $n \geq 1$, let p_n be the monic degree- n orthogonal polynomial. Define

$$H_{n-1} = [\mu_{i+j}]_{i,j=0}^{n-1}, \quad E_n = \mathbb{C}[z]/(p_n),$$

and the monic residue functional

$$\lambda_p(f) = [z^{n-1}] \text{rem}_{p_n} f.$$

Let J_p be its pairing matrix and C_p multiplication by z , in $1, z, \dots, z^{n-1}$. The anti-diagonal entries of J_p are one; the entries above that anti-diagonal are zero. Hence

$$\det J_p = (-1)^{n(n-1)/2} \neq 0.$$

This algebraic pairing is complex bilinear and can be indefinite on real vectors. It is not the native Hermitian form.

42.2 2. The multiplier with its original mass

Define the second-kind polynomial by the finite identity

$$q_{n-1}(z) = \int \frac{p_n(z) - p_n(x)}{z - x} d\rho(x).$$

The numerator is divisible by $z - x$, so this is a polynomial of degree $n - 1$ whose leading coefficient is exactly μ_0 . No normalization of the measure is needed.

For every polynomial f of degree less than n ,

$$\lambda_p(q_{n-1}f) = \int f(x) d\rho(x).$$

One proof is to use the simple real roots t_j of the orthogonal polynomial and the Lagrange cardinals $L_j(z) = p_n(z)/((z - t_j)p'_n(t_j))$. Evaluating the defining formula of q at t_j gives

$$\frac{q(t_j)}{p'_n(t_j)} = \int L_j(x) d\rho(x).$$

Lagrange interpolation of f then proves the identity. For completeness, the roots are simple and lie in the convex hull of the positive support by the usual elementary sign-change argument: if the real polynomial had fewer than n sign changes in that support, multiplying the factors at its sign changes would give a polynomial of degree below n whose product with p_n has a fixed nonzero sign almost everywhere, contradicting orthogonality.

For any product z^{i+j} , $0 \leq i, j < n$, divide by p_n . The quotient has degree at most $n - 2$. Its integral multiplied by p_n vanishes by orthogonality. Thus the same identity applies to all moment-Gram entries and gives

$$\boxed{H_{n-1} = J_p q_{n-1}(C_p).}$$

The order of the matrices is specified. Multiplication is self-adjoint for the residue pairing: $J_p C_p = C_p^T J_p$. Therefore the product is symmetric, as required. Since the original moments are real, that same real symmetric matrix is the native positive Hermitian form on complex coefficient vectors.

Both H_{n-1} and J_p are invertible, so $q(C_p)$ is a unit of this root-algebra representation. Equivalently p_n and q_{n-1} are coprime. The native mass is present in q : multiplying $d\rho$ by a positive scalar multiplies q and H by that scalar, while p_n and J_p stay unchanged.

42.3 3. Positivity of the finite weights and the resolvent identity

The polynomial $L_j^2 - L_j$ vanishes at every root of p_n . Its quotient by p_n has degree at most $n - 2$. Orthogonality therefore proves

$$w_j := \frac{q(t_j)}{p'_n(t_j)} = \int L_j d\rho = \int L_j^2 d\rho > 0.$$

The native Gram has the exact Gauss decomposition

$$H_{n-1} = \sum_j w_j (1, t_j, \dots, t_j^{n-1})^T (1, t_j, \dots, t_j^{n-1}).$$

This is a consequence of the native orthogonality, not an arbitrary choice of the ES roots as nodes.

For $a > 0$, partial fractions give

$$\sum_j \frac{w_j}{a + t_j} = -\frac{q(-a)}{p_n(-a)}.$$

The same quantity is the scalar Gauss lower approximation obtained from the native finite matrices:

$$c^T (D + aH)^{-1} c = -\frac{q(-a)}{p_n(-a)},$$

where $H = [\mu_{i+j}]_{0 \leq i, j < n}$, $D = [\mu_{i+j+1}]$, and $c = (\mu_0, \dots, \mu_{n-1})^T$. To see this without selecting new nodes, the Galerkin equation asks for Q with $\deg Q < n$ such that $1 - (x + a)Q$ is orthogonal to every polynomial of degree below n . Thus

$$1 - (x + a)Q(x) = p_n(x)/p_n(-a).$$

Integrating Q , using the defining divided difference of q , gives the rational expression. Expanding the squared residual in the original measure gives the standard nonnegative Gauss error. No unknown native integral is assigned a numerical value by this identity.

More explicitly, $p_n(-a) \neq 0$ because all roots of p_n lie in $[0, \infty)$, and the exact error is

$$\boxed{\int \frac{d\rho(x)}{x + a} - c^T (D + aH)^{-1} c = \frac{1}{p_n(-a)^2} \int \frac{p_n(x)^2}{x + a} d\rho(x) \geq 0.}$$

For a positive measure whose support is not contained in $[0, \infty)$, the same algebraic resolvent formula requires the separate hypothesis $p_n(-a) \neq 0$, and the displayed sign additionally requires $x + a > 0$ on the support.

42.4 4. Arbitrary root polynomials retain a full relation defect

Now take any monic real polynomial h of degree n , whether orthogonal or not. Let J_h be its perfect residue Gram and C_h its companion. Define the polynomial k , of degree below n , by

$$[k] = J_h^{-1}(\mu_0, \dots, \mu_{n-1})^T.$$

Then $\lambda_h(kf) = \int f d\rho$ for all f of degree below n , by the definition of k . For higher products, exact polynomial division gives

$$H_{ij} = [J_h k(C_h)]_{ij} + \mathcal{D}_{ij}, \quad \mathcal{D}_{ij} = \int h(x) \text{quo}_h(x^{i+j}) d\rho(x).$$

This is valid even if h has repeated roots, because no root evaluation or discriminant inverse is used. The entire defect is a finite linear combination of original moments. If $h = p_n$, orthogonality kills it, and $k = q_{n-1}$. For a generic ES quartic divided by its leading coefficient, no such orthogonality has been established, so the defect must remain.

For complex h , the coefficient-functional identity is still an exact bilinear statement on polynomial products. One must separately include conjugated coefficients when forming a Hermitian moment Gram; it is not legitimate to replace transpose by adjoint in the algebraic identity. The real ES literal-root polynomial and the real native orthogonal polynomials already give the directly applicable real case.

42.5 5. Programme-level meaning

The combined map is now explicit:

$$\text{root-algebra residue duality} \xrightarrow{M_q} \text{native Gauss Gram},$$

with full mass in q , and with the additional $\mathcal{D}$ whenever the root polynomial is not the native orthogonal polynomial. The ES/Fable algebra and the RH native moment calculation can use the same finite algebra machinery, but the map between their metrics contains a computed multiplier and, generally, a relation residual.

The next analytic task is to evaluate or enclose the actual moments determining q , k , and $\mathcal{D}$, rather than infer positivity from a complex residue pairing. Tracks IV and V give improved inverse margins and exact simultaneous propagation for that purpose.

42.6 6. An explicit inverse margin for the residue-to-native multiplier

Track IV proves $H_{n-1} \succeq (d_\rho/\tau_{n-1})I$ whenever the original measure has density at least d_ρ on the retained interval $[1, 2]$. The exact factorization above therefore gives

$$\|q(C_p)^{-1}\|_2 = \|H_{n-1}^{-1} J_p\|_2 \leq \frac{\tau_{n-1}}{d_\rho} \|J_p\|_2.$$

All entries of J_p are finite polynomial remainders determined by the actual native monic polynomial. No assertion of positivity of $q(C_p)$ in the Euclidean coefficient metric is needed. The bound is not valid for an arbitrary nonorthogonal ES polynomial after discarding $\mathcal{D}$; that difference is precisely why the full residual in Section 4 is retained.

43 Track IV — Exponentially better native monomial inverse margins

Status. This improves the explicit interval-Gram floor in the new repository source `NATIVE_MOMENT_PRECISION.tex`. The original native measure, monomials, interval and density floor are unchanged. Classical Legendre identities are credited to NIST DLMF §§18.5 and 18.9. The calculation below returns them to the exact original coordinates.

43.1 1. The source bound being improved

The repository derives a positive density floor d_λ for each original native parity measure or positive tilt on $[1, 2]$. Its monomial interval Gram is

$$J_t = \left[\frac{2^{i+j+1} - 1}{i + j + 1} \right]_{i,j=0}^t.$$

It uses the valid lower bound

$$J_t \succeq g_t I, \quad g_t = \det J_t / (\operatorname{tr} J_t)^t,$$

with

$$\det J_t = \prod_{j=0}^t \frac{(j!)^4}{(2j)!(2j+1)!}.$$

The complete source also gives a convenient lower bound for g_t with an exponent quadratic in t . The determinant identity is retained; the determinant-over-trace enclosure is what is sharpened here.

43.2 2. Exact inverse in the unmodified monomial coordinates

Set

$$L_j(x) = P_j(2x - 3) = \sum_{k=0}^j c_{jk} x^k,$$

where P_j is the classical Legendre polynomial. Rodrigues' formula, or its finite binomial expansion followed by $x \mapsto x - 1$, gives

$$c_{jk} = (-1)^{j-k} \sum_{m=k}^j \binom{j}{m} \binom{j+m}{m} \binom{m}{k}.$$

All coefficients are explicit integers. Let C_t contain these rows padded to length $t+1$. Classical orthogonality and the exact substitution $u = 2x - 3$, $dx = du/2$, give

$$C_t J_t C_t^T = \operatorname{diag}((2j+1)^{-1})_{j=0}^t.$$

The triangular C_t is invertible, so

$$J_t^{-1} = C_t^T \operatorname{diag}(2j+1) C_t.$$

This is an identity in the original monomial coordinates. It does not replace a native polynomial by a normalized Legendre coordinate without transporting its coefficients.

Define the positive integer

$$\tau_t = \text{tr } J_t^{-1} = \sum_{j=0}^t (2j+1) \sum_{k=0}^j c_{jk}^2.$$

The largest eigenvalue of a positive matrix lies between its trace divided by its dimension and its trace. Applying this to J_t^{-1} proves

$$\boxed{J_t \succeq \tau_t^{-1} I, \quad \tau_t^{-1} \leq \lambda_{\min}(J_t) \leq (t+1)\tau_t^{-1}.}$$

Thus the new lower margin is within a factor $t+1$ of the actual smallest eigenvalue of the interval Gram. This is not a claim about the smallest eigenvalue of the entire unknown native Gram; that Gram is only lower-bounded by the interval contribution.

It is also at least as strong as the earlier determinant-over-trace floor in every degree, not only in the checked finite table. If $\lambda_1, \dots, \lambda_{t+1} > 0$ are the eigenvalues of J_t , then

$$\tau_t = \sum_i \lambda_i^{-1} = \frac{e_t(\lambda_1, \dots, \lambda_{t+1})}{\det J_t}, \quad \tau_t^{-1} = \frac{\det J_t}{e_t(\lambda_1, \dots, \lambda_{t+1})}.$$

Every monomial occurring in the elementary symmetric polynomial e_t occurs with a positive coefficient in $(\sum_i \lambda_i)^t$. Hence

$$\boxed{\tau_t^{-1} \geq \frac{\det J_t}{(\text{tr } J_t)^t} = g_t.}$$

43.3 3. A simple exponential lower bound

The displayed coefficient signs alternate. Hence

$$\sum_k |c_{jk}| = |L_j(-1)| = P_j(5).$$

The three-term Legendre recurrence and positivity for $x > 1$ show $P_j(5) \leq 10^j$: the recurrence's subtraction term is nonnegative, and its leading multiplier is below 10; start from $P_0(5) = 1$, $P_1(5) = 5$. Therefore

$$\sum_k c_{jk}^2 \leq \left(\sum_k |c_{jk}| \right)^2 \leq 100^j,$$

and

$$\boxed{\tau_t \leq (t+1)(2t+1)100^t, \quad J_t \succeq \frac{I}{(t+1)(2t+1)100^t}.}$$

This removes the quadratic-in-degree exponent from this part of the conditioning bound. It does not remove other source constants, growing moment caps, quotient singular values or the computational expense of high-degree arithmetic moments.

Exact examples are

t	tau_t	new floor	approximate log10(new/old)
0	1	1	0
1	40	1/40	0
2	2685	1/2685	1.864
5	1556000328	1/1556000328	20.239

	t	tau_t	new floor	approximate log10(new/old)
	10	9826531238433072229	reciprocal of preceding integer	96.934
		5634990444070		
	20	6399022163411	reciprocal of preceding integer	427.757
		9557036635401		

The exact rational old/new values are recorded in `results/native_gram_floor.json`; decimals are descriptive only. `check_gram_floor.py` checks the inverse factorization through degree 7, all finite coefficient bounds through degree 20, and the exact floor comparison through degree 20. The proof above is for all degrees.

43.4 4. Native floors and finite moment tolerances

For a native measure with density at least $d_\lambda > 0$ on $[1, 2]$,

$$H_t^\lambda = \left[\int x^{i+j} d\lambda \right] \succeq d_\lambda J_t \succeq m_t I, \quad m_t = d_\lambda / \tau_t.$$

For every original positive pole $a > 0$, the same argument gives

$$H_t^{(1)} + aH_t \succeq (1+a)m_t I, \quad H_t^{(2)} + aH_t^{(1)} \succeq (1+a)m_t I.$$

Indeed $x \geq 1$ on that interval. Let all scalar moments through degree $2t+2$ have an absolute error at most ϵ , with real symmetric assembled approximations. A dimension- $(t+1)$ matrix with entry errors bounded by e has operator error at most $(t+1)e$. Thus the relative error is at most

$$\delta = \frac{(t+1)\epsilon\tau_t}{d_\lambda}$$

for each of these matrices; the common factor $1+a$ cancels in the weighted cases. A sufficient explicit prescription is

$$\epsilon \leq \frac{d_\lambda}{4(t+1)\tau_t},$$

which gives $\delta \leq 1/4$. If an individual true matrix has lower margin m and absolute operator error $e < m$, then

$$\|\tilde{H}^{-1} - H^{-1}\| \leq \frac{e}{m(m-e)}.$$

This follows from the resolvent identity $\tilde{H}^{-1} - H^{-1} = -\tilde{H}^{-1}(\tilde{H} - H)H^{-1}$. A relative certificate also yields

$$(1+\delta)^{-1}H^{-1} \preceq \tilde{H}^{-1} \preceq (1-\delta)^{-1}H^{-1}.$$

These finite bounds can replace the old g_t -based margin at each use in the repository's moment-to-current composition. All other terms in that composition must remain; the present calculation is not an evaluation of its native inputs.

43.5 5. Coupling with the previously derived auxiliary-degree rate

The earlier continuation proved, for the original tilted parity measures and poles $a_j = 4j^2$,

$$\lambda_r^{(b)}(4j^2) \leq \frac{\mathcal{K}_{k,j}}{(r+1)^{2j}}, \quad \mathcal{K}_{k,j} = \frac{\pi\sqrt{2\pi}}{4j} \bar{a}_k ((5/4)_{j-1})^2.$$

Here $\lambda_r^{(b)}$ is the native Gauss residual scalar; $\bar{a}_k$ is the unchanged source upper-envelope constant. The proof used the actual scalar residual minimum, domination by the original Gamma measure, and the Meixner–Pollaczek coefficient identity at $2ij$. Its finite coefficient inequalities have been rechecked in `check_previous_bounds.py`; no new rate is being claimed for the first time in this packet.

At a chosen pole, replace the existing finite-moment upper scalar γ_r by

$$\hat{\gamma}_r = \min\{\gamma_r, \mathcal{K}_{k,j}(r+1)^{-2j}\}.$$

This is a minimum of scalar bounds on the same rank-one ray. It is not a minimum of unrelated Loewner matrices. The required r follows from the prior source-error allocation; the improved margin above then prescribes sufficient accuracy for the moments used at that r . This provides a two-stage degree/precision prescription, still dependent on the stated original source constants and moments. Outer-tail truncation remains governed by the original TR/RR bounds.

43.6 6. General positive-coefficient weights in the newest precision source

For every degree- L polynomial weight $w(x) = \sum_{l=0}^L w_l x^l$ with $w_l \geq 0$ and $w(1) > 0$, its original weighted moment Gram has lower floor $w(1)d_\lambda/\tau_t$, because $w(x) \geq w(1)$ on $[1, 2]$. If every scalar moment needed by this weight has absolute error at most ϵ , its scalar entry error is at most $w(1)\epsilon$. Those data extend through degree $2t + L$; in particular, $x(x+a)(x+b)$ requires moments through degree $2t + 3$, rather than the degree $2t + 2$ sufficient for the matrices in Section 4. The same relative budget therefore applies with all original moment shifts present. This replaces only the interval factor g_t by τ_t^{-1} . The source’s separate Radau subtraction, reciprocal scalar margin, multiplication errors and quotient-coordinate constants still have to be included at their actual uses.

44 Track V — Simultaneous native moment uncertainties through both minima

Status. Exact simultaneous finite-rank formulas and a finite polynomial certification procedure. The parameters enclose errors generated by the same native measure; independence of those errors is not asserted. The fixed-column degree bounds are not reused unchanged when the native orthogonal-polynomial columns are recomputed. The original outer-tail bound and the newest projection-error bounds remain separate inputs.

44.1 1. A common source family

Let the original positive source Gram be enclosed by

$$H(\theta) = H_0 + V\Theta V^*, \quad \Theta = \text{diag}(\theta_1, \dots, \theta_m), \quad H_0 > 0.$$

The parameters are real, and the family is positive definite on the box being certified. Negative parameter intervals are allowed. This directly includes the RR finite resolvent family

$$H^{[J]} = H^{[J],+} - \frac{4}{\pi} \sum_j a_j \lambda_j z_j z_j^*, \quad 0 \leq \lambda_j \leq \gamma_j,$$

in both original parity blocks, followed by the exact complex coefficient congruence. The lower Radau source bound gives a positive floor for that whole enclosing box. The original infinite tail is not deleted; it must be attached through TR/RR or the latest PD bracket.

Let $\mathbf{A}$ be any one of the actual onto maps: source remainder, its composition with the original observation, or its composition with the full conductor observation. Put

$$P = H_0^{-1}, \quad G_0 = (\mathbf{A}P\mathbf{A}^*)^{-1}, \\ L = V^*PV, \quad Z = G_0\mathbf{A}PV, \quad K = L - Z^*G_0^{-1}Z.$$

Writing $T = \mathbf{A}P^{1/2}$, the matrix

$$Q := P - P\mathbf{A}^*G_0\mathbf{A}P = P^{1/2} \left[I - T^*(TT^*)^{-1}T \right] P^{1/2}$$

is positive semidefinite. The bracketed matrix is the Euclidean orthogonal projection onto $\ker T$, and the complete congruence by $P^{1/2}$ transports it back to the original source coordinates. Therefore $K = V^*QV \succeq 0$. Neither K nor Θ is assumed to commute with the other matrices.

44.2 2. Exact quotient, determinant and minimum representative

The whole attained quotient is

$$G(\theta) = G_0 + Z\Theta(I + K\Theta)^{-1}Z^*.$$

Its determinant and minimizing source lift are

$$\frac{\det G(\theta)}{\det G_0} = \frac{\det(I + L\Theta)}{\det(I + K\Theta)},$$

$$X_\theta = P\mathbf{A}^*G_0 - B\Theta(I + K\Theta)^{-1}Z^*, \quad B = (P - P\mathbf{A}^*G_0\mathbf{A}P)V.$$

Here $AB = 0$. Thus the correction remains inside the full original source kernel; $X_\theta u$ is the actual minimum representative of u .

Proof: the rank update identity, valid without Θ^{-1} , is

$$H(\theta)^{-1} = P - PV\Theta(I + L\Theta)^{-1}V^*P.$$

To make the noncommutative cancellation explicit, put

$$X_0 = P\mathbf{A}^*G_0, \quad B = QV, \quad \mathcal{R} = \Theta(I + K\Theta)^{-1}.$$

The definitions give the ordered identities

$$AB = 0, \quad V^*B = K, \quad V^*X_0 = Z^*.$$

For $X = X_0 - B\mathcal{R}Z^*$, first $\mathbf{A}X = I$. Also

$$H_0B = V - \mathbf{A}^*Z,$$

and $\mathcal{R} + \Theta K\mathcal{R} = \Theta$. Expanding every factor in its displayed order therefore gives

$$H(\theta)X = \mathbf{A}^*(G_0 + Z\mathcal{R}Z^*).$$

It follows that $X = H(\theta)^{-1}A^*G(\theta)$ and proves both the quotient and the minimizing lift without ever using Θ^{-1} . The identity $\Theta(I + K\Theta)^{-1} = (I + \Theta K)^{-1}\Theta$ also proves the Hermitian symmetry. Finally, since $Z^*G_0^{-1}Z = L - K$,

$$\frac{\det G}{\det G_0} = \det(I + \mathcal{R}(L - K)) = \det\left((I + \Theta K)^{-1}(I + \Theta L)\right) = \frac{\det(I + L\Theta)}{\det(I + K\Theta)}.$$

Both denominator determinants are strictly positive wherever $H(\theta) > 0$. For L , whiten by H_0 and use the determinant lemma on the positive source Gram. For K , compress that positive whitened source to $\ker(AP^{1/2})$, and use the same lemma. This argument covers indefinite Θ , zero parameters and linearly dependent columns of V .

44.3 3. Nested minima use the same parameters

Let A_E be the original remainder map and $\Lambda : E \rightarrow B$ the original onto observation map. Build $G_E(\theta)$ from A_E and $G_B(\theta)$ directly from ΛA_E . Then

$$G_B(\theta) = (\Lambda G_E(\theta)^{-1} \Lambda^*)^{-1}.$$

This follows either by substitution or by minimizing the source norm first over a remainder fibre and then over its observation fibre. The set of all allowed source representatives is the same composite fibre. There is no second independent moment error after the first minimum.

For the fixed original three-column matrix $W = [b_N, b_{N+1}, v_\lambda]$, define

$$F = W^* G_E W, \quad C = W^* \Lambda^* G_B \Lambda W, \quad R = F - C.$$

These are the full three-vector Gram, observed part and kernel part. The definitions retain all original complex entries. The equality $R = F - C$ follows from the orthogonal minimum decomposition in G_E ; it is not a subtraction of unrelated bounds.

44.4 4. Multiaffine entry numerators and bounded-degree current polynomials

For either actual map, write

$$\Delta(\theta) = \det(I + K\Theta), \\ N(\theta) = G_0 \Delta + Z\Theta \operatorname{adj}(I + K\Theta) Z^*.$$

Then $G = N/\Delta$. Every entry of N , and Δ , has degree at most one in each individual parameter. Indeed the i -th column of $I + K\Theta$ depends only affinely on θ_i . In $\Theta \operatorname{adj}(I + K\Theta)$, the cofactor paired with θ_i omits that column, so it contains no further θ_i .

Let the complete fixed-frame numerators be N_F over Δ_E and N_C over Δ_B . Then

$$R = \frac{N_R}{\Delta_E \Delta_B}, \quad N_R = N_F \Delta_B - N_C \Delta_E.$$

Each entry of N_R has degree at most two in each parameter. The original undivided currents are

$$I_C = 2 \operatorname{Re}(\overline{C_{13}} C_{23}), \quad I_R = 2 \operatorname{Re}(\overline{R_{13}} R_{23}), \\ I_\times = 2 \operatorname{Re}(\overline{R_{13}} C_{23} + \overline{C_{13}} R_{23}).$$

Consequently I_C has a real numerator of degree at most two in each parameter over $\Delta_B^2 > 0$. The other two have real numerators of degree at most four in each parameter over $(\Delta_E \Delta_B)^2 > 0$. Explicitly,

$$P_C = 2 \operatorname{Re}(\overline{(N_C)_{13}} (N_C)_{23}),$$

$$P_R = 2 \operatorname{Re}(\overline{(N_R)_{13}}(N_R)_{23}),$$

$$P_\times = 2 \operatorname{Re}[\overline{(N_R)_{13}}(N_C)_{23}\Delta_E + \overline{(N_C)_{13}}\Delta_E(N_R)_{23}].$$

Both mixed products are present. Their exact sum identity is

$$P_R + P_C\Delta_E^2 + P_\times = 2 \operatorname{Re}(\overline{(N_F)_{13}}(N_F)_{23})\Delta_B^2.$$

The positive original factor ω_N can then be divided out with its own exact value or positive enclosure. This step does not assign a constant ω_N while the underlying native polynomials are changing.

44.5 5. A finite sign certificate, not endpoint sampling

Let a real polynomial P have degree at most d_i in parameter θ_i , and transport a specified rational box to $[0, 1]^m$ by its stated affine coordinate map. Write its monomial coefficients there as a_β . Its tensor-product Bernstein coefficients are

$$b_\alpha = \sum_{\beta \leq \alpha} a_\beta \prod_i \frac{\binom{\alpha_i}{\beta_i}}{\binom{d_i}{\beta_i}}, \quad 0 \leq \alpha_i \leq d_i.$$

Expanding the Bernstein basis proves this conversion. The basis functions are nonnegative and sum to one, so

$$\min_\alpha b_\alpha \leq P(\theta) \leq \max_\alpha b_\alpha.$$

All positive coefficients certify $P > 0$ on the whole box. Otherwise subdivide and repeat. The code in `checks/bernstein.py` performs the coefficient conversion exactly over rational arithmetic and returns either a full positive covering certificate, a rational negative witness, or `inconclusive` at its prescribed depth. It does not claim that a negative point of the enclosing box is feasible for the original correlated native moments.

There is an explicit termination bound when a strict margin exists. Suppose $P \geq \mu > 0$ on the original box, let $J = \{i : d_i \geq 1\}$ be the active-coordinate set, and let L_i bound the absolute Bernstein coefficients of its physical-coordinate derivative $\partial_i P$ on that box. Restriction to a subbox preserves this bound because de Casteljau subdivision is convex averaging. For a subbox with widths h_i , the derivative-coefficient identity gives, for $i \in J$,

$$|b_{\alpha+e_i} - b_\alpha| \leq h_i L_i / d_i.$$

Summing at most d_i steps in each active coordinate shows that every coefficient is at least $P(\text{lower corner}) - \sum_{i \in J} h_i L_i$. Thus any subdivision satisfying

$$\sum_{i \in J} h_i L_i < \mu$$

certifies positivity. Uniform tensor refinement after s bisections in every one of m coordinates has widths $h_i = 2^{-s} \ell_i$, uses 2^{ms} subboxes, and has sequential branch depth ms ; it is sufficient once $2^{-s} \sum_{i \in J} \ell_i L_i < \mu$. For the implemented longest-side one-coordinate splitter, branch depth D is sufficient whenever

$$\boxed{\ell_{\max} 2^{-\lfloor D/m \rfloor} \sum_{i \in J} L_i < \mu.}$$

This is a conditional finite-completion theorem, not an assertion that the unknown native current has a positive margin. A smaller configured depth can return `inconclusive`, but cannot create a false positive because acceptance still requires an actual positive Bernstein covering.

Checking only vertices is invalid. The positive Gram family

$$C(s) = \begin{pmatrix} 2 & 0 & -1/4 \\ 0 & 2 & -3/4 \\ -1/4 & -3/4 & 2 \end{pmatrix} + s(1, 1, 1)^T(1, 1, 1), \quad 0 \leq s \leq 1,$$

has current numerator $2(s - 1/4)(s - 3/4)$. It is positive at both endpoints and negative at $s = 1/2$. This is an exact counterexample to that checking rule, not an arithmetic counterexample.

44.6 6. Native monic columns must move with the same moments

If the original monic polynomial itself is recomputed from $H(\theta)$, let q_0 be the original monic degree- n minimizer and let J insert all lower-degree coefficients. Put

$$M_0 = J^* H_0 J, \quad B_0 = J M_0^{-1} J^* V, \quad K_{\text{mon}} = V^* J M_0^{-1} J^* V, \quad b = V^* q_0,$$

and define the squared monic norms

$$h_0 = q_0^* H_0 q_0, \quad h_\theta = q_\theta^* H(\theta) q_\theta.$$

Then

$$q_\theta = q_0 - B_0 \Theta (I_m + K_{\text{mon}} \Theta)^{-1} b,$$

$$h_\theta = h_0 + b^* \Theta (I_m + K_{\text{mon}} \Theta)^{-1} b.$$

Here I_m is the identity on the m -parameter space. The first correction has no leading-degree coefficient. Multiplying $J^* H(\theta) q_\theta$ out gives zero, so it is the actual monic minimizer. Substitution in its squared norm, or completing the square in the lower-degree variables, gives the second identity. Orthonormal columns must also include $h_\theta^{-1/2}$.

The same parameters therefore propagate through the native polynomials, quotient, observation, class and current. The fixed- W per-parameter degree-four bound must not be claimed after replacing W by moving columns without substitution. For monic conventions the substituted entries remain rational; positive norm factors can be cleared with their proved positivity where applicable, or bounded separately. This note makes no blanket degree claim for that expanded expression.

44.7 7. What the certificate does and does not settle

This method processes a finite native moment enclosure without breaking phase or kernel dependencies. It supports original-map congruences, both attained minima and simultaneous pole errors. A successful box certificate proves a sign for the actual native parameter point inside that box. A failed box certificate may only mean that the enclosing box discarded useful moment correlations.

The still-needed inputs are actual certified native moments and source constants, control of the outer tail, and the growing-degree sign margin relevant to the closing proposition. The repository's newest finite tolerances and projection-radius estimates help certify those inputs; they are not replaced by our synthetic exact tests or by ES root samples. Neither ES occupancy nor RH follows from constructing this finite certificate alone.

45 Source and verification ledger

45.1 Reading boundary

This continuation read the relevant repository definitions and new finite-precision sections through the GitHub connector, and checked the original weighted-conductor provider from the conversation upload. It did not read every historical manuscript, replay either repository's complete verification suite, or verify a Lean build. Definitions already displayed in the preceding conversation were used only with their explicit formulas and receiving domains; the carried calculations have their own recheck suite.

The GitHub refs below are immutable snapshots, not a promise that their branches remain at those commits. The ES and RH branch refs were checked at the start of this continuation. No write action was performed.

45.2 Repository sources

45.2.1 ES-1 — Source identity and actual quartic

The Clankers. (2026, September 20). *The normalized Erdős–Straus quartic and its exact passage through the signed Fable cover to a fixed weighted conductor* [Research manuscript]. Erdős–Straus Foundation.

Repository: KokunoYumeto/erdos-strauss-foundation. Commit: 2b5ab3b63471092624b4fcfac2ab2a0fbafa23a3. Path: `research/incoming/es-fable-zeta-bridge-20260920/ES_FABLE_ZETA_CROSSWALK.tex`. Git blob: dd7aa82fed5dc8ea6dd903a8731ec475a85984be. Read scope: original polynomial EZ4, source ES equation and normalized coefficients, EZ6–EZ17, source lines 1–260. The whole raw file was not downloaded into this package.

https://github.com/KokunoYumeto/erdos-strauss-foundation/blob/2b5ab3b63471092624b4fcfac2ab2a0fbafa23a3/research/incoming/es-fable-zeta-bridge-20260920/ES_FABLE_ZETA_CROSSWALK.tex

The companion root README and bridge README were read for programme scope and exact locators, not treated as proof of every inherited claim.

45.2.2 ES-2 — Existing receiving-frame qualification

The Clankers. (2026, September 20). *Independent audit notes for the frozen upstream packet* [Research audit]. Same repository and commit. Path: `research/incoming/es-fable-zeta-bridge-20260920/AUDIT_NOTES.md`. Git blob: 99c6799fc1fd4c3a289913ea140fc682dc8bf9d8. Read completely through the connector. In particular, squarefreeness does not imply odd-frame invertibility; the source gives a distinct regular-target rank-three example. Our positive-real ES slice is an additional computation, not the first discovery of the general qualification.

45.2.3 RH-1 — Current programme frontier

Split-Zero programme. (2026, September 20). *Original theta transfer and certified complex-current precision* [Source release]. Zeta-function research reader. Repository: KokunoYumeto/zeta-function-research-reader. Commit: 1bcdcebb766d84c6edbb42bc7d4d845e960ec325. Its parent is f8b73284d2e669e981bf2c7609e99f4d2b3a50be. Read root README and workbenches/splitzero-tandem/continuations/20260920-theta-transfer-current-precision/RESULT_INDEX.json for the exact new release and remaining native moment/sign calculations. Source-reported test counts are not represented as our executions.

45.2.4 RH-2 — Native inverse margin being improved

Split-Zero programme. (2026, September 20). *Explicit native moment precision, inverse margins, and the original signed receiver* [Research manuscript]. Same RH commit. Path: `workbenches/splitzero-tandem/continuations/20260920-theta-transfer-current-precision/NATIVE_MOMENT_PRECISION.tex`. Git blob: `d7d4efac164faa684142fa4c38d9f58aab3fbcce`. Read source lines 1–460: original native objects, density floors, interval determinant/trace floor, weighted inverse margins, finite moment budget, outward quadratic estimates and Radau/phase formulas. Later sections were not exhaustively read in this continuation.

https://github.com/KokunoYumeto/zeta-function-research-reader/blob/1bcdcebb766d84c6edbb42bc7d4d845e960ec325/workbenches/splitzero-tandem/continuations/20260920-theta-transfer-current-precision/NATIVE_MOMENT_PRECISION.tex

45.2.5 RH-3 — Existing projection and phase bounds

Split-Zero programme. (2026, September 20). *Sharp original-metric projection control and disks for complex mixed pairings* [Research manuscript]. Same RH commit. Path: `workbenches/splitzero-tandem/continuations/20260920-theta-transfer-current-precision/PROJECTION_PHASE_RETURN.tex`. Git blob: `ce29a20ba43452d913682ce428708db4a12b1982`. Read source lines 1–240: PD1–PD17, including the sharp fixed-kernel projection bound and both complex pairing radii. Those are existing source results. The simultaneous rational-parameter update and polynomial certificate in Track V do not reattribute them as new results.

45.2.6 RH-4 — The actual arithmetic observation receiver

Split-Zero programme. (2026, September 20). *The ES–Fable inverse correspondence in the original weighted conductor* [Cumulative manuscript]. Same RH commit. Path: `workbenches/splitzero-tandem/continuations/20260920-marked-observation-illustrated/FABLE_TO_ORIGINAL_CONDUCTOR.tex`. Git blob: `4674db16bf242f95e0b800f8dac14514363d6d76`. Read the displayed cumulative definitions from the preceding conversation and re-read source lines 3835–3975 at the current pinned commit: MO4–MO11, original minimum residuals, the exact K-inverse marking, source/observation/conductor maps, original arithmetic action and its preserved scope. No whole-cumulative-manuscript reading claim is made.

45.2.7 UP-1 — Byte-preserved original weighted conductor

Split-Zero programme. (n.d.). *A polynomial forward bound for the full weighted conductor* [Additive LaTeX provider, WCF]. Provider body: `sources/WEIGHTED_CONDUCTOR_FORWARD.tex`. Original body size: **25,840 bytes**. Original SHA-256: `d86dcb3daf13c9b2d5a0aaa4d72cce35a1aff48066ad69c7d9846b9d097dc354`. Recovered from the automatically mounted conversation upload `02_Recent_Forward_Calculations.tex`. Its byte length and digest were checked against the provider header. The package includes the complete provider body, not the surrounding source-recovery collection or a guessed GitHub reconstruction. The operator, quotient, actual order, Gamma norm and polynomial definitions WCF1–WCF9 were inspected; this does not certify every downstream asymptotic in the complete provider.

45.3 Primary external sources

National Institute of Standards and Technology. (n.d.). *NIST Digital Library of Mathematical Functions*, §§18.5, 18.9 and equation 18.23.7. Retrieved September 20, 2026.

<https://dlmf.nist.gov/18.5>
<https://dlmf.nist.gov/18.9>
<https://dlmf.nist.gov/18.23.E7>

Use: classical Rodrigues/binomial representation and recurrence for the shifted Legendre calculation; Meixner–Pollaczek generating-function context for the retained WCF and prior auxiliary-degree rate. The new coordinate identities are derived in the proof notes, not copied from a purported programme-specific DLMF theorem.

The Stacks Project Authors. (n.d.). *Lemma 48.25.5* (Tag 0C15); *Integral and finite morphisms* (Tag 03ZN). Retrieved September 20, 2026.

<https://stacks.math.columbia.edu/tag/0C15>

<https://stacks.math.columbia.edu/tag/03ZN>

Use: standard finite-flat local-complete-intersection/Gorenstein and finite/proper background. The precise rank-eight algebra, perfect pairing, trace factor and all local constants used here are proved explicitly; no general theorem is used to hide a missing coordinate map.

Zimmerling, J., Druskin, V., & Simoncini, V. (2025). *Monotonicity, bounds and extrapolation of Block-Gauss and Gauss-Radau quadrature for computing $B^T \phi(A) B$* (Version 3) [Preprint]. arXiv:2407.21505.

<https://arxiv.org/abs/2407.21505>

Use: primary Gauss/Radau context cited by the new native-moment source. The abstract metadata were checked; this continuation does not claim a fresh complete reading of its full author source or import its finite-dimensional assumptions into the native measure problem. Track III contains its own finite orthogonality and residue derivation.

45.4 New versus carried statements

New relative to the inspected snapshots: the positive-real ES slice with exact frame-rank loss; the trace/residue factorization by multiplication by $2 \eta^3$ and complete local residue-unit comparison; the explicit residue-to-native multiplier and arbitrary-frame defect as used in this programme; the improved interval inverse-trace margin; simultaneous correlated-parameter quotient/current formulas and their finite sign-certificate interface.

Carried and rechecked: the general odd determinant factor and earlier one-point repair, the fixed-prime order-48 signed group calculation, the prior Gauss residual degree bound, and the original four-dimensional polynomial/Jacobian identities. Classical mathematics is credited at use. No global priority claim is made for elementary linear algebra, quadrature, residue identities or representation-theoretic facts.

Arithmetic-frame and signed-descent continuation

The final continuation keeps four proof tracks separate: arithmetic nonsingularity of the ES receiving frame, a sharp paired native determinant return, finite heat-action remainders in the retained metric, and an exact finite-field signed-cover descent calculation.

Arithmetic nonsingularity of the odd receiving frame

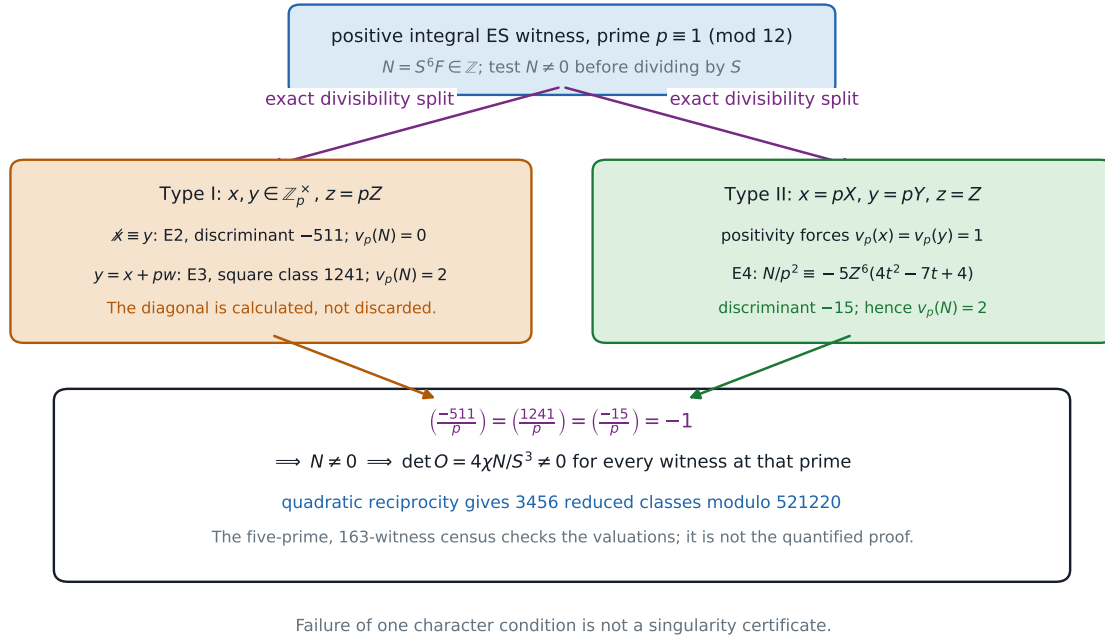


Figure 13: The exhaustive valuation split and the three quadratic obstructions used to force receiving-frame nonsingularity.

Native determinant returns and finite heat certificates

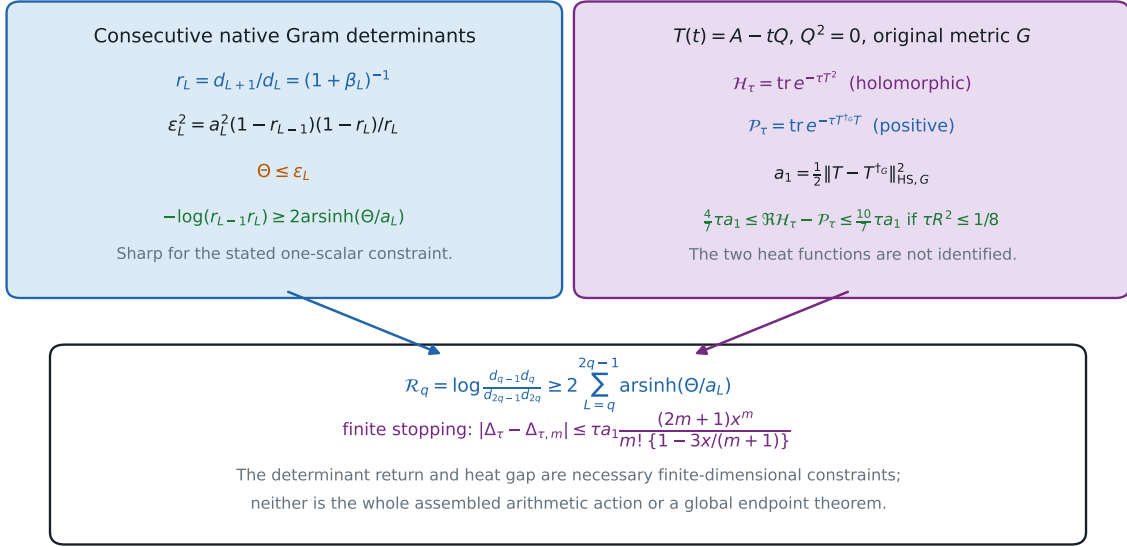
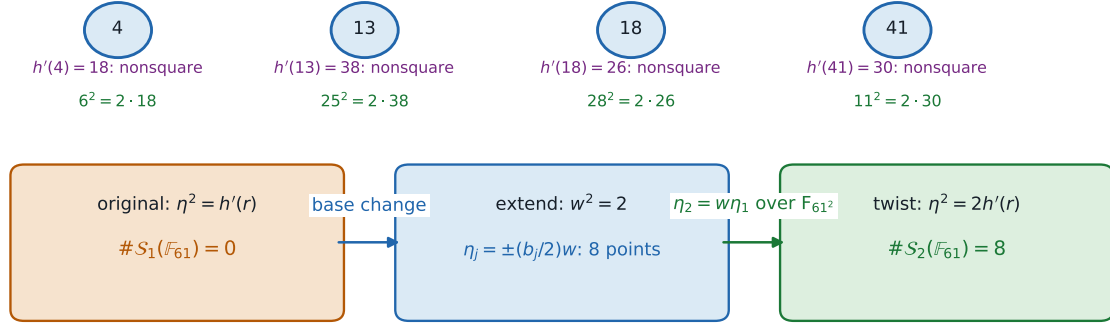


Figure 14: The exact determinant-ratio identity, its sharp two-ratio consequence, and the two distinct heat actions with their proved remainder boundary.

Signed-cover descent for the witness (13; 4, 18, 468) at $q = 61$

$$h(r) = 4r^4 + r^3 + 35r^2 + 8r + 28, \quad (t_j) = (4, 13, 18, 41)$$



$w^{61} = -w$: Frobenius is four disjoint transpositions, $N_n = 0$ for odd n and 8 for even n .

$$Z_{S_1}(T) = (1 - T^2)^{-4}; \text{Tr}(\text{Frob}) = 0 \text{ but } \text{Tr}_{S/\mathbb{F}_{61}}(1) = 8.$$

The extension and the twist are exact typed constructions; neither deletes the original ES witness.

Figure 15: The original signed cover and its nonsquare twist for (13; 4, 18, 468) at $q = 61$, including the quadratic base-extension map and Frobenius action.

46 ES–RH continuation B

46.1 Arithmetic nonsingularity, paired determinant bounds, finite heat actions, and arithmetic descent

20 September 2026. This continuation uses the delivered predecessor package and the repository’s newer original-metric spectral-action edition. It contains four complete proof tracks and reproducible exact diagnostics. It does not claim that time spent substitutes for proof, or that finite algebra alone closes either global conjecture.

The first result removes the extra receiving-frame singularity for every positive integral ES witness at primes in 3456 explicit residue classes modulo 521220. Three quadratic-character conditions are sufficient. The proof distinguishes all possible divisibility patterns and computes the exceptional diagonal term rather than discarding it. At the five diagnostic primes there are exactly 163 sorted witnesses; their enumeration checks, rather than proves, the quantified theorem.

The second result evaluates the updated repository’s existing full-multiplicity spectral allowance on the original quartet grid and carries it into a sharp two-step native determinant inequality. The four cutoff indices are retained. This contribution is not silently identified with the programme’s whole action or independent target minimum.

The third result completes the source’s stated finite-heat step: a genuinely convergent outgoing-parameter series, dimension-independent truncation remainders, and an explicit small-time comparison between holomorphic heat and singular-value heat in the original metric. The two heat functions remain different. A reusable rational certificate routine implements the finite remainder without numerically sampling an exponential.

The fourth result constructs an exact arithmetic descent test. The actual ES witness (13;4,18,468) has no signed rational lift over $\mathbb{F}_{61}$, although its base quartic has four distinct roots there. All eight original lifts appear over $\mathbb{F}_{61}^2$, and the nonsquare twist has eight over $\mathbb{F}_{61}$. The common integral algebra and both base changes are stated, so no nonexistent map from a complex native metric into a finite field is assumed.

46.1.1 Verification record

All **four new suites** pass in ordinary and optimized Python, with **1,121 exact checks and ten negative controls per mode** and identical output and receipt bytes. The **ten predecessor suites** also replay successfully: **539 checks and ten controls per mode**, for a combined **1,660 exact checks and twenty controls per mode**. The original predecessor ZIP is included unchanged. This is not a Lean build, independent human peer review, or evaluation of unknown native arithmetic moments.

46.1.2 Source basis

The ES source is pinned to 2b5ab3b63471092624b4fcfac2ab2a0fbafa23a3. The newer RH spectral-action source is pinned to 4594c6d7528472bb2c7998ced3a7db5d7ac72bd5; its preceding complete secular source was read at 34f960f3af2a9f9eb9f87f3a937051e2473507ea. MC19–20 already contains the aggregate spectral theorem. It is an input here, not counted as a new theorem. The source ledger records exact paths, Git blob identities, reading scope and primary-literature records. Neither repository was modified.

47 Track I. Arithmetic exclusion of the receiving-frame divisor

Status and scope. This track extends the preceding positive-real frame calculation to a sufficient theorem for **all positive integral witnesses at specified primes**. It does not prove existence of an ES witness at every prime, and it does not claim that the sufficient character conditions are necessary. The inherited receiving polynomial is the one in the preceding delivered package, Track I; its normalized ES input is the foundation's EZ6–EZ13 (The Clankers, 2026). The proof below uses the actual integral reciprocal equation, not just its complex coefficient hypersurface.

47.1 I.1 The literal integer to be tested

Retain

$$\frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}, \quad S = p + x + y + z,$$

with positive integers and prime $p \equiv 1 \pmod{12}$. Define

$$e_2 = p(x + y + z) + xy + xz + yz, \quad e_3 = 5xyz, \quad e_4 = pxyz.$$

The normalized literal-root quartic is

$$h(r) = -S^{-1}(r - p)(r - x)(r - y)(r - z) = Ar^4 + r^3 + Br^2 + Cr + D,$$

where $(A, B, C, D) = (-S^{-1}, -e_2/S, e_3/S, -e_4/S)$. For distinct roots r_j , set $d_j = h'(r_j)$, retain both signs of $\xi_j^2 d_j = 1$, and use the original odd columns

$$o_j = \xi_j(1, -id_j, Ad_j^2 + 2r_j d_j, i(7r_j^2 d_j - 13d_j^2))^T.$$

The already established coefficient calculation gives

$$\begin{aligned} \det O &= 4\chi F/A^3, \quad \chi \in \{1, -1\}, \\ F &= 20(3C - B^2) + A(87B^3 - 282BC - 51D) \\ &\quad + A^2(-28B^4 + 22B^2C + 431C^2 + 204BD) \\ &\quad + A^3(224B^2D - 168BC^2 - 856CD) - 448A^4D^2. \end{aligned}$$

Consequently the actual integer $N = S^6 F$ is

$$\begin{aligned} N &= 60S^5 e_3 - 20S^4 e_2^2 - 51S^4 e_4 - 282S^3 e_2 e_3 \\ &\quad + 87S^2 e_2^3 + 204S^2 e_2 e_4 + 431S^2 e_3^2 + 22S e_2^2 e_3 - 856S e_3 e_4 \\ &\quad - 28e_2^4 + 224e_2^2 e_4 - 168e_2 e_3^2 - 448e_4^2. \end{aligned} \tag{E1}$$

We work with E1 before reducing modulo p ; **no inverse of S modulo p is assumed**.

47.2 I.2 Exact divisibility alternatives

First, p, x, y, z are distinct. If $x = p$, then $(3y - p)(3z - p) = p^2$, with both factors positive. Each factor is a power of p , hence is $1 \pmod{3}$, whereas each is $-p \equiv 2 \pmod{3}$. This is impossible. If two denominators equal a , put $d = 2a - p > 0$. Clearing the equation gives $4dz = p(d + p)$, and thus $d \mid p^2$. The possibilities $d = 1, p, p^2$ respectively give $z = p(p + 1)/4, p/2, (p + 1)/4$, none integral for $p \equiv 1 \pmod{4}$.

The equation $p(xy + xz + yz) = 4xyz$ makes at least one denominator divisible by p . All three cannot be divisible by p , because their reciprocal sum would be at most $3/p$. There are precisely two cases, after relabelling.

Type I: x, y are units modulo p , and $z = pZ$ with $p \nmid Z$. Indeed, if $v_p(z) = r$, the valuation of the left side is one and that of the right side is r , forcing $r = 1$.

Type II: after relabelling, $x = pX, y = pY, z = Z$, where X, Y, Z are units. To prove equality of the two positive valuations, suppose they are $a < b$. Then the two sides of the cleared equation have valuations $a + 1$ and $a + b$, implying $b = 1$, contrary to $1 \leq a < b$. If their common valuation were $r \geq 2$, then the remaining denominator would obey

$$\frac{p}{4} < z \leq \frac{p^2}{4p-2} < \frac{p+1}{4}.$$

For $p \equiv 1 \pmod{4}$, this interval contains no integer. Hence the common valuation is exactly one.

These arguments cover every positive integral witness; no exterior/middle selector is imposed.

47.3 I.3 Type I, including its exceptional diagonal

At $e_3 = e_4 = 0$, E1 factors as

$$N = -e_2^2(4e_2 - S^2)(7e_2 - 20S^2).$$

For Type I this gives

$$N \equiv -x^2y^2(x-y)^2(20x^2 + 33xy + 20y^2) \pmod{p}. \quad (\text{E2})$$

The discriminant of the last quadratic is $-511 = -7 \cdot 73$. If $\left(\frac{-511}{p}\right) = -1$, that factor has no zero with $xy \neq 0$. Hence $x \not\equiv y \pmod{p}$ gives $v_p(N) = 0$.

The diagonal case must not be discarded. Write $y = x + pw$. The exact ES equation solves

$$z = \frac{pxy}{4xy - p(x+y)}.$$

Its denominator is a p -adic unit. Expanding the **integer polynomial E1 after this exact substitution** gives

$$N \equiv p^2x^6 \left(\frac{153}{16} - 73w^2 \right) \pmod{p^3}. \quad (\text{E3})$$

An independently reproducible way to check every coefficient is to put $T_0 = 4xy - p(x+y)$ and clear T_0^6 . After replacing y by $x + pw$, the constant and linear coefficients vanish, and the coefficient of p^2 is

$$-256x^{18}(1168w^2 - 153).$$

Dividing by the constant unit $(4x^2)^6$ gives E3. Terms of higher degree in p cannot contribute to this coefficient.

The square class of $153/1168$ is $1241 = 17 \cdot 73$, since $153 \cdot 1168 = 12^2 \cdot 1241$. If $\left(\frac{1241}{p}\right) = -1$, the coefficient in E3 is nonzero. Thus $v_p(N) = 2$ in this case.

47.4 I.4 Type II and a different quadratic obstruction

For the only possible witness valuation $r = 1$, substitution in E1 yields

$$\frac{N}{p^2} \equiv 20Z^6\{15XY - (1 + X + Y)^2\} \pmod{p}.$$

The ES relation itself gives $X + Y \equiv 4XY \pmod{p}$. With $t = X + Y$, this becomes

$$\frac{N}{p^2} \equiv -5Z^6(4t^2 - 7t + 4) \pmod{p}. \quad (\text{E4})$$

The discriminant is -15 . If $\left(\frac{-15}{p}\right) = -1$, E4 is nonzero, and $v_p(N) = 2$.

For reference, the same polynomial calculation on the formal, non-witness branch $r \geq 2$ would give $S \equiv Z$, $e_2/p \equiv Z$, $v_p(e_3) \geq 4$, $v_p(e_4) \geq 5$, and

$$\frac{N}{p^2} \equiv -20Z^6 \not\equiv 0 \pmod{p}. \quad (\text{E5})$$

Equation E5 is retained as an exact polynomial specialization, but the preceding positivity argument proves that no positive integral witness enters that branch. The exceptional primes dividing the displayed constants are excluded by the character conditions below.

47.5 I.5 Universal conditional-on-witness theorem

Theorem E. Suppose $p \equiv 1 \pmod{12}$ is prime and

$$\boxed{\left(\frac{-511}{p}\right) = \left(\frac{1241}{p}\right) = \left(\frac{-15}{p}\right) = -1.} \quad (\text{E6})$$

Then **every positive integral ES witness at this prime has** $N \neq 0$. Its four literal roots and all eight signed inverse states are distinct, and its original odd receiving matrix is invertible. In Type I, $v_p(N) = 0$ or 2 , according as the two unit denominators are distinct or equal modulo p . In Type II, $v_p(N) = 2$.

Proof. The exact divisibility alternatives exhaust the witnesses. Equations E2–E5 give the stated finite valuations. They exclude $N = 0$, whose valuation would not be finite. Root distinctness was proved before the case split. The inherited determinant identity now proves invertibility. This is a theorem about all witnesses at primes satisfying E6, not an inference from their enumeration.

Quadratic reciprocity, with $p \equiv 1 \pmod{12}$, writes the three conditions as

$$\left(\frac{p}{7}\right) \left(\frac{p}{73}\right) = -1, \quad \left(\frac{p}{17}\right) \left(\frac{p}{73}\right) = -1, \quad \left(\frac{p}{5}\right) = -1. \quad (\text{E7})$$

Thus they specify **3456 reduced residue classes modulo 521220**. To count them, fix the residue one modulo 12; there are two choices modulo 5, 72 choices modulo 73, three choices modulo 7 with the required sign, and eight modulo 17. Their product is 3456, out of $4 \cdot 6 \cdot 16 \cdot 72 = 27648$ reduced classes with residue one modulo 12. The complete CRT list is delivered. For the narrower $1 \pmod{24}$ domain, choose the unique suitable lift of each class modulo 1042440. Primes 97, 397, 613, 853 and 997 satisfy E6. No density theorem or new ES verification range is claimed.

47.6 I.6 Constants, exact original transport, and remaining arithmetic

The earlier entry bounds give $\|O\|_2 \leq 41S^3$, while $|\det O| = 4|N|/S^3$. Therefore Theorem E makes the previously conditional bound unconditional for these witnesses:

$$\boxed{\|O^{-1}\|_2 \leq \frac{41^3}{4} S^{12}.} \quad (\text{E8})$$

In the valuation-two cases the stronger $|N| \geq p^2$ gives an extra factor p^{-2} on the right. For original label and receiving metrics $R, G > 0$, multiply the coefficient bound by $\|R^{1/2}\|_2 \|G^{-1/2}\|_2$. These metrics have not been declared Euclidean.

For the unchanged invertible conductor L , the exact relationship remains

$$\ker(LO) = \ker O, \quad \det(LO) = \det L \det O, \quad (LO)^{-1} = O^{-1} L^{-1}.$$

The result removes the extra frame divisor on E6 without changing L , its leading moment, its quotient, or its source and target norms.

The complement of E6 is not a singularity theorem: the sufficient conditions may fail even at entirely regular witnesses. Conversely, the positive-real singular curve in the predecessor shows why dropping integrality from this argument is invalid. The complete integral zero-locus outside E6 and universal ES occupancy remain separate questions.

47.7 I.7 Verification boundary

`check_es_padic.py` verifies all polynomial expansions and CRT counts exactly. It also exhausts the sorted witnesses at the five displayed primes, finding 8, 29, 44, 34 and 48 respectively, for a total of **163**, and checks each predicted valuation.

Completeness of each finite enumeration is proved as follows. The smallest denominator satisfies $p/4 < a \leq 3p/4$. With $R = 4a - p$, $(Rb - pa)(Rc - pa) = (pa)^2$; for $b \leq c$, the first positive integer factor is at most pa . Enumerating all its divisors, imposing the two actual divisibilities and $a \leq b \leq c$, recovers every sorted witness. These finite checks do not establish occupancy for all primes.

48 Track II. Full spectral multiplicity and sharp paired determinant returns

Status and source distinction. The aggregate spectral inequality is already proved in the updated repository, MC19–MC20 (Split-Zero programme, 2026b). It is used here, not presented as a new discovery. The extensions in this track are its exact original-grid evaluation, its expression through consecutive native determinant ratios, and the sharp two-ratio optimization with the original four cutoffs retained.

48.1 II.1 The fixed original metric and outgoing relation

Retain one original packet and its actual even measure $m = w_h^{*k}$, with its mass unchanged. Let Q_n be its real monic orthogonal polynomials, $\omega_n = \int Q_n^2 m > 0$, and $a_n^2 = \omega_{n+1}/\omega_n$. Set

$$c = k/2, \quad p_n(S) = i^n Q_n((S - c)/i), \quad b_n = [p_n]_\chi,$$

where χ is the full original cyclic annihilator of degree q . In the same fixed remainder coordinates define

$$K_L = \sum_{n=0}^L \frac{b_n b_n^*}{\omega_n}, \quad G_L = K_L^{-1}, \quad L \geq q - 1. \quad (\text{D1})$$

The first q monic columns span the quotient, so these matrices are positive definite. They are the source's attained quotient metric, not a comparison weight.

NS6–NS9 and NS35–NS37 give

$$\begin{aligned} M &= (M_S - cI)/i, \quad A_L = M + Q_L, \quad Q_L = \frac{i}{\omega_L} b_{L+1} b_L^* G_L, \\ A_L^{\dagger G_L} &= A_L, \quad Q_L^2 = 0, \quad b_L^* G_L b_{L+1} = 0, \\ \epsilon_L^2 &= \|Q_L\|_{G_L}^2 = \frac{(b_L^* G_L b_L)(b_{L+1}^* G_L b_{L+1})}{\omega_L^2}. \end{aligned} \quad (\text{D2})$$

The zero cross pairing is the result of the proved original parity, not a term discarded from a Gram determinant.

The Hermitian defect $M_S + M_S^{\dagger G} - 2cI$ has eigenvalues $+\epsilon_L, -\epsilon_L$ and zeros. The repository's aggregate theorem therefore yields

$$\Theta := \sum_{\Re \kappa > c} e_\kappa (2\Re \kappa - 2c) \leq \epsilon_L. \quad (\text{D3})$$

For completeness, let W be the invariant sum of the generalized eigenspaces on the indicated side, and let P_W be its original-metric orthogonal projection. Then Θ is the trace of this projection against the Hermitian defect, and is at most the trace of its positive part, namely ϵ_L . Generalized eigenspaces retain algebraic multiplicities and require no diagonalizability of M_S .

48.2 II.2 Exact cost of the original quartet grid

For the stipulated quartet $1/2 \pm \delta \pm i\gamma$, retain the full common root multiplicity m_0 . The odd tensor degree has

$$\begin{aligned} e_k &= 1 + k(m_0 - 1), \quad q = e_k(k + 1)^2, \\ \kappa_{ab} &= k/2 + (2a - k)\delta + i(2b - k)\gamma, \quad 0 \leq a, b \leq k. \end{aligned}$$

The cyclic multiplicity is e_k , not the number of tensor words that give the same sum. Summing D3 gives

$$\begin{aligned}\Theta_k &= 2\delta e_k(k+1) \sum_{a>k/2} (2a-k) \\ &= \boxed{\frac{\delta e_k(k+1)^3}{2} = \frac{\delta q(k+1)}{2}}.\end{aligned}\tag{D4}$$

Indeed for $k = 2r+1$ the positive terms are $1, 3, \dots, 2r+1$, whose sum is $(r+1)^2$. No γ -dependent phase is removed from the actual operators; D4 only evaluates the real-part sum appearing in D3.

48.3 II.3 The exact consecutive determinant identity

Write

$$d_L = \det G_L, \quad r_L = \frac{d_{L+1}}{d_L}, \quad \beta_L = \frac{b_{L+1}^* G_L b_{L+1}}{\omega_{L+1}}.$$

The rank-one update of K_L , with the determinant lemma, gives

$$\boxed{r_L = (1 + \beta_L)^{-1} \in (0, 1]}.\tag{D5}$$

For $L \geq q$, the same update one index earlier yields

$$\frac{b_L^* G_L b_L}{\omega_L} = \frac{\beta_{L-1}}{1 + \beta_{L-1}} = 1 - r_{L-1}.$$

Substituting in D2 proves the full relation

$$\boxed{\epsilon_L^2 = a_L^2 \frac{(1 - r_{L-1})(1 - r_L)}{r_L}}.\tag{D6}$$

Every ratio uses the same packet, measure and remainder coordinates. The source mass cancels in this displayed relation only; it remains in each individual Gram.

Combining D3 with D6 gives

$$(1 - r_{L-1})(1 - r_L) \geq c_L r_L, \quad c_L = \Theta^2/a_L^2.\tag{D7}$$

This retains the preceding leverage rather than bounding it by one.

48.4 II.4 Sharp two-ratio optimization

Lemma D. If $0 < x, y \leq 1$, $c \geq 0$, and $(1-x)(1-y) \geq cy$, then

$$\boxed{xy \leq (\sqrt{1+c} - \sqrt{c})^2 = \exp[-2 \operatorname{arsinh} \sqrt{c}]}.\tag{D8}$$

For $c > 0$, put $b = (\sqrt{1+c} - \sqrt{c})^2$; then $c = (1-b)^2/(4b)$. The exact square completion

$$b(1-x+c) - x(1-x) = \left(x - \frac{1+b}{2}\right)^2$$

shows $xy \leq x(1-x)/(1-x+c) \leq b$. Equality is attained at $x = (1+b)/2$, $y = 2b/(1+b)$. For $c = 0$, the claim is simply $xy \leq 1$. Thus D8 is the best bound obtainable from that one scalar constraint, without a further original relation.

Applying the lemma to D7 gives

$$\boxed{-\log(r_{L-1}r_L) \geq 2 \operatorname{arsinh}(\Theta/a_L)}.\tag{D9}$$

48.5 II.5 The original four endpoint indices

Consider the **same-cyclic-quotient determinant contribution**

$$\mathcal{R}_q = \log \frac{d_{q-1}d_q}{d_{2q-1}d_{2q}}.$$

It has exactly the telescoping expansion

$$\mathcal{R}_q = - \sum_{L=q}^{2q-1} \log(r_{L-1}r_L).$$

The outer ratios occur once and the interior ratios twice; no endpoint is shifted. Hence

$$\boxed{\mathcal{R}_q \geq 2 \sum_{L=q}^{2q-1} \operatorname{arsinh}(\Theta/a_L).} \quad (\text{D10})$$

For the original quartet insert D4. This is not a claim that $\mathcal{R}_q$ equals the programme's entire assembled Gamma/arithmetic action. Any additional source Jacobians, target minima, boundary terms and signed budget corrections still have to be included through their actual identities.

For example, **conditional** on a proved bound $a_L \leq Cq$ throughout the window, D10 implies

$$\frac{\mathcal{R}_q}{q} \geq 2 \operatorname{arsinh} \left(\frac{\delta(k+1)}{2C} \right). \quad (\text{D11})$$

No such uniform C is asserted by this track.

There is nevertheless an explicit coarse input from the existing native envelope. If

$$\underline{a}_k(1+u^2)^{-M}\sigma(u) \leq m(u) \leq \bar{a}_k\sigma(u), \quad \sigma(u) = |\Gamma(1/4 + iu/2)|^2/(2\pi),$$

then

$$\boxed{a_L^2 \leq \frac{\bar{a}_k}{\underline{a}_k} [1 + 4(L+1)^2]^M (L+1)(L+1/2).} \quad (\text{D12})$$

To prove this, the exact Gamma recurrence bounds multiplication by u from degree L to degree $L+1$ by $2(L+1)$. For any degree- L polynomial, Jensen's inequality for $(1+x)^{-M}$ bounds its weighted Gamma norm below by $[1 + 4(L+1)^2]^{-M}$ times its Gamma norm. Minimizing over monic polynomials gives the lower bound for ω_L . Testing the native minimum at the monic Gamma polynomial gives $\omega_{L+1} \leq \bar{a}_k \omega_{L+1}^\Gamma$. Their exact Gamma ratio is $(L+1)(L+1/2)$, giving D12. The original constants and mass are unchanged.

48.6 II.6 A forced nonnormality scale in the same native metric

There is a quantitative consequence for interpreting the heat scale. The original imaginary part of M satisfies

$$\Im_G M = (M - M^\dagger)/(2i) = -\Im_G \mathcal{Q}_L.$$

For a square-zero rank-one map, its range vector and the Riesz vector of its defining covector are orthogonal. In the orthonormal basis of their two-plane its matrix is $\begin{pmatrix} 0 & \epsilon_L \\ 0 & 0 \end{pmatrix}$, with zeros on the complement. Thus $\Im_G M$ has eigenvalues $\pm \epsilon_L/2$ and zeros. This is a proof in an explicitly isometric basis, not a declaration that the original coordinate vectors are orthonormal. Since $\|\Im_G M\|_G \leq \|M\|_G$,

$$\boxed{\|M\|_{G_L} \geq \frac{\epsilon_L}{2} \geq \frac{\delta e_k(k+1)^3}{4}.} \quad (\text{D13})$$

The spectral radius of this original quartet companion is exactly $\rho(M) = k\sqrt{\gamma^2 + \delta^2}$. Consequently

$$\frac{\|M\|_{G_L}}{\rho(M)} \geq \frac{\delta e_k(k+1)^3}{4k\sqrt{\gamma^2 + \delta^2}}.$$

For simple roots, any eigenvector isomorphism from standard label space to this native space has condition number at least this ratio, by $M = V \operatorname{diag}(\lambda_j) V^{-1}$. At higher multiplicity no diagonalizer is asserted. The operator norm and the maximum eigenvalue modulus therefore cannot be interchanged in a heat remainder. This lower bound is compatible with the finite inverse-exterior identities; it quantifies the nonnormality those determinant identities do not by themselves control.

48.7 II.7 Meaning and invariance

For any fixed invertible coordinate map J , the transformations $M' = JM J^{-1}$, $G'_L = J^{-*} G_L J^{-1}$ preserve D2 and all determinant ratios. In particular the source's full physical-jet injection, used only on its image with its transported metric, carries these identities exactly. An original conductor with a different target metric requires the corresponding full pullback; it is not declared unitary.

D3–D10 hold for any positive even measure and the stated symmetric quotient, not only for a zeta measure. They are necessary structural constraints. A closing upper estimate would have to use additional actual arithmetic information. For instance a proved $\epsilon_{L(k)}/[e_k(k+1)^3] \rightarrow 0$ would exclude a fixed $\delta > 0$, but that decay has not been established here.

`check_spectral_returns.py` verifies the complete formulas in exact positive compact-support models, including repeated nonreal roots, real roots and nonunitary coordinate changes. Those models are explicitly not values of unknown native arithmetic moments. It also verifies the quartet sum and the sharp scalar equality case symbolically.

49 Track III. Genuine finite heat actions with explicit remainders

Status. The updated source proves polynomial trace coefficients, the exact square-zero outgoing relation, and quadratic metric curvature. Its stated next step is a finite heat action with convergent remainders (Split-Zero programme, 2026b, SA1–43 and MC1–35). This track supplies that step. Every space is finite-dimensional. No infinite spectral trace, limiting determinant, Schwartz functional calculus on a nonnormal infinite operator, or RH endpoint is asserted.

49.1 III.1 Original metric and the two different heat actions

Keep a specified positive Hermitian form G , the adjoint $X^\dagger = G^{-1}X^*G$, a selfadjoint $A^\dagger = A$, and

$$Q = r\ell, \quad \ell r = 0, \quad Q^2 = 0, \quad \epsilon = \|Q\|_G = \|Q\|_{\text{HS},G}.$$

The original native instance is the outgoing correction in D2. For real t , put $T(t) = A - tQ$. The repository's exact quadratic identity is

$$\text{tr}(T^\dagger T) - \Re \text{tr}(T^2) = t^2 \epsilon^2. \quad (\text{H1})$$

For $\tau > 0$, define separately

$$\begin{aligned} \mathcal{H}_\tau(t) &= \text{tr} e^{-\tau T(t)^2}, \\ \mathcal{P}_\tau(t) &= \text{tr} e^{-\tau T(t)^\dagger T(t)}. \end{aligned} \quad (\text{H2})$$

The first may be complex; the second is positive. Nilpotence does not identify them. At the original endpoint $T(1) = M = (M_S - cI)/i$, the first exponential is $e^{+\tau(M_S - cI)^2}$, not $e^{-\tau M_S^2}$. The second is $e^{-\tau(M_S - cI)^\dagger(M_S - cI)}$. These identities retain the original centre and the factor i .

49.2 III.2 Entire outgoing-parameter series and a dimension-independent remainder

Because $Q^2 = 0$,

$$T(t)^2 = A^2 - tD, \quad D = AQ + QA. \quad (\text{H3})$$

The rank of D is at most two, but it is not generally selfadjoint. Let $\|X\|_{1,G}$ be its trace norm in the original metric. The finite heat trace has the entire expansion

$$\mathcal{H}_\tau(t) = \text{tr} e^{-\tau A^2} + \sum_{n=1}^{\infty} t^n h_n(\tau), \quad (\text{H4})$$

$$h_n(\tau) = \tau^n \int_{\Delta_n} \text{tr} \left(e^{-\tau s_0 A^2} D e^{-\tau s_1 A^2} \dots D e^{-\tau s_n A^2} \right) ds. \quad (\text{H5})$$

Here $s_j \geq 0$, $\sum_{j=0}^n s_j = 1$, and the simplex measure has volume $1/n!$. In particular

$$\boxed{|h_n(\tau)| \leq \frac{\tau^n \|D\|_{1,G} \|D\|_G^{n-1}}{n!}}. \quad (\text{H6})$$

Proof. The operator Volterra equation is

$$U(\tau) = e^{-\tau A^2} + t \int_0^\tau e^{-(\tau-s)A^2} D U(s) ds.$$

Successive substitution gives the ordered-simplex products in H5. The original-metric norm of every e^{-sA^2} , $s \geq 0$, is at most one by selfadjointness. In the trace, bound one occurrence of D in trace norm and all other factors in operator norm. This proves H6, including the simplex volume. The analogous operator-norm bounds give a uniformly absolutely convergent series

on every bounded complex t -disk; its substitution in the Volterra equation proves equality and entire dependence.

Set $x = \tau|t|\|D\|_G$. Define the actual truncation error by

$$\mathcal{E}_m(t) = \mathcal{H}_\tau(t) - \text{tr } e^{-\tau A^2} - \sum_{n=1}^m t^n h_n(\tau).$$

Then

$$|\mathcal{E}_m(t)| \leq \tau|t|\|D\|_{1,G} \frac{e^x x^m}{(m+1)!}. \quad (\text{H7})$$

For $D = 0$ the remainder is zero and no division by its norm is used. For $D \neq 0$, sum H6 and use the scalar exponential tail. The original rank-one identity supplies the explicit estimates

$$\|D\|_G \leq 2\|A\|_G \epsilon, \quad \|D\|_{1,G} \leq 2\|A\|_G \epsilon. \quad (\text{H8})$$

There is no factor $\dim E$ in H6–H8. Dependence of the actual norms on degree is, of course, still present.

For completeness, the positive heat action has

$$T^\dagger T = A^2 + U_t, \quad U_t = -t(AQ + Q^\dagger A) + t^2 Q^\dagger Q, \quad (\text{H9})$$

$$\|U_t\|_{1,G}, \|U_t\|_G \leq 2|t|\|A\|_G \epsilon + t^2 \epsilon^2.$$

A second Volterra expansion, now using $-U_t$, gives the same form of convergent remainder at each real t . It is not the same perturbation as H3.

49.3 III.3 Exact extension of the source's cyclic coefficients

Let $A = \sum_\lambda \lambda P_\lambda$ use the complete original-metric spectral projections, including eigenspaces of dimension greater than one. Put $\eta_\lambda = \ell P_\lambda r$. For $F_\tau(z) = e^{-\tau z^2}$, the coefficient in H4 is also

$$h_n(\tau) = \frac{(-1)^n}{n} \sum_{\lambda_1, \dots, \lambda_n} F'_\tau[\lambda_1, \dots, \lambda_n] \prod_{j=1}^n \eta_{\lambda_j}. \quad (\text{H10})$$

All repeated nodes use their actual confluent divided differences. To prove the formula, first use the source's polynomial identity SA10, whose derivative factor is $(n-1)!$, so its coefficient factor is $1/n$. For rank one, each cyclic trace product equals the product of the η 's. Approximate F_τ by its Taylor polynomials uniformly on a disk containing the spectra of $A - tQ$ for a bounded t -disk. The finite matrix series and every fixed t -derivative converge uniformly. Divided differences on the finite node set converge as well, including derivative-defined repeated nodes. The polynomial identities therefore pass to the entire function. H7 supplies a separate explicit bound for summing these coefficients in t .

The proof preserves the phases of all η_λ . No absolute values are inserted into H10 itself. Bounds are taken only after the exact coefficient is formed. The two formulas H5 and H10 compute the same entire Taylor series.

49.4 III.4 A quantitative heat-gap theorem for any finite matrix

This part does not require rank one. For a finite matrix T in the original metric, define

$$a_n = \text{tr}((T^\dagger T)^n) - \Re \text{tr}(T^{2n}), \quad a_1 = \frac{1}{2}\|T - T^\dagger\|_{\text{HS},G}^2 \geq 0.$$

Let $R \geq \|T\|_G$. Then for every integer $n \geq 1$,

$$|a_n| \leq n(2n-1)R^{2n-2}a_1. \quad (\text{H11})$$

Proof. Write $T = H + iK$ with H, K selfadjoint and put $T_s = H + isK$, $0 \leq s \leq 1$. Since $T_s = (1+s)T/2 + (1-s)T^\dagger/2$, its norm is at most R . Set

$$f_n(s) = \text{tr}((T_s^\dagger T_s)^n) - \Re \text{tr}(T_s^{2n}).$$

Both terms agree at zero, and their first derivatives there have zero difference. The first trace is even in s , since $T_{-s} = T_s^\dagger$ and the two cyclic products have the same trace. The real part of the second is also even. Each second derivative is a sum of $2n(2n-1)$ ordered choices of two differentiated factors. Each trace term contains two factors K , with all remaining factors bounded by R . Cyclicity and Hilbert–Schmidt Cauchy–Schwarz bound its modulus by $\|K\|_{\text{HS},G}^2 R^{2n-2}$. There are two traces, so

$$|f_n''(s)| \leq 4n(2n-1)\|K\|_{\text{HS},G}^2 R^{2n-2}.$$

Integrating against $1-s$ gives H11 because $a_1 = 2\|K\|_{\text{HS},G}^2$. All uses of a Euclidean basis, if made, are through the explicit isometry $G^{1/2}$; the original metric is not changed.

Define the real finite heat gap

$$\Delta_\tau(T) = \Re \text{tr} e^{-\tau T^2} - \text{tr} e^{-\tau T^\dagger T}.$$

Absolute convergence and H11 give

$$\Delta_\tau(T) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} \tau^n a_n}{n!},$$

$$\boxed{|\Delta_\tau(T) - \tau a_1| \leq \tau a_1 \{(1+2x)e^x - 1\}, \quad x = \tau R^2.} \quad (\text{H12})$$

The scalar sum is exact: after putting $m = n-1$, it is $\sum_{m \geq 1} (2m+1)x^m/m!$.

In particular, $x \leq 1/8$ gives $e^x \leq (1-x)^{-1} \leq 8/7$, and hence

$$\boxed{\frac{4}{7}\tau a_1 \leq \Delta_\tau(T) \leq \frac{10}{7}\tau a_1, \quad \tau R^2 \leq \frac{1}{8}.} \quad (\text{H13})$$

For the original $T(t) = A - tQ$, H1 gives $a_1 = t^2 \epsilon^2$. At the actual quotient endpoint $T = M$, D3–D4 therefore give

$$\boxed{\Delta_\tau(M) \geq \frac{4}{7}\tau \Theta_k^2, \quad \tau R^2 \leq \frac{1}{8}.} \quad (\text{H14})$$

This is an exact finite heat comparison, not a proof of decay of ϵ or a sign for a different projected arithmetic current. D13 also shows that the sufficient condition $\tau R^2 \leq 1/8$, with $R \geq \|M\|_G$, can only hold at $\tau \leq 1/(2\Theta_k^2)$ when $\Theta_k > 0$. This restriction concerns the applicability of H13, not a necessary condition for positivity of the actual heat gap. The much smaller eigenvalue radius is not a substitute for the original operator norm. Since $\epsilon_L \leq 2R$, H13 also gives $0 \leq \Delta_\tau(M) \leq 5/7$ on this same small-time domain.

49.5 III.5 A rational finite stopping certificate

Let $\Delta_{\tau,m}(T) = \sum_{n=1}^m (-1)^{n+1} \tau^n a_n / n!$. After $m \geq 1$ terms, H11 gives

$$\boxed{|\Delta_\tau(T) - \Delta_{\tau,m}(T)| \leq \tau a_1 \frac{(2m+1)x^m}{m! \{1 - 3x/(m+1)\}}.} \quad (\text{H15})$$

The condition is $m+1 > 3x$. Indeed the positive majorant terms are $b_n = (2n-1)x^{n-1}/(n-1)!$. Their ratio satisfies $b_{n+1}/b_n = x(2n+1)/(n(2n-1)) \leq 3x/n$; summing the geometric majorant after $n = m$ proves H15. Thus rational matrix entries, a rational upper bound for R^2 , and

rational τ give an entirely rational outward interval. A convenient available upper bound is $R^2 \leq \text{tr}(T^\dagger T)$, though it may be coarse.

The small-time restriction is essential. Take

$$T = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}, \quad Q = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}, \quad A = T + Q = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Then $A = A^*$, $Q^2 = 0$, but $\Delta_\tau(T) = 2 \cos(2\tau) - 2e^{-2\tau}$, which is negative at $\tau = \pi/2$. No all-time positivity statement follows from H13.

49.6 III.6 The exact physical maps and the verification scope

For a fixed coordinate isomorphism J , transform $T' = JTJ^{-1}$ and $G' = J^{-*}GJ^{-1}$. Both heat traces and all constants in the appropriate original-metric norms are preserved. This includes the source's physical-jet injection onto its image with its transported metric.

An original conductor $L : U \rightarrow W$ need not be an isometry for its two separately given metrics. If $T_W = LT_UL^{-1}$ and $H = L^*G_WL$, the exact relation is instead

$$\text{tr} e^{-\tau T_W^{\dagger G_W} T_W} = \text{tr} e^{-\tau T_U^{\dagger H} T_U}. \quad (\text{H16})$$

Replacing H by the different original G_U requires an additional metric estimate. The holomorphic trace is conjugacy-invariant without this qualification. Finally the repository's selfadjoint dilation has $\text{tr} e^{-\tau \mathcal{D}_T^2} = 2\mathcal{P}_\tau(T)$; it is not the same as the holomorphic heat trace.

`check_heat.py` verifies repeated-node cyclic coefficients, the affine square identity, original nonidentity-metric conjugacies, the scalar majorant, and rational heat-gap intervals in exact diagnostic matrices. It uses 12 terms for its explicit intervals and includes false-inference controls. These are not native arithmetic moment evaluations. The original unknown moments and their outward tolerances must still be inserted before using these finite certificates on a hypothetical zero packet.

50 Track IV. Arithmetic descent of the signed cover and an exact twist correction

Status. This is a concrete arithmetic specialization of the original ES/Fable algebra, not a new conjecture about finite-field spectra. It shows that even an actual integral ES witness need not have a rational signed lift at a good residue prime. The missing signs are recovered by an explicit quadratic extension or a different quadratic twist. The analytic receiving metric is not silently reduced modulo a prime.

50.1 IV.1 A common integral model, not a map from complex numbers to a finite field

For a distinct integral ES witness with literal roots $t_1 = p, t_2 = x, t_3 = y, t_4 = z$, put

$$S = \sum_j t_j, \quad \Delta = \prod_{j < k} (t_k - t_j), \quad R = \mathbb{Z}[i, 1/(2S\Delta)].$$

The original normalized quartic lies in $R[r]$, with invertible leading coefficient. Its signed algebra is

$$\mathcal{S} = R[r, \eta]/(h(r), \eta^2 - h'(r)). \quad (\text{F1})$$

It is finite free of rank eight; the root algebra is rank four and the second relation is monic of degree two. The derivative is a unit because the discriminant is inverted. The relation therefore agrees with the original inverse chart by $\xi = \eta^{-1}$.

An embedding $R \rightarrow \mathbb{C}$ and a good reduction $R \rightarrow \mathbb{F}_q$ give two actual base changes of F1. They do **not** give a field homomorphism $\mathbb{C} \rightarrow \mathbb{F}_q$. A native receiving map with Gamma or arithmetic coefficients acts on the complex base change; no integral model of those extra coefficients is assumed. Thus this is a common algebraic source for two fibres, not an identification of their physical metrics.

50.2 IV.2 All rational signed points and both twists

Suppose the reduced quartic has four distinct roots in $\mathbb{F}_q$, where q is odd. Let $d_j = h'(t_j) \neq 0$, and write χ_q for its quadratic character. For any $a \in \mathbb{F}_q^*$, define the explicitly different twist

$$\mathcal{S}_a = \mathbb{F}_q[r, \eta]/(h(r), \eta^2 - ah'(r)).$$

Counting the two square roots or their absence gives

$$\boxed{\# \text{Spec } \mathcal{S}_a(\mathbb{F}_q) = 4 + \chi_q(a) \sum_{j=1}^4 \chi_q(d_j).} \quad (\text{F2})$$

The product of the derivatives is $A^4 \Delta^2$, a square. Consequently the number of nonsquare derivatives is even, and the count for either twist is 0, 4 or 8. For a nonsquare a ,

$$\boxed{\# \mathcal{S}_1(\mathbb{F}_q) + \# \mathcal{S}_a(\mathbb{F}_q) = 8.} \quad (\text{F3})$$

This is a point-count identity, not an identification of the two covers. Their disjoint union has sixteen geometric points. After adjoining a square root $b^2 = a$, the map $\eta_a = b\eta_1$ is an explicit isomorphism of the two base-changed covers.

50.3 IV.3 An actual integral witness with no rational signed lift

The witness

$$(p; x, y, z) = (13; 4, 18, 468)$$

satisfies $4/13 = 1/4 + 1/18 + 1/468$. Reduce at $q = 61$, taking $i \mapsto 11$, for which $11^2 = -1 \pmod{61}$. The complete normalized quartic is

$$h(r) = 4r^4 + r^3 + 35r^2 + 8r + 28 \quad \text{in } \mathbb{F}_{61}[r]. \quad (\text{F4})$$

Its roots and corresponding derivatives, in increasing residue order, are

$$(t_j) = (4, 13, 18, 41), \quad (d_j) = (18, 38, 26, 30). \quad (\text{F5})$$

All four derivatives are quadratic nonresidues. Therefore

$$\boxed{\#\mathcal{S}_1(\mathbb{F}_{61}) = 0.} \quad (\text{F6})$$

This does not remove the original ES witness. It only obstructs the additional choice of a rational normalized square root in this particular fibre.

The element 2 is a nonresidue modulo 61, while

$$(6^2, 25^2, 28^2, 11^2) = 2(18, 38, 26, 30) \pmod{61}.$$

Thus the twist $\eta^2 = 2h'(r)$ has eight rational points. In the extension $\mathbb{F}_{61^2} = \mathbb{F}_{61}[w]/(w^2 - 2)$, the eight original points have

$$\eta_j = \pm(b_j/2)w, \quad (b_j) = (6, 25, 28, 11), \quad \xi_j = \eta_j^{-1}. \quad (\text{F7})$$

The complete original source coordinates follow without selecting another polynomial:

$$\begin{aligned} a_j &= \eta_j^{-1}, & y_j &= -t_j - i\eta_j, \\ z_j &= A\eta_j^3 + 2t_j\eta_j + 3i\eta_j^2, \\ w_j &= 7it_j^2\eta_j + (B - 17t_j + At_j^2)\eta_j^2 - 13i\eta_j^3 - 2A\eta_j^4, \\ q_j &= (a_j, y_j, z_j, w_j). \end{aligned} \quad (\text{F8})$$

The exact checker evaluates the original four-component Fable polynomial at all eight F8 points and obtains F4's target coefficients. It does not substitute a fitted map.

50.4 IV.4 Frobenius, trace and the surviving algebra

Since $w^{61} = -w$, Frobenius exchanges the two signs above every root. Its permutation is four disjoint transpositions. Thus the point counts over extensions are

$$N_n = \begin{cases} 0 & n \text{ odd}, \\ 8 & n \text{ even}, \end{cases}$$

and the local point-count zeta function is

$$\boxed{Z_{\mathcal{S}_1}(T) = \exp \left(\sum_{n \geq 1} N_n T^n / n \right) = (1 - T^2)^{-4}.} \quad (\text{F9})$$

This is the zeta function of this finite fibre, not the Riemann zeta function. Its Frobenius characteristic polynomial is $(z^2 - 1)^4$. All eight complex permutation eigenvalues have modulus

one, but their trace is zero. The finite algebra itself is the product of four quadratic fields over $\mathbb{F}_{61}$, of dimension eight, not the zero algebra.

There are also two different traces here. The trace of multiplication by the algebra unit is $\text{Tr}_{\mathcal{S}/\mathbb{F}_{61}}(1) = 8$, while the trace of Frobenius on the eight-point permutation space is zero. In the predecessor's residue identity the multiplier $2\eta^3$ is a unit on this good-reduction fibre, so the multiplication trace pairing is nondegenerate. Zero Frobenius trace does not make that pairing or the algebra vanish.

The corrected arithmetic conclusion is explicit: the original complex signed cover may require a field extension after good reduction; alternatively the nonsquare twist has the complementary rational-point count. Any proposed ES inference requiring a rational original signed lift at every good prime is disproved by F4–F8. A native analytic metric or a purity statement alone supplies neither that rational lift nor the universal ES occupancy theorem.

`check_finite_field.py` verifies the literal reciprocal identity, all coefficients and derivative square classes, all eight source points in the explicit quadratic field, their Frobenius action, and the formal point-count zeta identity. No transcendental receiving coefficient is given an invented finite-field image.

51 Closure conditions and continuing research

The arithmetic theorem is universal over witnesses at its specified primes. It is not a witness-production theorem. The central ES occupancy question remains, as does the integral receiving divisor outside the sufficient character classes. The next arithmetic calculation should keep the same integer N and refine the residual cases where one or more quadratic obstructions becomes a square; a failed sufficient symbol test is not itself a singularity.

For RH, the paired return is a necessary inequality for the original quotient and native metric. The aggregate spectral charge and the recurrence coefficients enter on different sides of a fully explicit formula. A proof that closes the programme still needs additional native arithmetic information, not a generic positive-measure argument that would apply equally to the diagnostic nonreal quotients.

The finite heat step now has explicit convergent remainders and certified small-time signs for its stated difference. It does not evaluate the earlier, distinct phase-sensitive projected current. The source's moment intervals, full quotient kernels, moving orthogonal-polynomial columns and two minimum residuals must be carried into any numerical or asymptotic application. The reusable routine takes exact inputs; a numerical midpoint is not automatically the exact native operator.

The finite-field example supplies a precise obstruction to one possible arithmetic inference and an explicit replacement via a quadratic extension or twist. It neither identifies a denominator with a zeta zero nor imports finite-field purity into the original infinite arithmetic action. The local zeta function in F9 belongs to a finite fibre.

52 References

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53 Cumulative scope and remaining arithmetic problem

Every theorem in this reader starts either from an actual positive integral Erdős–Straus witness or from an explicitly stated algebraic, local or finite-dimensional input. The integral normalization and finite-field descent theorems classify what happens after such a witness is present. The three-character theorem proves receiving-frame nonsingularity on its stated residue classes conditional only on witness existence; failure of one of its characters does not imply singularity. The determinant and heat estimates are exact finite-dimensional constraints in their retained metric and are not substituted for the assembled arithmetic action.

The unresolved ES statement remains the original occupancy problem: for each hard prime, force at least one positive integral exterior or middle source. No finite signed fibre, normalization, nilpotent completion, conductor index, local Galois character, Gram determinant, heat remainder or CRT control family supplies that source by itself. Likewise, none of the proved root-algebra, weighted-conductor or finite-field maps identifies an ES root with a nontrivial zero of the Riemann zeta function.

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